PrepFusion

Control Systems

Detailed Solutions • 1988–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Control Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7^{th} question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Contents

SOLUTIONS

1

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

Q1.20.1 MCQ 2020 1M Ans: A ● ○ ○

Taking the Laplace transform on both sides of the given differential equation assuming zero initial conditions:

$$s^2Y(s) + 4Y(s) = 6R(s)$$

$$(s^2 + 4)Y(s) = 6R(s)$$

$$H(s) = \frac{Y(s)}{R(s)} = \frac{6}{s^2 + 4}$$

The poles of the system are the roots of the characteristic equation obtained by equating the denominator polynomial to zero:

$$s^2 + 4 = 0 \implies s^2 = -4$$

$$s = \pm\sqrt{-4} \implies s = \pm 2j$$

$$\boxed{\text{(A)} \quad +2j, -2j}$$

Q1.20.2 MCQ 2020 2M Ans: A ● ● ○

Given the transfer function:

$$H(s) = \frac{s^2 + 100}{s - p}$$

The DC gain is evaluated at $\omega = 0$ ($s = j0$):

$$H(j\omega) = \frac{-\omega^2 + 100}{j\omega - p} \implies H(j0) = \frac{100}{-p} = 5 \implies p = -20$$

Thus, the complete system transfer function is:

$$H(s) = \frac{s^2 + 100}{s + 20} \implies H(j\omega) = \frac{100 - \omega^2}{j\omega + 20}$$

To find the unity gain frequency, set the magnitude $|H(j\omega)| = 1$ at $\omega = \omega_1$:

Magnitude Equation:

$$\frac{100 - \omega_1^2}{\sqrt{\omega_1^2 + 400}} = 1$$

$$\sqrt{\omega_1^2 + 400} = 100 - \omega_1^2$$

Squaring both sides:

$$\omega_1^2 + 400 = \omega_1^4 - 200\omega_1^2 + 10000$$

$$\omega_1^4 - 201\omega_1^2 + 9600 = 0$$

Solving Quadratic (Let $x = \omega_1^2$):

$$x^2 - 201x + 9600 = 0$$

$$x = \frac{201 \pm \sqrt{(-201)^2 - 4(1)(9600)}}{2}$$

$$x = \frac{201 \pm \sqrt{40401 - 38400}}{2}$$

$$x = 122.86 \quad \text{or} \quad 78.15$$

$$\omega_1 = \sqrt{122.86} \approx 11.08\,\text{rad/s} \quad \text{or} \quad \omega_1 = \sqrt{78.15} \approx 8.84\,\text{rad/s}$$

The question asks for the **smallest positive frequency**, which is closest to 8.84 rad/s.

$$\boxed{(A) \;\; 8.84}$$

Key Points

Unity gain frequencies are the frequencies where the magnitude of the system response $|H(j\omega)| = 1$.

Q1.15.1 NAT 2015 1M Ans: 0.23 to 0.25 ● ○ ○

Given the impulse response of an open-loop control system: $g(t) = e^{-2t}(\sin 5t + \cos 5t)$
The transfer function $G(s)$ is the Laplace transform of the impulse response:

$$G(s) = \frac{5}{(s+2)^2 + 5^2} + \frac{s+2}{(s+2)^2 + 5^2}$$

The DC gain of the control system is defined as the value of the transfer function at $s = 0$:

$$\text{DC gain} = G(s)|_{s=0}$$

Substituting $s = 0$ into the equation:

$$G(0) = \frac{5}{2^2 + 5^2} + \frac{2}{2^2 + 5^2} = \frac{5}{29} + \frac{2}{29} = \frac{7}{29} \approx 0.241$$

$$\boxed{0.241}$$

Q1.15.2 MCQ 2015 2M Ans: A ● ○ ○

Method 1

Let the unit step response be $y(t)$.

$$G(s) = \frac{Y(s)}{U(s)}$$

$$Y(s) = \left(\frac{1}{s}\right)\left(\frac{1-2s}{1+s}\right)$$

$$Y(s) = \frac{1+s-3s}{s(1+s)} = \frac{1}{s} - \frac{3}{1+s}$$

Taking the inverse Laplace transform:

$$y(t) = \mathcal{L}^{-1}[Y(s)]$$

$$y(t) = (1 - 3e^{-t})u(t)$$

The waveform of $y(t)$ starts at -2 when $t = 0$, and reaches 1 as $t \to \infty$.

Method 2

The Laplace transform of the output response is:

$$Y(s) = \frac{1-2s}{s(1+s)}$$

Applying the Initial Value Theorem to find $y(0^+)$:

$$\lim_{t\to 0^+} y(t) = \lim_{s\to\infty} sY(s)$$

$$\lim_{t\to 0^+} y(t) = \lim_{s\to\infty} \left[\frac{1-2s}{1+s}\right] = -2$$

Only the waveform in option (A) satisfies this initial condition.

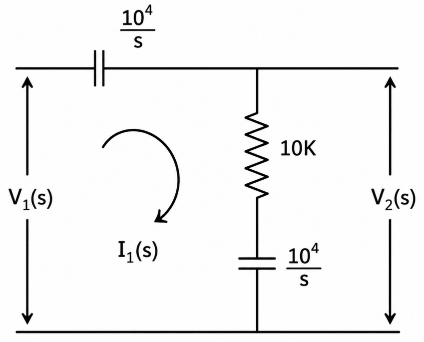
$$\boxed{(A)}$$

Q1.13.1 MCQ 2013 1M Ans: D ● ○ ○

Convert component values to Laplace impedances ($Z_C = \frac{1}{sC}$).

$$V_2(s) = V_1(s) \cdot \frac{R + \frac{1}{sC_2}}{\frac{1}{sC_1} + R + \frac{1}{sC_2}}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{10^4 + \frac{10^4}{s}}{\frac{10^4}{s} + 10^4 + \frac{10^4}{s}} = \frac{1 + \frac{1}{s}}{\frac{2}{s} + 1} = \frac{s+1}{s+2}$$



$$\boxed{(D)\ \frac{s+1}{s+2}}$$

Mistakes to be avoided

Incorrect Unit Conversion: Forgetting to convert $10\,\text{k}\Omega$ to $10^4\,\Omega$ or $100\,\mu\text{F}$ to 100×10^{-6} F leads to mismatched scaling factors in algebraic simplification.

Q1.04.1 MCQ | 2004 | 1M | Ans: C | ● ○ ○

A tachometer (or DC tachogenerator) is an electromechanical device that converts mechanical angular velocity into an electrical voltage signal.
The output voltage $e(t)$ generated by a tachometer is directly proportional to its angular speed $\omega(t)$:

$$e(t) = K_t \omega(t)$$

where $\omega(t)$ is the derivative of the rotor displacement $\theta(t)$ with respect to time: $\omega(t) = \frac{d\theta(t)}{dt}$

$$e(t) = K_t \frac{d\theta(t)}{dt}$$

Taking the Laplace transform on both sides under zero initial conditions: $E(s) = K_t \cdot s\Theta(s)$

$$\frac{E(s)}{\Theta(s)} = K_t s$$

$$\boxed{(C)\ K_t s}$$

Key Points

- When the input is rotor displacement $\theta(t)$, the transfer function is given by: $\frac{E(s)}{\Theta(s)} = K_t s$
- When the input is angular velocity $\omega(t)$, the transfer function is given by: $\frac{E(s)}{\Omega(s)} = K_t$

Q1.03.1 MCQ 2003 1M Ans: C

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 5x = 12\left(1 - e^{-2t}\right)$$

Taking the Laplace transform on both sides assuming zero initial conditions:

$$\left(s^2 + 6s + 5\right) X(s) = 12\left(\frac{1}{s} - \frac{1}{s+2}\right)$$

$$(s+1)(s+5)X(s) = 12\left(\frac{s+2-s}{s(s+2)}\right) = \frac{24}{s(s+2)}$$

$$X(s) = \frac{24}{s(s+1)(s+2)(s+5)}$$

To determine the steady-state response as $t \to \infty$, we apply the Final Value Theorem:

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) = \lim_{s\to 0} s\left[\frac{24}{s(s+1)(s+2)(s+5)}\right] = \frac{24}{(1)(2)(5)} = \frac{24}{10} = 2.4$$

Thus, the response of the system as $t \to \infty$ is $x = 2.4$.

$\boxed{\text{(C) x = 2.4}}$

Q1.03.2 MCQ 2003 2M Ans: B

The natural time constants of the response depend only on the poles of the system.

Taking the Laplace transform of the governing differential equation (with zero initial conditions for the characteristic behavior):

$$T(s) = \frac{C(s)}{R(s)} = \frac{1}{s^2 + \frac{s}{2} + \frac{1}{18}}$$

$$T(s) = \frac{18}{18s^2 + 9s + 1} = \frac{1}{(6s+1)(3s+1)}$$

Comparing this with the standard time-constant form:

$$\frac{1}{(1+sT_1)(1+sT_2)}, we\ get \quad T_1 = 6\,\text{sec}, \quad T_2 = 3\,\text{sec}$$

$\boxed{\text{(B) 3s and 6s}}$

Mistakes to be avoided

Confusing poles with time constants: Do not confuse the roots of the characteristic equation (poles, $s_1 = -1/6$ and $s_2 = -1/3$) with the time constants ($T = -1/s$). Selecting option (d) is a common error resulting from this confusion.

Q1.00.1 MCQ 2000 1M Ans: C

$$S_G^T = \frac{\partial T}{\partial G} \cdot \frac{G}{T} = \frac{1}{1 + G(s)H(s)}; \quad S_H^T = \frac{\partial T}{\partial H} \cdot \frac{H}{T} = \frac{-G(s)H(s)}{1 + G(s)H(s)}$$

Taking the magnitude of both expressions:

$$|S_G^T| = \frac{1}{|1 + G(s)H(s)|}; \quad |S_H^T| = \frac{|G(s)H(s)|}{|1 + G(s)H(s)|}$$

In a practical feedback control system, the loop gain magnitude is typically designed to be large ($|G(s)H(s)| > 1$) within the operating frequency range to track inputs accurately and reject noise. Consequently:

$$|S_H^T| > |S_G^T|$$

$$\boxed{\text{(C) less sensitive to forward-path parameter changes than to feedback path parameter changes}}$$

Key Points

Feedback configuration substantially reduces sensitivity to forward path variations but makes the system critically reliant on the stability of the feedback path components.

Q1.98.1 MCQ 1998 1M Ans: C ● ○ ○

Case-(i):

$$C(s) = F(s)R(s)$$

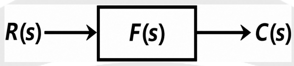


Fig. Solution Q1.98.1

Case-(ii):

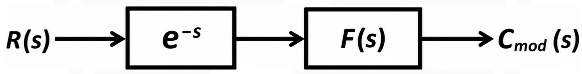


Fig. Solution Q1.98.1

$$C_{\text{mod}}(s) = R(s) \cdot e^{-s}F(s)$$

$$C_{\text{mod}}(s) = e^{-s}C(s)$$

Taking the inverse Laplace transform using the time-shifting property:

$$C_{\text{mod}}(t) = c(t-1)u(t-1)$$

$$\boxed{(C)\;\; c(t-1)u(t-1)}$$

Key Points

- **Time Shifting Property**: $\mathcal{L}^{-1}\{F(s)e^{-as}\} = f(t-a)u(t-a)$
- Multiplication by e^{-s} in the s-domain corresponds to an ideal time delay of 1s in the time domain.

Q1.96.1 MCQ 1996 1M Ans: B ● ○ ○

The closed-loop transfer function (C.L.T.F.) is the Laplace transform of the unit-impulse response:

$$\text{C.L.T.F.} = \mathcal{L}\{c(t)\} = \mathcal{L}\{-te^{-t} + 2e^{-t}\} = \frac{-1}{(s+1)^2} + \frac{2}{s+1} = \frac{2s+1}{(s+1)^2}$$

For a unity feedback system ($H(s) = 1$):

$$\text{C.L.T.F.} = \frac{G(s)}{1+G(s)} \implies G(s) = \frac{\text{C.L.T.F.}}{1-\text{C.L.T.F.}}$$

$$G(s) = \frac{\frac{2s+1}{(s+1)^2}}{1 - \frac{2s+1}{(s+1)^2}} = \frac{2s+1}{(s^2+2s+1)-(2s+1)} = \frac{2s+1}{s^2}$$

$$\boxed{(B)\quad \frac{2s+1}{s^2}}$$

Key Points

For unity feedback, the open-loop transfer function $G(s)$ is related to the closed-loop transfer function $T(s)$ by $G(s) = \dfrac{T(s)}{1-T(s)}$.

Q1.95.1

MCQ | 1995 | 1M | Ans: A

For a unit step input, $R(s) = \frac{1}{s}$. Therefore, the output response in the s-domain is:

$$C(s) = \frac{2(s-1)}{s(s+2)(s+1)}$$

Using partial fraction expansion:

$$C(s) = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1} = \frac{-1}{s} + \frac{-3}{s+2} + \frac{4}{s+1}$$

Taking the inverse Laplace transform to obtain the time-domain output $c(t)$:

$$c(t) = -3e^{-2t} + 4e^{-t} - 1$$

$$\boxed{(A)\quad -3e^{-2t} + 4e^{-t} - 1}$$

DEBUG INFO

Chapter I

Basics of Control Systems

Total Questions : 12

MCQ : 11

MSQ : 0

NAT : 1

SOLUTIONS

II

Block Diagram and Signal Flow Graphs

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

Q2.24.1 MCQ 2024 1M Ans: D ● ○ ○ CLICK FOR VIDEO SOLUTION

Method 1

Block Diagram Reduction:

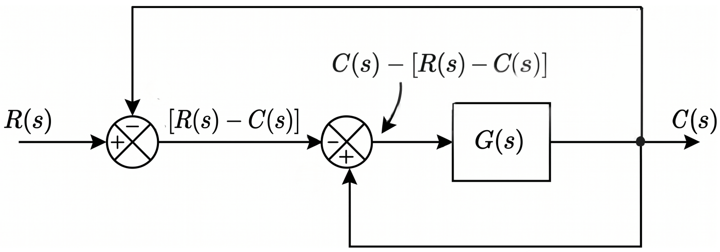


Fig. Solution Q2.24.1

$$C(s) = [C(s) - [R(s) - C(s)]]G(s)$$

$$C(s) = -R(s)G(s) + 2C(s)G(s)$$

$$C(s)[1 - 2G(s)] = -R(s)G(s)$$

$$\frac{C(s)}{R(s)} = -\frac{G(s)}{1 - 2G(s)}$$

Method 2

Mason's Gain Formula:

- Forward Path (P_1):

$$P_1 = (1) \times (-1) \times G(s) = -G(s)$$

- Individual Loops (L):

$$L_1 = (1) \times G(s) \times (1) = G(s)$$

$$L_2 = (-1) \times (-1) \times G(s) \times (1) = G(s)$$

- Path Determinant (Δ): Both loops touch each other and the forward path:

$$\Delta = 1 - (L_1 + L_2) = 1 - 2G(s)$$

$$\Delta_1 = 1$$

$$T.F = \frac{P_1\Delta_1}{\Delta} = \frac{-G(s)}{1 - 2G(s)}$$

$$\boxed{\text{(D)} \quad -\frac{G(s)}{1 - 2G(s)}}$$

Key Points

- Mason's Gain Formula: $\frac{C(s)}{R(s)} = \frac{\sum P_k \Delta_k}{\Delta}$
  - P_k: Gain of the k^{th} forward path
  - $\Delta = 1 -$ (sum of individual loop gains) + (sum of gain products of two non-touching loops) $- \ldots$
  - Δ_k: The cofactor of the k^{th} forward path, calculated by removing the loops touching the k^{th} path.

Q2.23.1 MCQ 2023 1M Ans: B ● ○ ○

By Mason's gain formula,

$$\frac{Y(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

Here, the forward path gains and their respective cofactor determinants are:

$$P_1 = 3,\ \Delta_1 = 1 \quad ; \quad P_2 = \frac{2}{s},\ \Delta_2 = 1$$

The system determinant (Δ) considering the feedback loop is: $\Delta = 1 - \frac{1}{s}$

$$\frac{Y(s)}{R(s)} = \frac{3 + \frac{2}{s}}{1 - \frac{1}{s}} = \frac{3s + 2}{s - 1}$$

$$\boxed{(B)\ \frac{3s+2}{s-1}}$$

Q2.21.1 MCQ 2021 1M Ans: C ● ● ○

Method 1

We can find the error transfer function $\frac{E(s)}{R(s)}$ by applying Mason's Gain Formula:

$$T = \frac{\sum P_k \Delta_k}{\Delta}$$

1. Forward Path for $E(s)$: Taking $R(s)$ as the input and $E(s)$ as the output node,

$$P_1 = 1,\ \Delta_1 = 1$$

2. Loop Gain and System Determinant:

$$L_1 = -GH$$

$$\Delta = 1 - L_1 = 1 - (-GH) = 1 + GH$$

Substituting these values into Mason's Gain Formula:

$$\frac{E(s)}{R(s)} = \frac{P_1\Delta_1}{\Delta} = \frac{1\times 1}{1+GH} = \frac{1}{1+GH}$$

Method 2

From the block diagram, the error signal $E(s)$ is the input to the plant G, and $C(s)$ is its output:

$$\frac{C(s)}{E(s)} = G \implies C(s) = G\cdot E(s) \quad \text{— (i)}$$

The standard closed-loop transfer function for a negative feedback system is:

$$\frac{C(s)}{R(s)} = \frac{G}{1+GH} \quad \text{— (ii)}$$

Substituting equation (i) into equation (ii):

$$\frac{G\cdot E(s)}{R(s)} = \frac{G}{1+GH}$$

$$\frac{E(s)}{R(s)} = \frac{1}{1+GH}$$

$$\boxed{\text{(C)}\ \frac{1}{1+GH}}$$

Key Points

The ratio $\frac{C(s)}{R(s)}$ represents the system transfer function, whereas $\frac{E(s)}{R(s)}$ represents the error transfer function.

Q2.20.1 MCQ 2020 1M Ans: C ● ○ ○

First, let's find the transfer function TF_{original} of the given signal flow graph using Mason's Gain Formula:

- Forward path: $P_1 = ad$, $\Delta_1 = 1$.
- Individual loops: $L_1 = cd$; $L_2 = ade$.
- $\Delta = 1 - (L_1 + L_2) = 1 - cd - ade = 1 - d(c + ae)$

$$TF_{\text{original}} = \frac{P_1\Delta_1}{\Delta} = \frac{ad}{1-d(c+ae)} \quad \text{— (i)}$$

Option (A):

- Forward Path: $P_1 = a(d+c)$
- Feedback Loop: $L_1 = a(d+c)e$

$$TF_A = \frac{a(d+c)}{1-ae(d+c)} \neq TF_{\text{original}}$$

Option (B):

- Forward Path: $P_1 = (a+c)d$
- Feedback Loop: $L_1 = (a+c)de$

$$TF_B = \frac{d(a+c)}{1-de(a+c)} \neq TF_{\text{original}}$$

Option (C):

- Forward Path: $P_1 = a\left(\frac{d}{1-cd}\right) = \frac{ad}{1-cd}$
- Feedback Loop: $L_1 = a\left(\frac{d}{1-cd}\right)e = \frac{ade}{1-cd}$

$$TF_C = \frac{\frac{ad}{1-cd}}{1-\frac{ade}{1-cd}} = \frac{ad}{1-cd-ade}$$

$$TF_C = \frac{ad}{1-d(c+ae)} = TF_{\text{original}}$$

Option (D):

- Forward Path: $P_1 = a\left(\frac{c}{1-cd}\right) = \frac{ac}{1-cd}$
- Feedback Loop: $L_1 = a\left(\frac{c}{1-cd}\right)e = \frac{ace}{1-cd}$

$$TF_D = \frac{\frac{ac}{1-cd}}{1-\frac{ace}{1-cd}} = \frac{ac}{1-cd-ace}$$

$$TF_D = \frac{ac}{1-c(d+ae)} \neq TF_{\text{original}}$$

$$\boxed{\text{(C)}}$$

Q2.17.1 MCQ 2017 2M Ans: A ● ○ ○

To find the transfer function $\frac{Y(s)}{U_1(s)}$, we apply the **principle of superposition** by setting the other independent input $U_2(s) = 0$.

- Forward path gain: $P_1 = 1 \cdot \frac{1}{L} \cdot \frac{1}{s} \cdot k_1 \cdot \frac{1}{J} \cdot \frac{1}{s} \cdot 1 = \frac{k_1}{LJs^2}$; $\Delta_1 = 1$
- Individual Feedback loops gain:

$$L_1 = -\frac{R}{Ls} \quad ; \quad L_2 = -\frac{k_1k_2}{LJs^2}$$

- Determinant (Δ):

$$\Delta = 1-(L_1+L_2) = 1+\frac{R}{Ls}+\frac{k_1k_2}{JLs^2}$$

Applying Mason's gain formula:

$$\frac{C(s)}{R(s)} = \frac{\sum_k P_k\Delta_k}{\Delta}$$

$$\frac{Y(s)}{U_1(s)} = \frac{\frac{k_1}{LJs^2}}{1+\frac{R}{Ls}+\frac{k_1k_2}{LJs^2}} = \frac{k_1}{LJs^2+RJs+k_1k_2}$$

$$\boxed{\text{(A)} \quad \frac{k_1}{JLs^2+JRs+k_1k_2}}$$

Q2.15.1 MCQ 2015 1M Ans: B ● ○ ○

Using Mason's Gain Formula with input $X_1(s) = 0$:

- Forward Path: $P_1 = G_2$ and $\Delta_1 = 1$
- Individual Loops: $L_1 = -G_1G_2$; $L_2 = -G_1$
- Graph Determinant (Δ): $\Delta = 1-(L_1+L_2) = 1-(-G_1G_2-G_1) = 1+G_1G_2+G_1 = 1+G_1(1+G_2)$

Transfer Function:

$$\frac{Y(s)}{X_2(s)} = \frac{P_1\Delta_1}{\Delta} = \frac{G_2}{1+G_1(1+G_2)}$$

$$\boxed{(B) \quad \frac{G_2}{1+G_1(1+G_2)}}$$

Q2.15.2 MCQ 2015 2M Ans: C ● ● ○

Method 1

Analytical Method

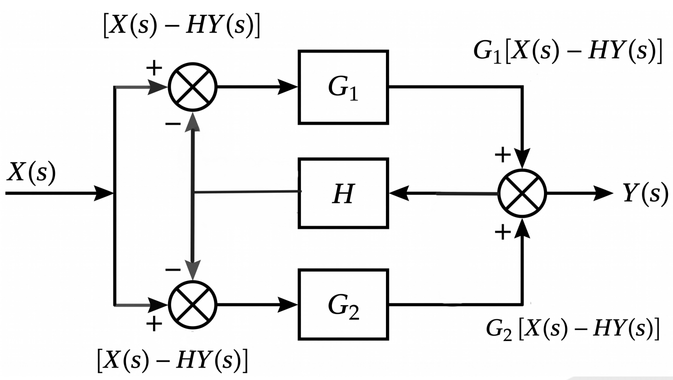


Fig. Solution Q2.15.2

From the block diagram, the system equations are:

$$Y(s) = [X(s) - HY(s)]G_1 + [X(s) - HY(s)]G_2$$

$$Y(s) = [X(s) - HY(s)][G_1 + G_2]$$

$$Y(s)[1 + H(G_1 + G_2)] = X(s)[G_1 + G_2]$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 + G_2}{1 + H(G_1 + G_2)}$$

Method 2

Signal Flow Graph (SFG)

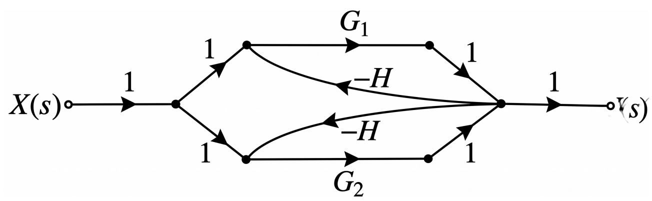


Fig. Solution Q2.15.2

- Forward paths: $P_1 = G_1, \quad P_2 = G_2$
- Individual loops: $L_1 = -HG_1, \quad L_2 = -HG_2$
- Path factors: $\Delta_1 = 1, \quad \Delta_2 = 1$

Applying Mason's gain formula:

$$\frac{Y(s)}{X(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{1 - (L_1 + L_2)}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 + G_2}{1 - (-HG_1 - HG_2)}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 + G_2}{1 + H(G_1 + G_2)}$$

$$\boxed{(C) \quad \frac{G_1 + G_2}{1 + H(G_1 + G_2)}}$$

Q2.14.1 MCQ 2014 1M Ans: C ● ○ ○

Given Signal Flow Graph has:

- Forward Paths: $P_1 = \frac{h_0}{s^2}$, $\Delta_1 = 1$ *and* $P_2 = \frac{h_1}{s}$, $\Delta_2 = \left(1 + \frac{a_1}{s}\right)$
- Individual Loops: $L_1 = -\frac{a_1}{s}$ *and* $L_2 = -\frac{a_0}{s^2}$
- Graph Determinant : $\Delta = 1 - (L_1 + L_2) = 1 - \left(-\frac{a_1}{s} - \frac{a_0}{s^2}\right) = \frac{s^2 + a_1 s + a_0}{s^2}$

Transfer Function ($G(s)$):

$$G(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{\left[\frac{h_0}{s^2}(1) + \frac{h_1}{s}\left(\frac{s + a_1}{s}\right)\right]}{\frac{s^2 + a_1 s + a_0}{s^2}}$$

$$G(s) = \frac{h_0 + h_1(s + a_1)}{s^2 + a_1 s + a_0} = \frac{(b_0 - b_1 a_1) + b_1(s + a_1)}{s^2 + a_1 s + a_0} = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$$

$$\boxed{(C) \quad G(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}}$$

Q2.14.2 MCQ 2014 2M Ans: B

Method 1

Block diagram Reduction :

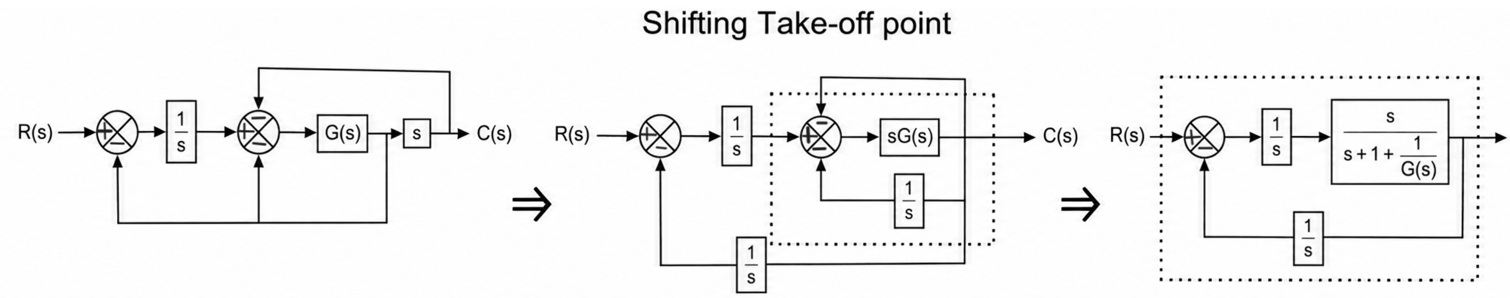


Fig. Solution Q2.14.2

$$\frac{C(s)}{R(s)} = \frac{\dfrac{1}{\dfrac{1}{G(s)}+s+1}}{1+\dfrac{1}{s\left[\dfrac{1}{G(s)}+s+1\right]}} = \frac{s}{\dfrac{s}{G(s)}+s^2+s+1}$$

After comparing the given desired system transfer function equation $\dfrac{C(s)}{R(s)} = \dfrac{s}{s^2+s+2}$, with equation:

$$\frac{s}{G(s)}+s^2+s+1 = s^2+s+1 \implies \frac{s}{G(s)} = 1$$

$$G(s) = s$$

Method 2

Using Mason's gain formula from the block diagram:

- Forward path: $P_1 = G(s) \quad ; \; \Delta_1 = 1-0 = 1$
- Individual feedback loop gains: $L_1 = -G(s), \;\; L_2 = -\dfrac{G(s)}{s}, \;\; L_3 = -sG(s)$
- Determinant (Δ):

$$\Delta = 1-(L_1+L_2+L_3) = 1-\left(-G(s)-sG(s)-\frac{G(s)}{s}\right) = 1+G(s)+sG(s)+\frac{G(s)}{s}$$

Applying Mason's gain formula:

$$\frac{C(s)}{R(s)} = \frac{P_1\Delta_1}{\Delta} = \frac{G(s)}{1+G(s)\left[1+s+\dfrac{1}{s}\right]} = \frac{1}{\dfrac{1}{G(s)}+1+s+\dfrac{1}{s}} = \frac{s}{\dfrac{s}{G(s)}+s^2+s+1}$$

Comparing this with the given desired transfer function $\dfrac{C(s)}{R(s)} = \dfrac{s}{s^2+s+2}$:

$$\frac{s}{G(s)}+s^2+s+1 = s^2+s+2$$

$$\frac{s}{G(s)} = 1 \implies G(s) = s$$

$$\boxed{(B)\ s}$$

Q2.13.1 MCQ 2013 2M Ans: A ● ○ ○

Using Mason's gain formulae:

- Forward Paths: $P_1 = s^{-2} = \frac{1}{s^2}$ and $P_2 = s^{-1} = \frac{1}{s}$
- Feedback Loops: $L_1 = -2s^{-1}$, $L_2 = -2s^{-2}$, $L_3 = -4$, $L_4 = -4s^{-1}$
- System Determinant (Δ):

$$\Delta = 1 - [-2s^{-1} - 2s^{-2} - 4 - 4s^{-1}]$$

$$\Delta = 1 + \frac{2}{s} + \frac{2}{s^2} + 4 + \frac{4}{s} = \frac{5s^2 + 6s + 2}{s^2}$$

- Path Cofactors: $\Delta_1 = 1$, $\Delta_2 = 1$

Applying the transfer function formula:

$$\frac{Y(s)}{U(s)} = \frac{\sum P_k \Delta_k}{\Delta} = \frac{\frac{1}{s^2} \times 1 + \frac{1}{s} \times 1}{\frac{5s^2 + 6s + 2}{s^2}} = \frac{s + 1}{5s^2 + 6s + 2}$$

$$\boxed{(A)\ \frac{s+1}{5s^2 + 6s + 2}}$$

Q2.10.1 MCQ 2010 1M Ans: A ● ● ○

Given:

- Forward path gain, $G = 100 \pm 10\%$
- Gain variation, $\frac{\Delta G}{G} = 10\% = 0.1$
- Feedback path value, $H = \frac{9}{100}$

The overall closed-loop system gain (T) for a negative feedback system is given by:

$$T = \frac{G}{1 + GH}$$

Substituting the nominal values:

$$T = \frac{100}{1 + 100 \times \left(\frac{9}{100}\right)} = \frac{100}{1 + 9} = 10$$

The sensitivity of overall gain (T) with respect to forward path gain (G) is:

$$S_G^T = \frac{\partial T/T}{\partial G/G} = \frac{1}{1 + GH}$$

$$S_G^T = \frac{1}{1 + 100 \times \left(\frac{9}{100}\right)} = \frac{1}{10}$$

Now, find the fractional change in the overall gain ($\frac{\Delta T}{T}$):

$$\frac{\Delta T}{T} = S_G^T \times \left(\frac{\Delta G}{G}\right) = \frac{1}{10} \times 10\% = 1\%$$

Therefore, the overall system gain with its tolerance is:

$$T = 10 \pm 1\%$$

$$\boxed{(A)\ 10 \pm 1\%}$$

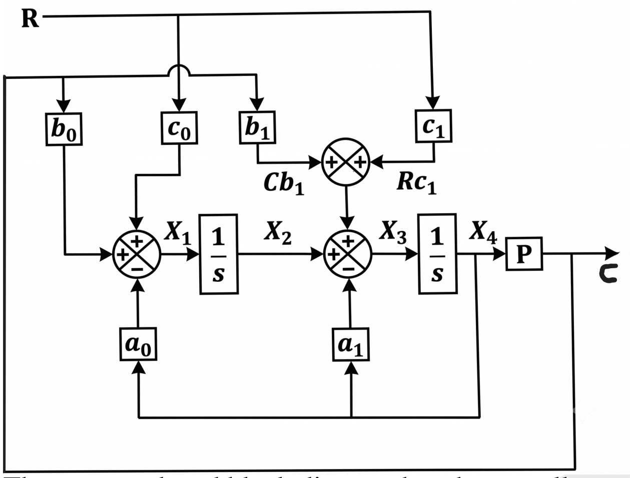
Key Points

Negative feedback reduces the variation in overall gain caused by parameter changes in the forward path by a factor of $(1 + GH)$.

Q2.07.1 MCQ 2007 2M Ans: D ● ● ●

Method 1

Block Diagram Reduction:



The target reduced block diagram has the overall transfer function:

$$\frac{C}{R} = \frac{XYP}{1 - YPZ}$$

From the above block diagram :

$$C = PX_4 \implies X_4 = \frac{C}{P}$$

$$X_1 = Rc_0 + Cb_0 - X_4a_0 \implies X_1 = Rc_0 + Cb_0 - \frac{Ca_0}{P}$$

$$X_2 = \frac{X_1}{s} = \frac{Rc_0}{s} + \frac{Cb_0}{s} - \frac{Ca_0}{Ps}$$

$$X_3 = X_2 - X_4a_1 + Cb_1 + Rc_1$$

$$X_3 = \frac{Rc_0}{s} + \frac{Cb_0}{s} - \frac{Ca_0}{Ps} - \frac{Ca_1}{P} + Cb_1 + Rc_1$$

$$X_4 = \frac{X_3}{s}$$

$$\frac{C}{P} = \frac{Rc_0}{s^2} + \frac{Cb_0}{s^2} - \frac{Ca_0}{Ps^2} - \frac{Ca_1}{Ps} + \frac{Cb_1}{s} + \frac{Rc_1}{s}$$

Rearranging terms to separate C and R:

$$\frac{C}{R} = \frac{P[c_1s + c_0]}{s^2 + a_1s + a_0 - P[b_1s + b_0]}$$

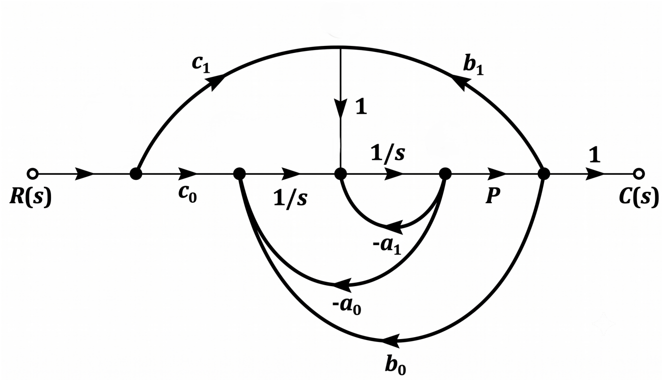
$$\frac{C}{R} = \frac{\dfrac{P[c_1s + c_0]}{s^2 + a_1s + a_0}}{1 - \dfrac{P[b_1s + b_0]}{s^2 + a_1s + a_0}}$$

Comparing this with the target format $\frac{XYP}{1-YPZ}$:

$$X = c_1s + c_0 \quad Y = \frac{1}{s^2 + a_1s + a_0} \quad Z = b_1s + b_0$$

Method 2

Signal Flow Graph (SFG) Method:



Using Mason's Gain Formula, the forward path gains are:

$$P_1 = \frac{Pc_0}{s^2}, \quad P_2 = \frac{Pc_1}{s}$$

The individual feedback loop gains are:

$$L_1 = -\frac{a_1}{s}, \quad L_2 = -\frac{a_0}{s^2} \quad L_3 = \frac{Pb_0}{s^2}, \quad L_4 = \frac{Pb_1}{s}$$

The system determinant (Δ) with no non-touching loops:

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4)$$

$$\Delta = 1 - \left(-\frac{a_1}{s} - \frac{a_0}{s^2} + \frac{Pb_0}{s^2} + \frac{Pb_1}{s}\right)$$

Since all loops touch both forward paths, $\Delta_1 = 1$ and $\Delta_2 = 1$. The overall transfer function is:

$$\frac{C}{R} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$\frac{C}{R} = \frac{\dfrac{Pc_0}{s^2} + \dfrac{Pc_1}{s}}{1 + \dfrac{a_1}{s} + \dfrac{a_0}{s^2} - \dfrac{Pb_1}{s} - \dfrac{Pb_0}{s^2}}$$

$$\frac{C}{R} = \frac{P(c_1s + c_0)}{s^2 + a_1s + a_0 - P(b_1s + b_0)}$$

$$\frac{C}{R} = \frac{(c_1s + c_0)\cdot\left(\dfrac{1}{s^2 + a_1s + a_0}\right)\cdot P}{1 - \left(\dfrac{1}{s^2 + a_1s + a_0}\right)\cdot P\cdot(b_1s + b_0)}$$

Comparing with $\frac{XYP}{1-YPZ}$, we directly obtain:

$$X = c_1s + c_0, \quad Y = \frac{1}{s^2 + a_1s + a_0}, \quad Z = b_1s + b_0$$

$$\boxed{(D)\ \ X = c_1s + c_0, Y = \frac{1}{(s^2 + a_1s + a_0)}, Z = b_1s + b_0}$$

Q2.04.1 MCQ 2004 2M Ans: B

- Forward paths: $P_1 = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$; $P_2 = 1 \cdot \frac{1}{s} = \frac{1}{s}$; $P_3 = 1$
- The graph determinant is: $\Delta = 1$
- Since there are no loops, all forward paths touch all parts of the graph : $\Delta_1 = \Delta_2 = \Delta_3 = 1$

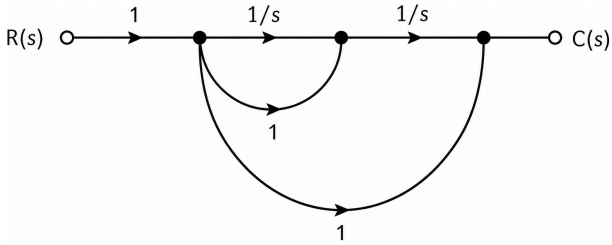


Fig. Solution Q2.04.1

Applying Mason's gain formula:

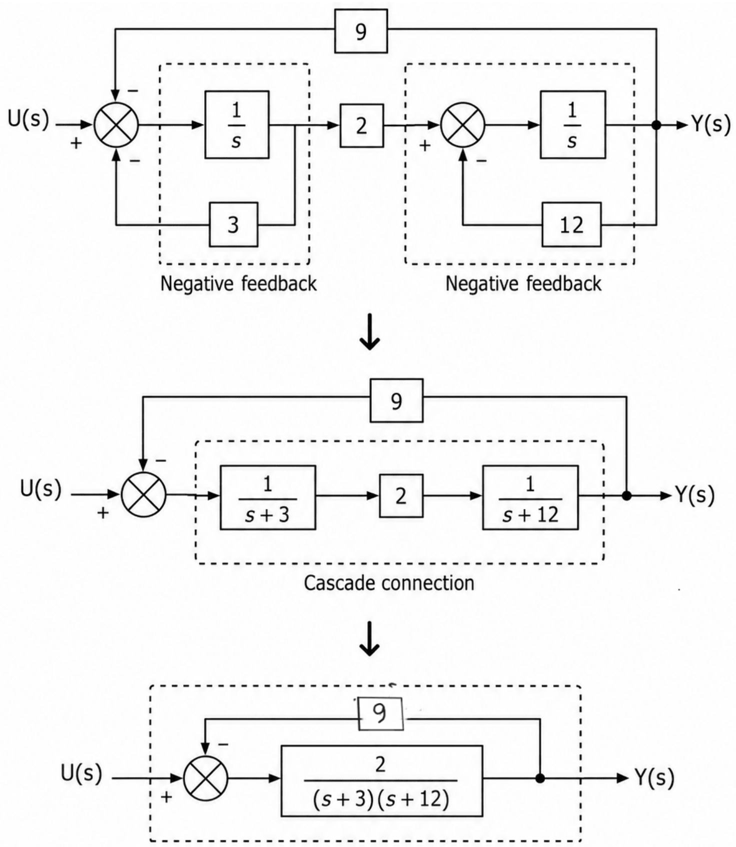
$$\frac{C(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2 + P_3\Delta_3}{\Delta} = \frac{\frac{1}{s^2} \cdot 1 + \frac{1}{s} \cdot 1 + 1 \cdot 1}{1} = \frac{s^2 + s + 1}{s^2}$$

$$\boxed{(B) \quad \frac{s^2 + s + 1}{s^2}}$$

Q2.03.1 MCQ 2003 2M Ans: B

Method 1

Block Diagram Reduction Technique:



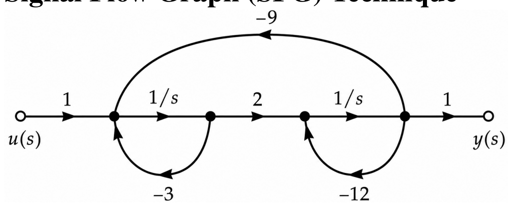
$$G(s) = \frac{\dfrac{2}{(s+3)(s+12)}}{1 + \dfrac{18}{(s+3)(s+12)}} = \frac{2}{(s+3)(s+12)+18}$$

$$= \frac{2}{s^2+15s+54} = \frac{2}{(s+9)(s+6)} = \frac{1}{27\left(1+\frac{s}{6}\right)\left(1+\frac{s}{9}\right)}$$

$$\boxed{(B)\quad \frac{1}{27\left(1+\dfrac{s}{6}\right)\left(1+\dfrac{s}{9}\right)}}$$

Method 2

Signal Flow Graph (SFG) Technique



- Forward path: $P_1 = \dfrac{2}{s^2}, \Delta_1 = 1$
- Individual loops:

$$L_1 = -\frac{3}{s}, \quad L_2 = -\frac{12}{s}, \quad L_3 = -\frac{18}{s^2}$$

- Non-touching loops (L_1 and L_2):

$$L_1 L_2 = \frac{36}{s^2}$$

The graph determinant (Δ) is:

$$\Delta = 1 - (L_1 + L_2 + L_3) + L_1 L_2 = 1 + \frac{3}{s} + \frac{12}{s} + \frac{18}{s^2} + \frac{36}{s^2}$$

Applying Mason's Gain Formula:

$$G(s) = \frac{P_1 \Delta_1}{\Delta} = \frac{\dfrac{2}{s^2}}{1 + \dfrac{15}{s} + \dfrac{54}{s^2}} = \frac{2}{(s+9)(s+6)}$$

$$= \frac{2}{54\left(1+\frac{s}{9}\right)\left(1+\frac{s}{6}\right)} = \frac{1}{27\left(1+\frac{s}{6}\right)\left(1+\frac{s}{9}\right)}$$

Q2.98.1 MCQ | 1998 | 1M | Ans: A | ● ○ ○

Using Mason's Gain Formula on the given signal flow graph:

- **Forward path:** $P_1 = G_1 G_2 G_3; \quad \Delta_1 = 1$
- **Individual loops:** $L_1 = -G_1 G_2 H_1; \quad L_2 = -G_2 G_3 H_2$

$$\Delta = 1 - (L_1 + L_2) = 1 + G_1 G_2 H_1 + G_2 G_3 H_2$$

$$\text{T.F.} = \frac{P_1 \Delta_1}{\Delta} = \frac{G_1 G_2 G_3 (1)}{1 + G_1 G_2 H_1 + G_2 G_3 H_2}$$

$$\boxed{(A)\quad \frac{G_1 G_2 G_3}{1 + H_2 G_2 G_3 + H_1 G_1 G_2}}$$

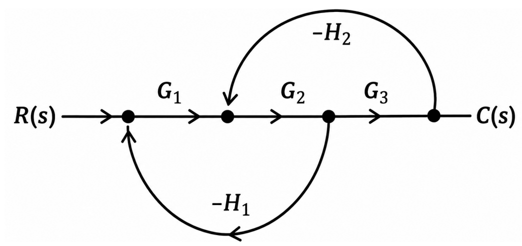


Fig. Solution Q2.98.1

Q2.93.1 MCQ 1993 1M Ans: B ● ○ ○

A signal flow graph (SFG) is a graphical representation of a set of simultaneous linear algebraic equations. It provides a visual tool to simplify complex control loop architectures and systematically compute the **overall closed-loop transfer function** between the input node and the output node using Mason's Gain Formula.

$$\boxed{\text{(B) transfer function of a system}}$$

Q2.92.1 MCQ 1992 1M Ans: B ● ○ ○

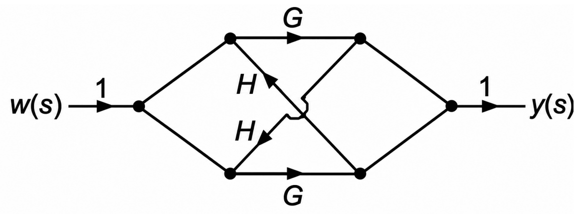


Fig. Solution Q2.92.1

- The forward are: $P_1 = G, \quad P_2 = G, \quad P_3 = G^2H, \quad P_4 = G^2H \ \ and \ \ \Delta_1 = 1, \quad \Delta_2 = 1, \quad \Delta_3 = 1, \quad \Delta_4 = 1$
- The system has one feedback loop with gain: Loop $= G^2H^2$
- The system determinant Δ is given by: $\Delta = 1 - G^2H^2$

$$\text{T.F.} = \frac{P_1\Delta_1 + P_2\Delta_2 + P_3\Delta_3 + P_4\Delta_4}{\Delta} = \frac{G + G + G^2H + G^2H}{1 - G^2H^2} = \frac{2G(1 + GH)}{(1 + GH)(1 - GH)} = \frac{2G}{1 - GH}$$

$$\boxed{(B) \ \frac{2G}{1 - GH}}$$

Q2.91.1 NAT 1991 2M Ans: 4,4 ● ○ ○

- Number of forward paths = 4 (i.e., abdfg, ahfg, aklfg, akmg)
- Number of loops = 4 (i.e., c, de, lfn, mn)

$$\boxed{4, 4}$$

Key Points

- **Forward Path:** A path from the input node to the output node that does not cross any node more than once.
- **Feedback Loop:** A closed path that starts and ends at the same node, without visiting any other node more than once.

DEBUG INFO

Chapter II

Block Diagram and Signal Flow Graphs

Total Questions : 18

MCQ : 17

MSQ : 0

NAT : 1

SOLUTIONS

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters and their variation
6. Steady State Error - Unity and Non-Unity Feedback
7. Error Constants and System Types

Q3.26.1 MCQ 2026 1M Ans: A ● ● ○

The step response of the system in the Laplace domain is given by:

$$Y(s) = \frac{100}{s(s+100)} = \frac{1}{s} - \frac{1}{s+100}$$

Taking the inverse Laplace transform, the time-domain response is:

$$y(t) = 1 - e^{-100t}$$

Comparing this with the standard first-order system response $y(t) = 1 - e^{-t/\tau}$:

- **Time Constant (τ):** $\tau = \dfrac{1}{100} = 0.01\,\text{s}$
- **Rise Time (T_r):** The time taken to go from 0.1 to 0.9 of its final value:

$$t_1 : 1 - e^{-100t_1} = 0.1 \implies e^{-100t_1} = 0.9 \implies t_1 = \frac{\ln(1/0.9)}{100}$$

$$t_2 : 1 - e^{-100t_2} = 0.9 \implies e^{-100t_2} = 0.1 \implies t_2 = \frac{\ln(1/0.1)}{100}$$

$$T_r = t_2 - t_1 = \frac{\ln(9)}{100} = \frac{2.197}{100} \approx 0.022\,\text{s}$$

- **Settling Time (T_s):** The time taken to reach 0.98 of its final value (2% tolerance):

$$1 - e^{-100T_s} = 0.98 \implies e^{-100T_s} = 0.02 \implies T_s = \frac{\ln(50)}{100} = \frac{3.912}{100} \approx 0.04\,\text{s}$$

$$\boxed{\text{(A)}\ T_r = 0.022,\ T_s = 0.04,\ T_c = 0.01}$$

Key Points

- For a first-order system response $y(t) = 1 - e^{-t/\tau}$, the time constant τ is the time taken to reach 63.2% of the steady-state value.
- **Shortcut Formulations:**
  - Rise time (10% to 90%): $T_r = \tau \ln(9) \approx 2.2\tau$
  - Settling time (2% tolerance): $T_s = \tau \ln(50) \approx 4\tau$

Q3.25.1 MCQ 2025 1M Ans: B ● ○ ○

The step response of a stable first-order linear time-invariant (LTI) system is represented as:

$$y(t) = K_{ss}(1 - e^{-t/\tau})$$

From the given data table, as $t \to \infty$, the output reaches its final value: $K_{ss} = 3$

$$y(t) = 3(1 - e^{-t/\tau}) \implies e^{-t/\tau} = 1 - \frac{y(t)}{3}$$

Taking the data point at $t = 0.6$ sec where $y(t) = 0.78$:

$$e^{-0.6/\tau} = 1 - \frac{0.78}{3}$$

$$-\frac{0.6}{\tau} = \ln(0.74) \approx -0.3011$$

$$\tau = \frac{0.6}{0.3011} \approx 1.993\,\text{sec}$$

$$\boxed{\text{(B)}\ \ 2}$$

Q3.24.1 MCQ 2024 1M Ans: B ● ○ ○ CLICK FOR VIDEO SOLUTION

We know that,

$$\zeta = \cos\theta$$

The standard 2% settling time expression is defined as:

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{(\cos\theta)\omega_n}$$

For System 1: Given $\omega_{n1} = 3\,\text{rad/sec}$ and $\theta_1 = 60°$:

$$\zeta_1 = \cos(60°) = 0.5$$

$$T_{s1} = \frac{4}{3 \times 0.5} \approx 2.67\,\text{sec}$$

For System 2: Given $\omega_{n2} = 1\,\text{rad/sec}$ and $\theta_2 = 70°$:

$$\zeta_2 = \cos(70°) \approx 0.342$$

$$T_{s2} = \frac{4}{1 \times 0.342} \approx 11.69\,\text{sec}$$

Comparing the two settling times, we find that $T_{s2} > T_{s1}$.

$$\boxed{\text{(B) Settling time of System 2 is more than that of System 1}}$$

Key Points

Pole Position and Transient Metrics: The settling time is inversely proportional to the horizontal distance ($\sigma = \zeta\omega_n$) of the poles from the imaginary axis. The closer the poles are to the $j\omega$-axis, the larger the settling time.

Q3.24.2 MCQ 2024 1M Ans: B ● ● ○

Before approximation: The overall transfer function $T(s)$ is:

$$T(s) = \frac{C(s)}{R(s)} = \frac{s+40}{(s+1)(s+20)}$$

Calculating the original DC gain at $s = 0$:

$$\text{DC gain } (s=0) = \frac{40}{1 \times 20} = 2$$

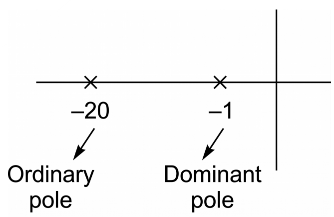


Fig. Solution Q3.24.2

After approximation: Converting to time-constant form to neglect the faster transient component (insignificant pole at $s = -20$):

$$T(s) = \frac{40\left(\frac{s}{40}+1\right)}{(s+1)20\left(\frac{s}{20}+1\right)}$$

Neglecting the higher frequency terms $\frac{s}{40}$ and $\frac{s}{20}$ preserves the steady-state performance; DC gain $(s=0) = 2$

$$T(s) = \frac{2}{s+1}$$

$$\boxed{\text{(B)}\ \frac{2}{s+1}}$$

Mistakes to be avoided

The value of DC gain must remain the same after approximation.: Do not simply remove the $(s + 20)$ term from the denominator without accounting for its constant scaling effect. Forgetting to convert to time-constant form will incorrectly yield a DC gain of 40.

Q3.22.1 MCQ 2022 2M Ans: A

To find the closed-loop transfer function, we first reduce the inner minor feedback loop.

$$G_{\text{inner}}(s) = \frac{G_1(s)}{1 + G_1(s)H_1(s)} = \frac{\frac{10}{s}}{1 + \frac{10}{s}} = \frac{10}{s + 10}$$

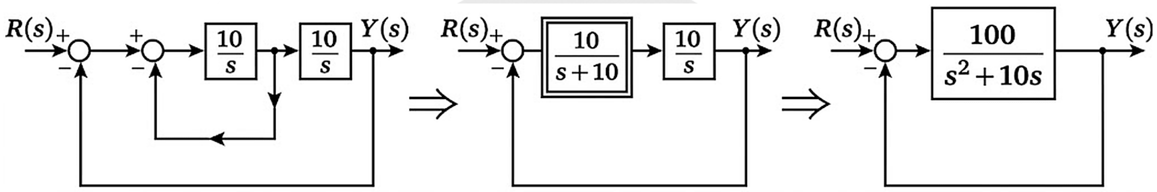


Fig. Solution Q3.22.1

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{\frac{100}{s^2+10s}}{1 + \frac{100}{s^2+10s}} = \frac{100}{s^2 + 10s + 100}$$

Comparing this with the standard second-order characteristic equation: $\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

We get:

$$\omega_n^2 = 100 \implies \omega_n = 10\,\text{rad/s}$$

$$2\zeta\omega_n = 10 \implies 2\zeta(10) = 10 \implies \zeta = 0.5$$

$$\boxed{\text{(A)}\ \ \zeta = 0.5\ \ and\ \ \omega_n = 10\,\text{rad/s}}$$

Q3.20.1 MCQ 2020 2M Ans: B

Method 1

Given input $V(s) = \frac{5}{s}$. The steady-state speed is 10 rad/s and at $t = 0.5$ s, speed is 6.32 rad/s. For a first-order system response:

$$\omega(t) = \omega(\infty)\left(1 - e^{-t/\tau}\right)$$

$$6.32 = 10\left(1 - e^{-0.5/\tau}\right)$$

$$-\frac{0.5}{\tau} = \ln(0.368) \approx -1 \implies \tau = 0.5\,\text{s}$$

Standard transfer function layout:

$$H(s) = \frac{\Omega(s)}{V(s)} = \frac{K}{1 + \tau s}$$

By final value theorem:

$$\omega(\infty) = \lim_{s\to 0} s \cdot H(s) \cdot \frac{5}{s} = 5K$$

$$10 = 5K \implies K = 2$$

Substituting K and τ:

$$H(s) = \frac{2}{0.5s + 1}$$

Method 2

We can check the given options using the Final Value Theorem condition:

$$\omega(\infty) = \lim_{s\to 0} s \cdot H(s) \cdot \frac{5}{s} = 5 \cdot H(0) = 10\,\text{rad/s}$$

Thus, $H(0)$ must equal 2.

Checking Option (A):

$$H(0) = \frac{10}{0.5(0) + 1} = 10 \neq 2$$

Checking Option (B):

$$H(0) = \frac{2}{0.5(0) + 1} = 2 \quad \text{(Satisfied)}$$

Checking Option (C):

$$H(0) = \frac{2}{0 + 0.5} = 4 \neq 2$$

Checking Option (D):

$$H(0) = \frac{10}{0 + 0.5} = 20 \neq 2$$

$$\boxed{\text{(B)}\ \frac{2}{0.5s + 1}}$$

Key Points

Final Value Theorem states $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$, applicable only when all poles of $sF(s)$ lie strictly in the left half of the s-plane.

Mistakes to be avoided

Ensure to include the step input magnitude (5 V) when computing the system's steady state response value. Do not evaluate $H(0)$ directly as the target response value (10 rad/s) without accounting for input gain.

Q3.18.1 MCQ 2018 1M Ans: C ● ○ ○

The standard transfer function of a second-order system is: $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

System P:

$$P(s) = \frac{15}{s^2 + 5s + 15}$$

$$\omega_n^2 = 15 \implies \omega_n = \sqrt{15}\,\text{rad/s}$$

$$2\zeta\omega_n = 5$$

$$\zeta = \frac{5}{2\sqrt{15}} \approx 0.645$$

Since $0 < \zeta < 1$:

Underdamped

System Q:

$$Q(s) = \frac{25}{s^2 + 10s + 25}$$

$$\omega_n^2 = 25 \implies \omega_n = 5\,\text{rad/s}$$

$$2\zeta\omega_n = 10$$

$$\zeta = \frac{10}{2(5)} = 1$$

Since $\zeta = 1$:

Critically damped

System R:

$$R(s) = \frac{35}{s^2 + 18s + 35}$$

$$\omega_n^2 = 35 \implies \omega_n = \sqrt{35}\,\text{rad/s}$$

$$2\zeta\omega_n = 18$$

$$\zeta = \frac{18}{2\sqrt{35}} \approx 1.521$$

Since $\zeta > 1$:

Overdamped

$$\boxed{(C)\ \ P-III,\ Q-II,\ R-I}$$

Key Points

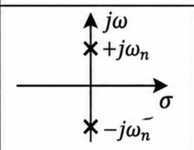
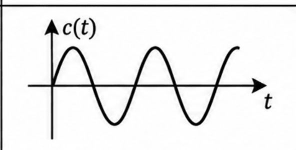
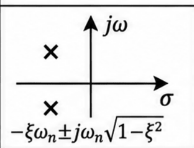
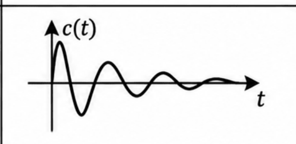
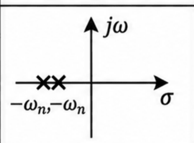
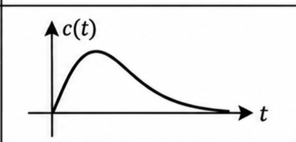
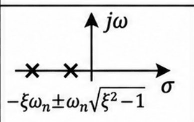
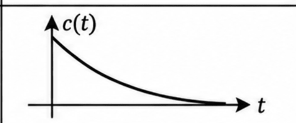
ξ	Damping	s-plane	Impulse Response	Poles	Stability
$\xi = 0$	Undamped			Imaginary Conjugate	Marginally stable
$0 < \xi < 1$	Underdamped			Complex conjugate roots on the LHS of s-plane	Stable
$\xi = 1$	Critically damped			Negative Equal Real Roots	Stable
$\xi > 1$	Overdamped			Negative Unequal Real Roots	Stable

Fig. Solution Q3.18.1

Q3.18.2 NAT 2018 1M Ans: 0.65 to 0.69

Given a unity feedback system ($H(s) = 1$) with the forward path transfer function:

$$G(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s^2 + 3s + 2}$$

Method 1

Standard Limit Approach:
Using the standard definition of the positional error constant:

$$K_p = \lim_{s \to 0} G(s)H(s)$$

$$K_p = \lim_{s \to 0} \frac{1}{(s+1)(s+2)}$$

$$K_p = \frac{1}{(0+1)(0+2)} = \frac{1}{2} = 0.5$$

Method 2

Shortcut Constant Method:
For any Type-0 system expressed in polynomial form, K_p can be directly computed as:

$$K_p = \frac{\text{Numerator constant}}{\text{Denominator constant}}$$

$$K_p = \frac{1}{2} = 0.5$$

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+0.5} = \frac{2}{3} \approx 0.67$$

$$\boxed{0.67}$$

Q3.18.3 NAT 2018 2M Ans: 8 ● ○ ○

Given the open-loop transfer function of a unity feedback system:

$$G(s)H(s) = \frac{K}{(s+1)^2(s+2)}$$

For a closed-loop system, the response $C(s)$ in terms of the reference input $R(s)$ is given by:

$$\frac{C(s)}{R(s)} = \frac{G(s)H(s)}{1+G(s)H(s)}$$

$$C(s) = \frac{R(s)K}{(s+1)^2(s+2)+K}$$

Given that the input is a unit step, $R(s) = \frac{1}{s}$; $C(s) = \frac{K}{s\left[(s+1)^2(s+2)+K\right]}$

Using the Final Value Theorem, the steady-state value of the response is:

$$\lim_{t \to \infty} c(t) = \lim_{s \to 0} sC(s)$$

From the given step response curve, the steady-state value is 0.8. Therefore:

$$\lim_{s \to 0} sC(s) = 0.8$$

$$\frac{K}{(0+1)^2(0+2)+K} = 0.8 \implies \frac{K}{2+K} = 0.8$$

$$\boxed{K = 8}$$

Mistakes to be avoided

Do not apply final value theorem blindly as it is not applicable for unstable systems: Always ensure the closed-loop system is stable for the calculated range of K.

Q3.17.1 NAT 2017 2M Ans: 0.550 to 0.556

The input is a unit step, so $X(s) = \frac{1}{s}$. The Laplace transform of the output response $Y(s)$ is:

$$Y(s) = G(s) \cdot \frac{1}{s} = \frac{-s+1}{s(s+1)}$$

Using partial fraction expansion:

$$Y(s) = \frac{A}{s} + \frac{B}{s+1} = \frac{1}{s} - \frac{2}{s+1}$$

Taking the inverse Laplace transform to find the time-domain response $y(t)$ for $t \geq 0$:

$$y(t) = u(t) - 2e^{-t}u(t) = 1 - 2e^{-t}$$

We need to calculate the value of the response at $t = 1.5$ sec:

$$y(1.5) = 1 - 2e^{-1.5}$$

$$y(1.5) = 1 - 2(0.22313) = 0.55374$$

$$\boxed{0.554}$$

Key Points

All-Pass Filter Response: The system $G(s) = \frac{-s+1}{s+1}$ is a first-order all-pass filter. It introduces phase shift while keeping the magnitude constant across frequencies.

Q3.17.2 MCQ 2017 1M Ans: D

Given the closed-loop transfer function:

$$\frac{C(s)}{R(s)} = \frac{Ks+b}{s^2+as+b}$$

Method 1

We know,

$$e(t) = r(t) - c(t)$$

Applying the Final Value Theorem:

$$e_{ss} = \lim_{s\to0} s[R(s) - C(s)]$$

$$e_{ss} = \lim_{s\to0} sR(s)\left[1 - \frac{C(s)}{R(s)}\right]$$

Substituting $R(s) = \frac{1}{s^2}$:

$$e_{ss} = \lim_{s\to0} s \cdot \frac{1}{s^2}\left[1 - \frac{Ks+b}{s^2+as+b}\right]$$

$$e_{ss} = \lim_{s\to0} \frac{1}{s}\left[\frac{s^2+as+b-Ks-b}{s^2+as+b}\right]$$

$$e_{ss} = \lim_{s\to0} \frac{1}{s}\left[\frac{s(s+a-K)}{s^2+as+b}\right]$$

$$e_{ss} = \lim_{s\to0} \frac{s+a-K}{s^2+as+b} = \frac{a-K}{b}$$

Method 2

The open-loop transfer function $G(s)$ is :

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \implies G(s) = \frac{\frac{C(s)}{R(s)}}{1 - \frac{C(s)}{R(s)}}$$

$$G(s) = \frac{Ks+b}{(s^2+as+b) - (Ks+b)} = \frac{Ks+b}{s(s+a-K)}$$

Since the system is a Type-1 system,

$$e_{ss} = \frac{1}{K_v}$$

Where the velocity error constant K_v is defined as:

$$K_v = \lim_{s\to0} sG(s) = \lim_{s\to0} s\left[\frac{Ks+b}{s(s+a-K)}\right]$$

$$K_v = \frac{b}{a-K}$$

Thus, $e_{ss} = \frac{1}{K_v} = \frac{a-K}{b}$

$$\boxed{\text{(D)} \quad \frac{a-K}{b}}$$

Mistakes to be avoided

- Do not blindly apply $e_{ss} = \dfrac{1}{K_v}$ directly to the closed-loop denominator without converting it into the open-loop form $G(s)$ first.
- **Stability Check Constraint**: The conditions $a > 0$ and $b > 0$ are required to ensure the system remains fundamentally stable, because **steady state error exists only for stable systems**.

Q3.17.3 MCQ 2017 2M Ans: C ● ○ ○

The standard form of a second-order characteristic equation is:$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$
The maximum peak overshoot (M_p) for a unit step input is given by:

$$M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100\% \quad ; \quad M_p \propto \frac{1}{\xi}$$

Comparing each option with the standard form ($\omega_n^2 = 100 \implies \omega_n = 10\,\text{rad/s}$):

- Option (A): $2\xi\omega_n = 10 \implies 2\xi(10) = 10 \implies \xi = 0.5$
- Option (B): $2\xi\omega_n = 15 \implies 2\xi(10) = 15 \implies \xi = 0.75$
- Option (C): $2\xi\omega_n = 5 \implies 2\xi(10) = 5 \implies \xi = 0.25$
- Option (D): $2\xi\omega_n = 20 \implies 2\xi(10) = 20 \implies \xi = 1.0$

The minimum value of ξ is 0.25, which corresponds to option (C).

$$\boxed{(C) \quad \frac{100}{s^2 + 5s + 100}}$$

Q3.16.1 MCQ 2016 2M Ans: B ● ○ ○

The general transfer function for a second-order system with DC gain A is:

$$T(s) = \frac{C(s)}{R(s)} = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Substituting the standard characteristic equation terms:

$$\omega_n^2 = 10^2 = 100$$

$$2\zeta\omega_n = 2(0.5)(10) = 10 \implies s^2 + 10s + 100 = 0$$

Method 1

Using the final value theorem to match the steady-state output condition:

$$c(\infty) = \lim_{s\to 0} sC(s) = \lim_{s\to 0} s\left[T(s)\cdot\frac{1}{s}\right] = T(0)$$

$$T(0) = \frac{A(100)}{100} = A \implies A = 1.02$$

Substitute A back into the numerator:

$$\text{Numerator} = 1.02 \times 100 = 102$$

$$T(s) = \frac{102}{s^2 + 10s + 100}$$

Method 2

Alternatively, apply option elimination based on the system criteria:

- The denominator must have a linear term coefficient of $2\zeta\omega_n = 10$. This eliminates options (A) and (D).
- For option (B):

$$c(\infty) = \lim_{s\to 0} \frac{102}{s^2 + 10s + 100} = 1.02$$

- For option (C):

$$c(\infty) = \lim_{s\to 0} \frac{100}{s^2 + 10s + 100} = 1.00$$

Thus, option (B) matches perfectly.

$$\boxed{(B)\ \frac{102}{s^2 + 10s + 100}}$$

Key Points

The steady-state value of a system responding to a unit step input is equivalent to its DC gain $T(0)$.

Q3.14.1 | NAT | 2014 | 1M | Ans: 0 | ● ○ ○

Given the closed-loop transfer function:

$$T(s) = \frac{4}{s^2 + 0.4s + 4} = \frac{G(s)}{1 + G(s)}$$

We can determine the open-loop transfer function $G(s)$ assuming a unity feedback system:

$$G(s) = \frac{4}{s^2 + 0.4s + 4 - 4} = \frac{4}{s^2 + 0.4s} = \frac{4}{s(s + 0.4)}$$

Since the system has one pole at the origin ($s = 0$), it is a **Type-1 system**.

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{4}{s(s+0.4)} = \infty$$

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = 0$$

$$\boxed{0}$$

Key Points

Condition	e_{ss}
Type = Input	Constant
Type > Input	0
Type < Input	∞

- **Type-0 & Step:** $e_{ss} = \dfrac{A}{1+K_p}$
- **Type-1 & Ramp:** $e_{ss} = \dfrac{A}{K_v}$
- **Type-2 & Parabolic:** $e_{ss} = \dfrac{A}{K_a}$

Q3.13.1 MCQ 2013 2M Ans: C ● ○ ○

Given the open-loop transfer function of a dc motor:

$$\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s} = \frac{10}{1+\tau s}$$

$$\tau = 10$$

The closed-loop transfer function is given by:

$$\frac{\omega(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\dfrac{10K_a}{1+10s}}{1+\dfrac{10K_a}{1+10s}} = \frac{10K_a}{1+10K_a+10s}$$

To find the closed-loop time constant, we rearrange the denominator into the standard form $(1+\tau_{cl}s)$:

$$\frac{\omega(s)}{R(s)} = \frac{\dfrac{10K_a}{1+10K_a}}{1+\left(\dfrac{10}{1+10K_a}\right)s} \implies \tau_{cl} = \frac{10}{1+10K_a}$$

According to the problem statement, the closed-loop time constant is reduced by one hundred times compared to the open-loop time constant:

$$\tau_{cl} = \frac{\tau}{100}$$

$$\frac{10}{1+10K_a} = \frac{1}{100} \times 10$$

$$K_a = 9.9 \approx 10$$

$$\boxed{(C)\ \ 10}$$

Key Points

- The standard form of a first-order system transfer function is $\dfrac{K}{1+\tau s}$, where τ represents the time constant.
- Negative feedback reduces the time constant of a system by a factor of $(1+GH)$, thereby increasing the speed of response (bandwidth).

Mistakes to be avoided

Incorrect Time Constant Extraction: Do not mistake the coefficient of s as the time constant before ensuring the constant term in the denominator is normalized to 1. Always bring the expression to the form $1+\tau s$.

Q3.11.1 MCQ 2011 1M Ans: A ● ○ ○

Given a unity feedback linear system with an open-loop transfer function $G(s)$ and $H(s) = 1$. The error signal in the s-domain is expressed as:

$$E(s) = \frac{R(s)}{1+G(s)}$$

Case 1: Unit Step Input

For a unit step input, $r_1(t) = u(t) \implies R_1(s) = \frac{1}{s}$. The steady-state error e_{ss1} is given as 0.1:

$$e_{ss1} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s \cdot \frac{1/s}{1+G(s)} = 0.1$$

$$\lim_{s\to 0} \frac{1}{1+G(s)} = 0.1$$

Case 2: Pulse Input

The input pulse of height 10 and duration 1 s is:

$$r_2(t) = 10[u(t) - u(t-1)]$$

$$R_2(s) = \frac{10}{s}(1-e^{-s})$$

The steady-state error e_{ss2} becomes:

$$e_{ss2} = \lim_{s\to 0} s \cdot \frac{\frac{10}{s}(1-e^{-s})}{1+G(s)}$$

$$e_{ss2} = \lim_{s\to 0} \frac{10(1-e^{-s})}{1+G(s)}$$

Substituting the boundary limit from Case 1 into the expression for Case 2:

$$e_{ss2} = 10\left(1-e^{0}\right) \cdot \left[\lim_{s\to 0} \frac{1}{1+G(s)}\right] = 10 \cdot (1-1) \cdot 0.1 = 0$$

$$\boxed{(A)\ 0}$$

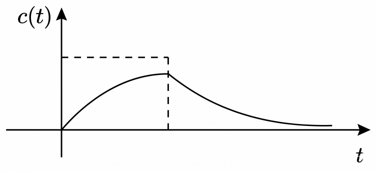


Fig. Solution Q3.11.1

Key Points

Pulse Signals: A pulse of finite length possesses a total net area but drops completely back to zero value as $t \to \infty$. Consequently, any stable linear system tracks it perfectly over long periods, yielding zero steady-state error.

Q3.11.2 MCQ 2011 2M Ans: A ● ○ ○

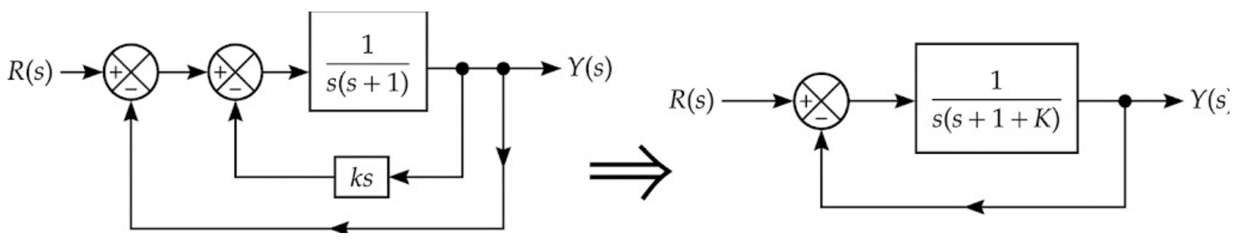


Fig. Solution Q3.11.2

The overall system has unity negative feedback ($H(s) = 1$). The closed-loop characteristic equation is:

$$1 + G(s) \cdot 1 = 0$$

$$1 + \frac{1}{s(s+1+k)} = 0$$

$$s^2 + (1+k)s + 1 = 0$$

Comparing this to the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 1 \implies \omega_n = 1 \text{ rad/s} \quad \text{(Constant, independent of } k\text{)}$$

$$2\xi\omega_n = 1 + k \implies \xi = \frac{1+k}{2}$$

Since the peak overshoot is explicitly parameterized by the damping ratio via $M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$, varying the tacho-generator gain k alters ξ and thus mainly influences the **peak overshoot**.

$$\boxed{\text{(A) peak overshoot}}$$

Key Points

Tachometer Feedback: Tacho-generator feedback yields a derivative-like effect (ks), which introduces localized inner-loop damping to smooth out transient oscillations.

Q3.10.1 MCQ 2010 1M Ans: C ● ● ○

Method 1

$$\frac{C(s)}{R(s)} = \frac{2}{s+1}$$

For a unit step input, $R(s) = \frac{1}{s}$:

$$C(s) = \frac{2}{s(s+1)} = 2\left[\frac{1}{s} - \frac{1}{s+1}\right]$$

$$C(t) = 2\left(1 - e^{-t}\right)$$

The steady-state final value is:

$$C_{ss} = \lim_{t\to\infty} C(t) = 2$$

To reach 98% of its final value at $t = T$:

$$0.98 \times 2 = 2\left(1 - e^{-T}\right)$$

$$T = -\ln(0.02) \approx 3.91 \text{ s} \approx 4 \text{ s}$$

Method 2

Comparing the system with the standard first-order system transfer function:

$$G(s) = \frac{K}{\tau s + 1} = \frac{2}{s+1}$$

Equating the denominators gives the time constant:

$$\tau = 1 \text{ s}$$

By definition, the time taken for a first-order system response to reach 98% of its final steady-state value corresponds to its settling time (t_s) within a 2% tolerance band.
For a first-order system:

$$t_s = 4\tau$$

Substituting $\tau = 1$ s:

$$t_s = 4 \times 1 \text{ s} = 4 \text{ s}$$

$$\boxed{(C)\ \ 4 \text{ s}}$$

Q3.09.1 MCQ 2009 2M Ans: D ● ● ○

Method 1

From the given response curve, the steady-state final value of the system response is: $C_{ss} = 0.75$

Since the input is a unit-step input ($R = 1$), the steady-state error (e_{ss}) is:

$$e_{ss} = 1 - C_{ss} = 1 - 0.75 = 0.25$$

$$e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)H(s)}$$

Substituting $R(s) = \frac{1}{s}$ and $G(s) = \frac{K}{(s+1)(s+2)}$:

$$e_{ss} = \lim_{s \to 0} \frac{s \cdot \frac{1}{s}}{1 + \frac{K}{(s+1)(s+2)}} = \frac{1}{1 + \frac{K}{2}}$$

Equating this to the observed value:

$$\frac{1}{1 + \frac{K}{2}} = 0.25 \implies K = 6$$

Method 2

The closed-loop transfer function $T(s)$ is:

$$T(s) = \frac{G(s)}{1 + G(s)} = \frac{\frac{K}{(s+1)(s+2)}}{1 + \frac{K}{(s+1)(s+2)}}$$

$$T(s) = \frac{K}{s^2 + 3s + (2 + K)}$$

Using the final value theorem, the steady-state value for a unit-step input is:

$$C_{ss} = \lim_{s \to 0} s \cdot C(s) = \lim_{s \to 0} s \cdot \left[T(s) \cdot \frac{1}{s}\right]$$

$$C_{ss} = T(0) = \frac{K}{2 + K}$$

From the given curve, $C_{ss} = 0.75 = \frac{3}{4}$:

$$\frac{K}{2+K} = \frac{3}{4}$$

$$4K = 6 + 3K \implies K = 6$$

$$\boxed{(D)\ 6}$$

Q3.08.1 MCQ 2008 2M Ans: A ● ○ ○

Given input $r(t)$ is a unit impulse applied at $t = 1$:

$$r(t) = \delta(t - 1)$$

$$R(s) = \mathcal{L}[r(t)] = e^{-s}$$

The transfer function is:

$$G(s) = \frac{C(s)}{R(s)} = \frac{1}{s^2 + 3s + 2}$$

The output response in s-domain is:

$$C(s) = R(s)G(s) = \frac{e^{-s}}{s^2 + 3s + 2}$$

The steady-state value of the output can be found using the Final Value Theorem (FVT):

$$C_{ss} = \lim_{s \to 0} sC(s) = \lim_{s \to 0} \frac{se^{-s}}{s^2 + 3s + 2} = \frac{0 \cdot 1}{0 + 0 + 2} = 0$$

$$\boxed{(A)\ 0}$$

Key Points

A time-delay factor e^{-s} does not affect the stability or the application of the final value theorem as $lim_{s \to 0} e^{-s} = 1$.

Q3.07.1 MCQ 2007 2M Ans: A ● ● ○

From the given block diagram, the error signal is:

$$E(s) = R(s) - C(s)$$

The relationship for the variable $Z(s)$ is:

$$Z(s) = \left(\frac{k_i}{s}\right) E(s)$$

The overall open-loop transfer function $G(s)$ is:

$$G(s) = \left(k_p + \frac{k_i}{s}\right)\left(\frac{\omega^2}{s^2 + 2\xi\omega s + \omega^2}\right)$$

Since $H(s) = 1$, the closed-loop expression for $E(s)$ is:

$$E(s) = \frac{R(s)}{1 + G(s)H(s)} = \frac{R(s)}{1 + G(s)}$$

Substituting $E(s)$ into the expression for $Z(s)$:

$$Z(s) = \frac{k_i}{s} \cdot \frac{R(s)}{1 + G(s)}$$

Given a unit step input, $R(s) = \frac{1}{s}$. Thus:

$$Z(s) = \frac{k_i}{s^2[1 + G(s)]}$$

Applying the Final Value Theorem to find the steady-state value z_{ss}:

$$z_{ss} = \lim_{s \to 0} sZ(s) = \lim_{s \to 0} \frac{s \cdot \frac{k_i}{s} \cdot \frac{1}{s}}{1 + \left(k_p + \frac{k_i}{s}\right)\left(\frac{\omega^2}{s^2 + 2\xi\omega s + \omega^2}\right)}$$

Multiplying the numerator and denominator by s:

$$z_{ss} = \lim_{s \to 0} \frac{k_i}{s + (sk_p + k_i)\left(\frac{\omega^2}{s^2 + 2\xi\omega s + \omega^2}\right)}$$

$$z_{ss} = \frac{k_i}{0 + (0 + k_i)\left(\frac{\omega^2}{0 + 0 + \omega^2}\right)} = \frac{k_i}{k_i(1)} = 1$$

$$\boxed{(A)\ \ 1}$$

Q3.07.2 MCQ 2007 2M Ans: C ● ● ○

Given a series $R - L - C$ circuit with parameters: $R = 10\,\Omega$, $L = 1$ mH, and $C = 10\,\mu$F. The loop equation for the circuit is:

$$e_i = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt$$

Taking the Laplace transform on both sides (assuming zero initial conditions):

$$E_i(s) = \left(R + Ls + \frac{1}{Cs}\right) I(s) \implies I(s) = \frac{E_i(s)}{R + Ls + \frac{1}{Cs}}$$

The output voltage e_o across the capacitor is given by:

$$e_o = \frac{1}{C}\int i\,dt$$

$$E_o(s) = \frac{1}{Cs} I(s) = \frac{1}{Cs}\left[\frac{E_i(s)}{R + Ls + \frac{1}{Cs}}\right]$$

The transfer function of the system is:

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{LC\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

The characteristic equation is obtained by setting the denominator to zero:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \frac{1}{\sqrt{LC}}, \quad \xi = \frac{R}{2L\omega_n} = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$\xi = \frac{10}{2}\sqrt{\frac{10\times10^{-6}}{1\times10^{-3}}} = 5\times\sqrt{10^{-2}} = 0.5$$

Since $\xi = 0.5$ (where $0 < \xi < 1$), the system is **under-damped**.
The peak overshoot (M_p) for a step input is given by:

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} = e^{-\pi\times0.5/\sqrt{1-0.5^2}} = e^{-1.8138} \approx 0.163 \text{ or } 16.3\%$$

$$\boxed{(C)\ \ 16\%}$$

Key Points

For a standard second-order system, the damping ratio for a series $R-L-C$ circuit is given by $\xi = \frac{R}{2}\sqrt{\frac{C}{L}}$.

Mistakes to be avoided

Radian vs Degree Mode: When computing the exponent value $\frac{\pi\xi}{\sqrt{1-\xi^2}}$, ensure your calculator handles π as a mathematical constant (≈ 3.14159) rather than treating trigonometric expressions in degree mode incorrectly.

Q3.07.3 MCQ 2007 2M Ans: C ● ● ○

From the characteristic parameters of the given series $R-L-C$ network layout:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1\times10^{-3}\times10\times10^{-6}}} = 10^4 \text{ rad/s}$$

Given that the damping ratio $\xi = 0.5$, the settling time (t_s) for a 2% tolerance band is:

$$t_s = \frac{4}{\xi\omega_n} = \frac{4}{0.5\times10^4} = 0.8 \text{ ms}$$

To view a transient step response cleanly on a non-storage Cathode Ray Oscilloscope (CRO), a single step function is unsuitable because the trace fades away immediately. Instead, a periodic square wave input is applied to repeatedly trigger the response.

For an optimal visual presentation of the entire transient period up to stabilization, the duration of each half-cycle ($\frac{T}{2}$) must be strictly greater than the settling time (t_s), but not excessively large:

$$\frac{T}{2} > t_s \implies \frac{1}{2f} > 0.8 \text{ ms}$$

Let us evaluate the available frequency choices:

- **For $f = 50$ Hz:**

  $$\frac{T}{2} = \frac{1}{2\times50} = 10 \text{ ms} \gg 0.8 \text{ ms}$$

  Since 10 ms is much larger than 0.8 ms, the transient phase finishes within a tiny fraction (8%) of the display time, leaving the remaining 9.2 ms as a flat line.

- **For $f = 300$ Hz:**

  $$\frac{T}{2} = \frac{1}{2\times300} \approx 1.67 \text{ ms} > 0.8 \text{ ms}$$

  The transient is completely visible and takes up a balanced portion of the sweep screen profile.

- **For $f = 2.0$ kHz:**

  $$\frac{T}{2} = \frac{1}{2\times2\times10^3} = 0.25 \text{ ms} < 0.8 \text{ ms}$$

  The square wave switches before the transient settles down completely.

Therefore, a square wave of 50 Hz or 300 Hz allows complete settlement, but 300 Hz provides the best representation of the transient details without a long flat line.

$$\boxed{\text{(C) square wave of frequency 300 Hz}}$$

Key Points

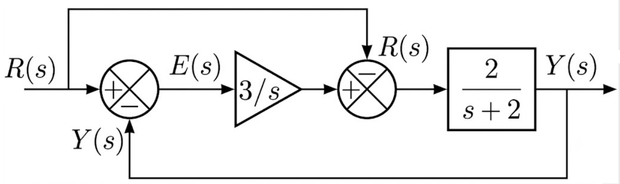
- Non-storage CROs need periodic signals like square waves to persistently display transient behaviors via repeated sweeps.
- The half-period interval constraint criteria requires $\frac{T}{2} \geq t_s$ to avoid truncation of transient information.

Q3.05.1

MCQ | 2005 | 2M | Ans: C | ● ● ○

Method 1

Block Diagram Reduction :



$$E(s) = R(s) - Y(s)$$

The output response $Y(s)$ is given by:

$$Y(s) = \left[\{R(s) - Y(s)\}\frac{3}{s} - R(s)\right]\frac{2}{s+2}$$

$$Y(s)\left[1 + \frac{6}{s(s+2)}\right] = R(s)\left[\frac{6-2s}{s(s+2)}\right]$$

$$Y(s) = R(s)\frac{6-2s}{s^2+2s+6}$$

$$E(s) = R(s) - \frac{6-2s}{s^2+2s+6}R(s)$$

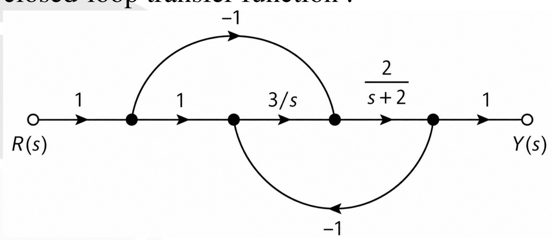
$$E(s) = R(s)\left[\frac{s^2+4s}{s^2+2s+6}\right]$$

For a unit step input $R(s) = \frac{1}{s}$, the steady-state error e_{ss} via Final Value Theorem is:

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\left[\frac{1}{s}\frac{s^2+4s}{s^2+2s+6}\right] = 0$$

Method 2

Using **Mason's Gain Formula**, we determine the closed-loop transfer function :



- Forward Paths:

$$P_1 = \frac{6}{s(s+2)}, \quad \Delta_1 = 1; \qquad P_2 = \frac{-2}{(s+2)}, \quad \Delta_2 = 1$$

- Feedback Loop: $L_1 = \dfrac{-6}{s(s+2)}$

$$\Delta = 1 - L_1 = 1 + \frac{6}{s(s+2)} = \frac{s^2+2s+6}{s(s+2)}$$

$$\frac{Y(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{6-2s}{s^2+2s+6}$$

The error expression is $E(s) = R(s)[1 - \frac{Y(s)}{R(s)}]$

$$E(s) = R(s)\left[\frac{s^2+4s}{s^2+2s+6}\right]$$

For $R(s) = \frac{1}{s}$: $e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{s}\left[\frac{s^2+4s}{s^2+2s+6}\right] = 0$

$$\boxed{(C)\ \ 0}$$

Q3.04.1 MCQ 2004 2M Ans: D ● ○ ○

The unit impulse response of the system is given by: $c(t) = 12.5e^{-6t} \sin 8t \cdot u(t)$

The transfer function $G(s)$ of a system is equal to the Laplace transform of its unit impulse response under zero initial conditions. Using the standard Laplace transform pairs and property $\mathcal{L}\{e^{-at} \sin \omega t\} = \dfrac{\omega}{(s+a)^2 + \omega^2}$:

$$G(s) = 12.5 \times \frac{8}{(s+6)^2 + 8^2} = \frac{100}{s^2 + 12s + 100}$$

For a unit step input, $R(s) = \frac{1}{s}$, the output response in the s-domain is:

$$C(s) = R(s) \cdot G(s) = \frac{1}{s} \cdot \frac{100}{s^2 + 12s + 100}$$

Using the Final Value Theorem, the steady-state value of the response is:

$$C_{ss} = \lim_{s \to 0} sC(s) = \lim_{s \to 0} s \left[\frac{1}{s} \cdot \frac{100}{s^2 + 12s + 100} \right] = \frac{100}{0 + 0 + 100} = 1$$

$$\boxed{(D)\ \ 1}$$

Q3.04.2 MCQ 2004 2M Ans: D ● ○ ○

Given parameters from the problem statement:

$$\text{Damping ratio} : \zeta = 0.7 \text{ and } \quad \text{Undamped natural frequency} : \omega_n = 5, \text{rad/sec}$$

The characteristic equation (C.E.) of a closed-loop system with negative feedback is given by:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+2)}(1 + sP) = 0 \implies [s^2 + s(2 + KP) + K = 0$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = K \implies K = 5^2 = 25$$

$$2\zeta\omega_n = 2 + KP$$

$$2(0.7)(5) = 2 + 25P$$

$$25P = 5 \implies P = \frac{5}{25} = 0.2$$

Thus, the values are $K = 25$ and $P = 0.2$.

$$\boxed{\text{(D) 25 and 0.2}}$$

Q3.03.1 MCQ 2003 2M Ans: A ● ○ ○

The forward path transfer function is:

$$G(s) = \left(\frac{3}{s+15}\right) \times \left(\frac{15}{s+1}\right) = \frac{45}{(s+1)(s+15)}$$

For a unity feedback system, $H(s) = 1$. The open-loop transfer function is:

$$G(s)H(s) = \frac{45}{(s+1)(s+15)}$$

Since there is no pole at the origin, the system is a **type-0 system**.

The steady-state error e_{ss} for a unit step input is given by: $e_{ss} = \dfrac{1}{1+K_p}$

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{45}{(s+1)(s+15)} = \frac{45}{1\times 15} = 3$$

$$e_{ss} = \frac{1}{1+3} = 0.25 = 25\%$$

$$\boxed{(A)\ \ 25\%}$$

Key Points

Shortcut Constant Method: For any Type-0 system expressed in polynomial form, K_p can be directly computed as:

$$K_p = \frac{\text{Numerator constant}}{\text{Denominator constant}}$$

Q3.03.2 MCQ 2003 2M Ans: D ● ○ ○

The characteristic equation of the closed-loop system is given by:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{45}{(s+15)(s+1)} = 0$$

$$s^2 + 16s + 15 + 45 = 0$$

$$s^2 + 16s + 60 = 0$$

$$(s+10)(s+6) = 0$$

Thus, the roots of the closed-loop characteristic equation (closed-loop poles) are:

$$s = -6 \quad \text{and} \quad s = -10$$

$$\boxed{(D)\ \ \text{-6}\ \ and\ \ \text{-10}}$$

Q3.00.1 MCQ 2000 1M Ans: A

The steady-state error E_{ss} for a unity feedback system is given by the final value theorem: $E_{ss} = \lim_{s\to 0} \dfrac{s \cdot R(s)}{1 + G(s)H(s)}$

- **(a) Step input and type-1 $G(s)$:** $E_{ss} = \lim_{s\to 0} \dfrac{s\left(\frac{1}{s}\right)}{1+\frac{K}{s}} = \lim_{s\to 0} \dfrac{1}{1+\frac{K}{s}} = \dfrac{1}{\infty} = 0$
- **(b) Ramp input and type-1 $G(s)$:** $E_{ss} = \lim_{s\to 0} \dfrac{s\left(\frac{1}{s^2}\right)}{1+\frac{K}{s}} = \lim_{s\to 0} \dfrac{1}{s+K} = \dfrac{1}{K} \neq 0$
- **(c) Step input and type-0 $G(s)$:** $E_{ss} = \lim_{s\to 0} \dfrac{s\left(\frac{1}{s}\right)}{1+K} = \dfrac{1}{1+K} \neq 0$
- **(d) Ramp input and type-0 $G(s)$:** $E_{ss} = \lim_{s\to 0} \dfrac{s\left(\frac{1}{s^2}\right)}{1+K} = \lim_{s\to 0} \dfrac{1}{s(1+K)} = \infty \neq 0$

$$\boxed{\text{(A) step input and type } -1 \ G(s)}$$

Q3.00.2 MCQ 2000 2M Ans: B

The closed-loop characteristic equation is given by $1 + G(s) = 0$:

$$1 + \frac{25}{s(s+6)} = 0 \implies s^2 + 6s + 25 = 0$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 25 \implies \omega_n = 5\,\text{rad/s}$$

$$2\xi\omega_n = 6 \implies 2\xi(5) = 6 \implies \xi = 0.6$$

$$M_p = e^{-\left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)} = e^{-\left(\frac{0.6\times\pi}{\sqrt{1-0.6^2}}\right)} \approx e^{-2.356} \approx 0.0948$$

$$\%M_p = 0.0948 \times 100\% \approx 9.48\% \approx 10\%$$

$$\boxed{(B)\ \ 10\%}$$

Q3.97.1 NAT 1997 5M Ans: 5

The step response of a first-order system initially at rest is given by the standard formula:

$$c(t) = C_{ss}\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$1.1 = 2\left(1 - e^{-\frac{4}{\tau}}\right)$$

$$e^{-\frac{4}{\tau}} = 0.45$$

$$\tau = \frac{4}{\ln(0.45)} \approx \boxed{5\,\text{s}}$$

Q3.96.1 MCQ 1996 1M Ans: C ● ○ ○

For a Type 2 system, the open-loop transfer function can be written as: $G(s)H(s) = \dfrac{k}{s^2(s+a)}$

The steady-state error E_{ss} for a ramp input $R(s) = \frac{1}{s^2}$ is given by:

$$E_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)H(s)} = \lim_{s\to 0} \frac{s\left(\frac{1}{s^2}\right)}{1+\dfrac{k}{s^2(s+a)}} = \lim_{s\to 0} \frac{1}{s+\dfrac{k}{s(s+a)}} = 0$$

$\boxed{\text{(C) zero}}$

Q3.95.1 MCQ 1995 5M Ans: A-T, B-S, C-P, D-R, E-Q ● ● ○

- **A → T:** Real, distinct poles on the negative real half of the s-plane indicate an critically/overdamped stable system whose step response smoothly approaches a steady-state value without any oscillations.
- **B → S:** Complex conjugate poles in the left-half of the s-plane characterize an underdamped stable system, where the response exhibits decaying (damped) oscillations towards 1.0.
- **C → P:** Purely imaginary poles (on the $j\omega$-axis) correspond to an undamped system yielding sustained, fixed-amplitude periodic oscillations.
- **D → R:** Complex conjugate poles located in the right-half of the s-plane cause an underdamped unstable response characterized by exponentially growing oscillations.
- **E → Q:** Real poles situated on the positive real axis mean the system is completely unstable, and its output increases exponentially toward infinity without oscillating.

$\boxed{\text{A-T, B-S, C-P, D-R, E-Q}}$

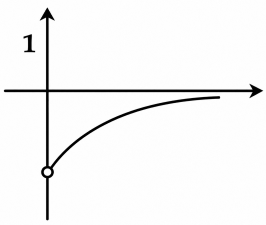
Key Points

- Poles in the left-half s-plane produce bounded, stable transient behaviors.
- Poles on the imaginary axis correspond to marginally stable boundaries with undamped periodic outputs.
- Poles in the right-half s-plane lead to unbounded, unstable growing response characteristics.

Q3.92.1 MCQ 1992 4M Ans: MTA ● ○ ○

Given the transfer functions and their corresponding impulse responses:

(a) $G(s) = \dfrac{s}{s+1} = 1 - \dfrac{1}{s+1} \implies g(t) = \delta(t) - e^{-t}u(t)$



It does not match any of the given options.

(b) $G(s) = \dfrac{1}{(s+1)^2} \implies g(t) = te^{-t}u(t)$ This represents a critically damped response, **(P)**.

(c) $G(s) = \dfrac{1}{s(s+1)+1} = \dfrac{1}{s^2+s+1}$

Comparing with the standard second-order transfer function $G(s) = \dfrac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$:

$$\omega_n^2 = 1 \implies \omega_n = 1\,\text{rad/sec}; \quad 2\xi\omega_n = 1 \implies \xi = \frac{1}{2} = 0.5$$

Since $\xi < 1$, it is an underdamped system,**(S)**.

(d) $G(s) = \dfrac{1}{s^2+1}$

The system has poles at $s = \pm j$. Hence, it is an undamped system ($\xi = 0$), **(R)**

Q3.92.2 MCQ 1992 1M Ans: A ● ○ ○

The error signal in frequency domain is: $E(s) = \dfrac{R(s)}{1+G(s)H(s)}$

For a unit step input, $R(s) = \frac{1}{s}$. Using the Final Value Theorem, the steady-state error is given by:

$$\lim_{t\to\infty} E(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{s \cdot \frac{1}{s}}{1+\frac{s+1}{s^2+5s+a}\cdot\frac{1}{s+4}} = \frac{1}{1+\frac{1}{4a}} = \frac{4a}{4a+1}$$

For the steady-state error to be zero:

$$\frac{4a}{4a+1} = 0 \implies 4a = 0 \implies a = 0$$

$$\boxed{\text{(A) a=0}}$$

Q3.91.1 NAT 1991 2M Ans: 5,10 ● ○ ○

From the given unit step response curve of the first-order system:

- The initial slope line intersects the final steady-state value line at $t = 0.2$ sec. Therefore, the time constant of the system is $\tau = 0.2$ sec.
- The steady-state final value of the response is $c(\infty) = 2.0$.

The transfer function of the system is given as:

$$G(s) = \frac{K}{s+a} = \frac{K/a}{1+\frac{s}{a}}$$

Comparing this with the standard form of a first-order system $G(s) = \dfrac{A}{1+s\tau}$:

$$\tau = \frac{1}{a} = 0.2\,\text{sec} \implies a = \frac{1}{0.2} = 5$$

For a unit step input $R(s) = \dfrac{1}{s}$, the output response is $C(s) = G(s)\cdot R(s)$. Applying the Final Value Theorem:

$$c(\infty) = \lim_{s\to 0} sC(s) = \lim_{s\to 0} s\left(\frac{K}{s+a}\cdot\frac{1}{s}\right) = \frac{K}{a}$$

From the figure, the final value is 2:

$$\frac{K}{a} = 2 \implies K = 2\times 5 = 10$$

$$\boxed{a = 5, K = 10}$$

DEBUG INFO

Chapter III

Time Response Analysis

Total Questions : 36

MCQ : 30

MSQ : 0

NAT : 6

SOLUTIONS

IV

Routh's Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

Q4.25.1 MCQ | 2025 | 2M | Ans: B | ● ○ ○

The open-loop transfer function of the system is given by:

$$G_{OL}(s) = K(s-1)G(s) = \frac{K(s-1)}{(s+1)(s+2)}$$

The characteristic equation of the closed-loop system is:

$$1 + G_{OL}(s) = 0 \implies 1 + \frac{K(s-1)}{(s+1)(s+2)} = 0 \implies s^2 + (K+3)s + (2-K) = 0$$

RH Table:

$$\begin{array}{c|cc} s^2 & 1 & 2-K \\ s^1 & K+3 & \\ s^0 & 2-K & \end{array}$$

For the system to be stable, all coefficients in the first column of the Routh table must be strictly positive

1. $K+3 > 0 \implies K > -3$
2. $2-K > 0 \implies K < 2$

Thus, the system is stable in the range:

$$-3 < K < 2$$

The system becomes unstable when $K > 2$ or $K < -3$.

$$\boxed{\text{(B) unstable for all } K > 2}$$

Key Points

Shortcut: For a second-order characteristic equation $as^2 + bs + c = 0$, the system is stable if and only if all coefficients a, b, and c have the same sign ($a > 0, b > 0, c > 0$).

Q4.22.1 MCQ | 2022 | 1M | Ans: C | ● ○ ○

The given open loop transfer function of the unity feedback system is:

$$G(s) = \frac{k}{s^2 + 4s - 5}$$

The characteristic equation of the closed loop system is given by:$1 + G(s)H(s) = 0$
Since it is a unity gain negative feedback system ($H(s) = 1$):

$$1 + \frac{k}{s^2 + 4s - 5} = 0 \implies s^2 + 4s + (k-5) = 0$$

For a second-order system $as^2 + bs + c = 0$ to be stable, all coefficients must be real and positive ($a > 0$, $b > 0$, $c > 0$). Therefore, for system stability:

$$k - 5 > 0 \implies k > 5$$

$$\boxed{\text{(C) } k > 5}$$

Q4.20.1 MCQ | 2020 | 2M | Ans: C | ● ○ ○

Given the transfer functions from the system block diagram:

$$G(s) = \frac{1}{s+1} \cdot \frac{1}{s^2} = \frac{1}{s^2(s+1)} \quad and \quad H(s) = \frac{20}{s+20}$$

The characteristic equation of the closed-loop system is given by:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{1}{(s+1)s^2} \cdot \frac{20}{(s+20)} = 0$$

$$s^2(s+1)(s+20) + 20 = 0$$

$$s^4 + 21s^3 + 20s^2 + 20 = 0$$

Routh-Hurwitz Array:

$$\begin{array}{c|ccc} s^4 & 1 & 20 & 20 \\ s^3 & 21 & 0 & \\ s^2 & 20 & 20 & \\ s^1 & -21 & 0 & \\ s^0 & 20 & & \end{array}$$

Since the highest power of s is 4, the system is a 4^{th} **order** system.

In the simplified characteristic equation, the coefficient of the s^1 term is missing (0). A system cannot be stable if any coefficient of the characteristic equation is zero or negative.

Alternatively, looking at the first column of the Routh array, there are two sign changes, which confirms that the system is **unstable**.

$$\boxed{(C)\ \ 4^{th}\text{order and unstable}}$$

Key Points

The total number of sign changes in the first column of a Routh array equals the total number of roots located in the right-half plane.

Q4.20.2 MCQ | 2020 | 2M | Ans: B | ● ○ ○

The characteristic equation for a unity feedback system ($H(s) = 1$) is given by:

$$1 + G(s) = 0$$

$$1 + \frac{s^2 + s + 1}{s^3 + 2s^2 + 2s + K} = 0 \implies s^3 + 3s^2 + 3s + (K+1) = 0$$

Method 1

Using Routh-Hurwitz Criteria:

Constructing the Routh array for the characteristic equation:

$$\begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 3 & K+1 \\ s^1 & \dfrac{9-(K+1)}{3} & 0 \\ s^0 & K+1 & 0 \end{array}$$

For the system to have poles on the imaginary axis (marginally stable condition), the row of s^1 must become a row of zeros:

$$\frac{9-(K+1)}{3} = 0$$

$$9 - K - 1 = 0$$

$$K = 8$$

Method 2

Using Inner/Outer Product Rule:

For a standard 3^{rd}-order characteristic equation:

$$as^3 + bs^2 + cs + d = 0$$

The condition for the system to have non-repeated poles on the imaginary axis (marginally stable condition) is:

$$\text{Inner Product} = \text{Outer Product}$$

$$b \times c = a \times d$$

Comparing with our characteristic equation:

$$a = 1, \quad b = 3, \quad c = 3, \quad d = K+1$$

Applying the product rule:

$$3 \times 3 = 1 \times (K+1)$$

$$K = 8$$

$$\boxed{(B)\ \ 8}$$

Key Points

- A row of all zeros in the Routh array indicates the presence of symmetrically located roots about the origin.
- For any stable third-order system, $bc > ad$. When $bc = ad$, the system becomes marginally stable with poles on the $j\omega$-axis.

Q4.19.1 MCQ 2019 1M Ans: C ● ○ ○

Given the characteristic equation of the linear time-invariant (LTI) system:

$$\Delta(s) = s^4 + 3s^3 + 3s^2 + s + K = 0$$

For the system to be bounded-input bounded-output (BIBO) stable, all elements in the first column of the Routh-Hurwitz array must be strictly positive.

Routh-Hurwitz Array:

$$\begin{array}{c|ccc}
s^4 & 1 & 3 & K \\
s^3 & 3 & 1 & \\
s^2 & \frac{8}{3} & K & \\
s^1 & \dfrac{\left(\frac{8}{3}\right)(1) - 3(K)}{\frac{8}{3}} & & \\
s^0 & K & &
\end{array}$$

1. From the s^0 row condition: $K > 0$
2. From the s^1 row condition: $\dfrac{\frac{8}{3} - 3K}{\frac{8}{3}} > 0$

$$3K < \frac{8}{3} \implies K < \frac{8}{9}$$

Combining:

$$0 < K < \frac{8}{9}$$

$$\boxed{(C)\ \ 0 < k < \frac{8}{9}}$$

Q4.18.1 MCQ 2018 2M Ans: A ● ● ○

The characteristic equation of the system is given by:

$$s^7 + s^6 + 7s^5 + 14s^4 + 31s^3 + 73s^2 + 25s + 200 = 0$$

Routh Table:

s^7	1	7	31	25
s^6	1	14	73	200
s^5	−7	−42	−175	
s^4	8	48	200	
s^3	(0)32	(0)96		
s^2	24	200		
s^1	−170			
s^0	200			

The auxiliary equation $A(s)$ is formed from the s^4 row:

$$A(s) = 8s^4 + 48s^2 + 200 = 0$$

$$\frac{d}{ds}A(s) = 32s^3 + 96s$$

Sign Changes in the 1^{st} Column:

- **Above $A(s)$ (Rows s^7 to s^4):** There are 2 sign changes. This signifies 2 roots in the right-half plane (RHP).
- **Below $A(s)$ (Rows s^4 to s^0):** There are 2 sign changes. Since the 4 roots of $A(s)$ are symmetric, these 2 sign changes indicate 2 RHP roots, meaning the 2 corresponding symmetric roots of $A(s)$ lie in the left-half plane (LHP).

A row of zeros indicates that there are symmetric roots about the origin.

Total roots in Right Half Plane (RHP) = 2 + 2 = 4

Total roots on Imaginary Axis ($j\omega$-axis) = 0

Total roots in Open Left Half Plane (LHP) = 7 - 4 = 3

$$\boxed{(A)\ \ 3}$$

Mistakes to be avoided

Do not assume an ROZ always means roots are purely on the imaginary axis. Check for sign changes below the auxiliary row to see if roots leave the $j\omega$-axis.

Q4.17.1 MCQ 2017 1M Ans: D

The characteristic equation is: $s^3 + Ks^2 + (K+2)s + 3 = 0$

Method 1

Constructing the Routh's array:

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & K & 3 \\ s^1 & \dfrac{K(K+2)-3}{K} & \\ s^0 & 3 & \end{array}$$

For stability, there should be no sign change in the first column:

$$K > 0 \quad \text{and} \quad \frac{K^2+2K-3}{K} > 0$$

$$K^2 + 2K - 3 > 0 \implies (K-1)(K+3) > 0$$

Since $K > 0$, we must have:

$$K > 1$$

Method 2

For a third-order system $as^3 + bs^2 + cs + d = 0$ with positive coefficients to be stable, the product of internal coefficients must be greater than the product of external coefficients:

$$bc > ad$$

Here, $a = 1$, $b = K$, $c = K + 2$, and $d = 3$.

$$K(K+2) > 1 \cdot 3$$

$$K^2 + 2K - 3 > 0$$

$$(K-1)(K+3) > 0$$

Since all coefficients must be positive for stability ($K >$ 0):

$$K > 1$$

$$\boxed{(D)\ \ K > 1}$$

Q4.17.2 MCQ 2017 2M Ans: A ● ○ ○

The characteristic equation of the system is given by: $s^3 + 3s^2 + 2s + K = 0$

Method 1

Forming the Routh's array:

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 3 & K \\ s^1 & \dfrac{6-K}{3} & 0 \\ s^0 & K & 0 \end{array}$$

For all roots to lie in the left half of the complex s-plane, the system must be stable,

$$K > 0 \quad \text{and} \quad \frac{6-K}{3} > 0$$

$$6 - K > 0 \implies K < 6$$

Combining the conditions:

$$0 < K < 6$$

Method 2

For a standard third-order characteristic equation $as^3 + bs^2 + cs + d = 0$ with all positive coefficients ($a, b, c, d > 0$), the system is stable if:

$$bc > ad$$

$$a = 1, \quad b = 3, \quad c = 2, \quad d = K$$

For all coefficients to be positive:

$$K > 0$$

Applying the stability criterion:

$$(3)(2) > (1)(K) \implies 6 > K$$

Combining both constraints gives:

$$0 < K < 6$$

$$\boxed{(A)\ \ 0 < K < 6}$$

Q4.16.1 NAT 2016 2M Ans: 2 ● ● ○

The given third-order polynomial equation is: $s^3 + 5.5s^2 + 8.5s + 3 = 0$

To find the number of roots with real parts strictly less than -1, we shift the imaginary axis to the left by substituting $s = z - 1$:

$$(z-1)^3 + 5.5(z-1)^2 + 8.5(z-1) + 3 = 0$$

$$(z^3 - 3z^2 + 3z - 1) + 5.5(z^2 - 2z + 1) + 8.5(z - 1) + 3 = 0$$

$$z^3 + 2.5z^2 + 0.5z - 1 = 0$$

Forming the Routh's array for the transformed polynomial:

$$\begin{array}{c|cc} z^3 & 1 & 0.5 \\ z^2 & 2.5 & -1 \\ z^1 & \dfrac{(2.5)(0.5) - (1)(-1)}{2.5} = 0.9 & \\ z^0 & -1 & \end{array}$$

There is exactly 1 sign change (from $+0.9$ to -1). This indicates that 1 root lies to the right of the shifted axis ($\text{Re}(z) > 0$), meaning 1 root has a real part strictly greater than -1.

Since the total number of roots is 3, the number of roots with real parts strictly less than -1 is:

$$N = 3 - 1 = 2$$

$$\boxed{2}$$

Q4.16.2 MCQ 2016 2M Ans: C ● ○ ○

The characteristic equation for the unity feedback system is $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s+1)}{s(1+Ts)(1+2s)} = 0$$

$$s(1 + 2s + Ts + 2Ts^2) + Ks + K = 0$$

$$2Ts^3 + (2+T)s^2 + (1+K)s + K = 0$$

Forming the Routh's array:

$$\begin{array}{c|cc} s^3 & 2T & 1+K \\ s^2 & 2+T & K \\ s^1 & \dfrac{(2+T)(1+K)-2TK}{2+T} & 0 \\ s^0 & K & 0 \end{array}$$

For the closed-loop system to be stable, all elements in the first column must be strictly positive

since $T > 0$ and $K > 0$:

$$\frac{(2+T)(1+K)-2TK}{2+T} > 0$$

Since $(2+T) > 0$, the numerator must be positive:

$$(2+T)(1+K) - 2TK > 0$$

$$2 + 2K + T - TK > 0$$

$$T < \frac{2(K+1)}{K-1} \quad or \quad K < \frac{T+2}{T-2}$$

$$\boxed{(C)\ \ 0 < K < \frac{T+2}{T-2}}$$

Q4.14.1 MCQ 2014 1M Ans: B ● ○ ○

In the formation of the Routh-Hurwitz array, a row of all zeros (premature termination) indicates the existence of an auxiliary polynomial $A(s)$.
The roots of the auxiliary polynomial $A(s)$ are always symmetric with respect to the origin of the s-plane. This radial symmetry means the roots can occur as:

- Pairs of purely imaginary roots $(\pm j\omega)$.
- Pairs of real roots equal in magnitude but opposite in sign $(\pm\sigma)$.
- Complex conjugate quadruplets located symmetrically around the origin $(\pm\sigma \pm j\omega)$.

Since purely imaginary roots are symmetric about the origin, a row of zeros occurs whenever the characteristic polynomial contains imaginary roots.

$$\boxed{\text{(B) imaginary roots}}$$

Q4.14.2 NAT 2014 2M Ans: 0.61 to 0.63 ● ○ ○

Given the open-loop transfer function from the block diagram:

$$G(s) = \frac{s+\alpha}{s^3 + (1+\alpha)s^2 + (\alpha-1)s + (1-\alpha)}$$

For unity negative feedback, the characteristic equation is $1 + G(s) = 0$:

$$s^3 + (1+\alpha)s^2 + (\alpha-1)s + (1-\alpha) + (s+\alpha) = 0$$

$$s^3 + (1+\alpha)s^2 + \alpha s + 1 = 0$$

Constructing the Routh array:

$$\begin{array}{c|cc} s^3 & 1 & \alpha \\ s^2 & 1+\alpha & 1 \\ s^1 & \dfrac{\alpha(1+\alpha)-1}{1+\alpha} & 0 \\ s^0 & 1 & \end{array}$$

For the system to be stable, all elements in the first column must be strictly positive

1. $1+\alpha > 0 \implies \alpha > -1$
2. $\alpha^2+\alpha-1 > 0$

Solving the quadratic boundary $\alpha^2+\alpha-1=0$:

$$\alpha = \frac{-1 \pm \sqrt{1-4(1)(-1)}}{2} = \frac{-1\pm\sqrt{5}}{2}$$

$$\alpha \approx 0.618 \quad \text{or} \quad \alpha \approx -1.618$$

Intersecting with $\alpha > -1$, we chose -0.618

$$\boxed{0.618}$$

Q4.14.3 NAT 2014 2M Ans: 5

Given the open-loop transfer function:

$$G(s) = \frac{K}{s(s+2)(s^2+2s+2)} = \frac{K}{s^4+4s^3+6s^2+4s}$$

For a unity negative feedback system, the characteristic equation is given by $1+G(s)=0$:

$$s^4+4s^3+6s^2+4s+K=0$$

Constructing the Routh array:

$$\begin{array}{c|ccc} s^4 & 1 & 6 & K \\ s^3 & 4 & 4 & \\ s^2 & 5 & K & \\ s^1 & \frac{20-4K}{5} & & \\ s^0 & K & & \end{array}$$

For the closed-loop system to be marginally stable, the elements of the s^1 row must equal zero

$$\frac{20-4K}{5} = 0$$

$$20-4K=0$$

$$4K=20 \implies K=5$$

$$\boxed{5}$$

Q4.12.1 MCQ 2012 2M Ans: A

The characteristic equation of the unity negative feedback system is:

$$1+G(s)H(s)=0 \implies s^3+as^2+(K+2)s+(K+1)=0$$

RH Table:

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & a & K+1 \\ s^1 & \dfrac{a(K+2)-(K+1)}{a} & \\ s^0 & K+1 & \end{array}$$

For sustained oscillations, the s^1 row must be a row of zeros:

$$a(K+2)-(K+1)=0$$

$$a = \frac{K+1}{K+2} \quad \text{— (i)}$$

The auxiliary equation from the s^2 row is:

$$A(s) = as^2+(K+1)=0$$

Given the oscillation frequency is $\omega = 2$ rad/s, we substitute $s=j\omega=j2$ into $A(s)=0$:

$$a(j2)^2+(K+1)=0 \implies -4a+K+1=0 \implies K=4a-1 \quad \text{— (ii)}$$

Substituting equation (i) into equation (ii):

$$K = 4\left(\frac{K+1}{K+2}\right) - 1 \implies K^2 + 2K = 3K + 2 \implies (K-2)(K+1) = 0$$

For system stability configurations, the open-loop gain must be positive ($K > 0$), which yields:

$$K = 2$$

Substituting $K = 2$ back into equation (i) determines the parameter a:

$$a = \frac{2+1}{2+2} = \frac{3}{4} = 0.75$$

$$\boxed{(A)\ \ K = 2 \text{ and } a = 0.75}$$

Key Points

- A sustained oscillation condition implies a system is marginally stable.
- The roots of the auxiliary equation $A(s) = 0$ specify the exact frequency of these sustained oscillations on the imaginary axis.

Q4.11.1 MCQ 2011 1M Ans: B ● ○ ○

To determine the nature of the open-loop system, we analyze its stability and its phase type based on the given transfer function:

$$G(s) = \frac{s-1}{(s+2)(s+3)}$$

1. Phase Type Analysis:

- **Poles:** $s = -2$ and $s = -3$ (Both lie on the left-half of the s-plane).
- **Zeros:** $s = 1$ (Lies on the right-half of the s-plane).
- Since there is a zero in the right-half of the s-plane (RHS), the system belongs to the **non-minimum phase type**.

2. Open-Loop Stability Analysis:

- An open-loop system is stable if all its open-loop poles lie strictly in the left-half of the s-plane (LHS).
- As both poles ($s = -2, -3$) have negative real parts, the open-loop system is completely **stable**.

$$\boxed{\text{(B) stable and of the non-minimum phase type}}$$

Key Points

- **Minimum Phase System:** A system is minimum-phase if all its poles and zeros lie in the left-half of the s-plane (excluding the $j\omega$-axis for zeros).
- **Non-Minimum Phase System:** If any pole or zero lies in the right-half of the s-plane, it introduces additional phase lag and is classified as a non-minimum phase system.

Q4.09.1 MCQ 2009 1M Ans: D ● ○ ○

RH Array:

$$\begin{array}{c|cc} s^3 & 2 & 2 \\ s^2 & 4 & 4 \\ s^1 & 0(8) & \\ s^0 & 4 & \end{array}$$

Auxiliary Equation: $A(s) = 4s^2 + 4 = 0$
Differentiating with respect to s:

$$\frac{d}{ds}A(s) = \frac{d}{ds}(4s^2 + 4) = 8s + 0$$

Roots of the auxiliary equation:

$$4s^2 + 4 = 0 \implies s = \pm j1$$

Therefore, there are two roots at $s = \pm j$. Since there is no sign change in the first column of the Routh array, the remaining one root will lie in the left half of the s-plane.

$$\boxed{\text{(D) two roots at } s = \pm j \text{ and one root in left half s-plane}}$$

Q4.08.1 | MCQ | 2008 | 2M | Ans: C | ● ○ ○

The characteristic equation is given by: $1 + G(s)H(s) = 0$

$$1 + \frac{k}{s(s+3)(s+10)} = 0 \implies s^3 + 13s^2 + 30s + k = 0$$

Routh-array:

$$\begin{array}{c|cc} s^3 & 1 & 30 \\ s^2 & 13 & k \\ s^1 & \dfrac{13 \times 30 - k}{13} & \\ s^0 & k & \end{array}$$

According to the RH criterion, for a stable system, the signs of the elements in the first column must be positive:

$$k > 0$$

and $\dfrac{13 \times 30 - k}{13} > 0 \implies 390 - k > 0 \implies k < 390$

$$\boxed{\text{(C) } 0 < \text{k} < 390}$$

Key Points

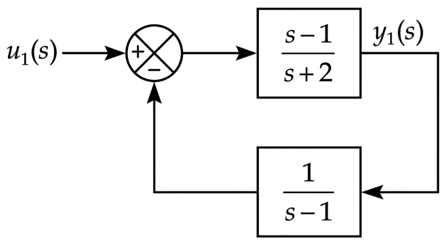
- **Shortcut:** For a third-order characteristic equation $as^3 + bs^2 + cs + d = 0$ with all positive coefficients, the system is stable if the inner product is greater than the outer product ($bc > ad$).
- Here, $13 \times 30 > 1 \times k \implies k < 390$.

Q4.07.1 | MCQ | 2007 | 1M | Ans: D | ● ○ ○

To determine the stability of the multi-input multi-output (MIMO) system, we analyze its stability under individual inputs using the superposition principle.

Case 1: Considering input $u_1(s)$ alone :

The block diagram reduces to a standard negative feedback loop configuration:



The corresponding closed-loop transfer function is given by:

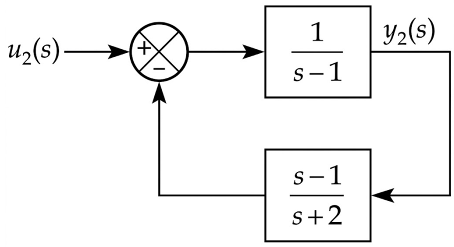
$$TF_1 = \frac{Y_1(s)}{U_1(s)} = \frac{\frac{s-1}{s+2}}{1 + \left(\frac{s-1}{s+2}\right)\left(\frac{1}{s-1}\right)}$$

$$TF_1 = \frac{\frac{s-1}{s+2}}{1 + \frac{1}{s+2}} = \frac{s-1}{s+3}$$

The pole is located at $s = -3$, which lies entirely on the left-hand side (LHS) of the s-plane. Hence, the system is **stable** with respect to input u_1.

Case 2: Considering input $u_2(s)$ alone :

The block diagram reduces to a standard negative feedback loop configuration:



The corresponding closed-loop transfer function is given by:

$$TF_2 = \frac{Y_2(s)}{U_2(s)} = \frac{\frac{1}{s-1}}{1 + \left(\frac{1}{s-1}\right)\left(\frac{s-1}{s+2}\right)}$$

$$TF_2 = \frac{\frac{1}{s-1}}{\frac{(s+2)+1}{s+2}} = \frac{s+2}{(s-1)(s+3)}$$

The system has poles at $s = -3$ and $s = 1$. The pole at $s = 1$ lies on the right-hand side (RHS) of the s-plane. Hence, the system is **unstable** with respect to input u_2.

(D) stable for input u_1, but unstable for input u_2

Key Points

MIMO Stability Criterion: For a multi-input system to be stable, the transfer functions corresponding to every single input channel must be stable (all poles must reside strictly in the LHS of the s-plane).

Q4.07.2 MCQ | 2007 | 2M | Ans: B

The characteristic equation of the closed-loop system is determined by setting $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s+3)}{(s+8)^2} = 0 \implies (s+8)^2 + K(s+3) = 0$$

$$s^2 + 16s + 64 + Ks + 3K = 0$$

$$s^2 + (16+K)s + (64+3K) = 0$$

Routh-Hurwitz array:

$$\begin{array}{c|cc} s^2 & 1 & 64+3K \\ s^1 & 16+K & \\ s^0 & 64+3K & \end{array}$$

For sustained oscillations to be induced, a row of zeros must be formed in the Routh array, which forces the system to have purely imaginary poles on the $j\omega$-axis.

This requires setting the s^1 row coefficient to zero:

$$16 + K = 0 \implies K = -16$$

Substituting $K = -16$ into the s^0 row element gives:

$$64 + 3(-16) = 64 - 48 = 16$$

Since $16 > 0$, the s^0 term remains valid and positive.

$$A(s) = 1 \cdot s^2 + (64 + 3K) = 0$$

$$s^2 + 16 = 0 \implies s^2 = -16$$

$$s = \pm j4$$

Comparing this with $s = \pm j\omega$, the frequency of sustained oscillation is $\omega = 4$ rad/s.

$$\boxed{\text{(B) The frequency of this oscillation must be must be 4 rad/s}}$$

Q4.06.1 MCQ 2006 2M Ans: C ● ○ ○

Given algebraic equation:

$$F(s) = s^5 - 3s^4 + 5s^3 - 7s^2 + 4s + 20 = 0$$

Applying the Routh-Hurwitz criteria, we construct the Routh array:

$$\begin{array}{c|ccc} s^5 & 1 & 5 & 4 \\ s^4 & -3 & -7 & 20 \\ s^3 & \frac{8}{3} & \frac{32}{3} & \\ s^2 & 5 & 20 & \\ s^1 & 0(2) & & \\ s^0 & 20 & & \end{array}$$

Since we encounter a row of zeros at the s^1 row, we form the auxiliary equation $A(s)$ from the coefficients of the s^2 row:

$$A(s) = s^2 + 4 = 0 \implies s = \pm j2$$

Differentiating $A(s)$ with respect to s:

$$\frac{dA(s)}{ds} = 2s$$

- There are 2 sign changes in the first column, which means there are 2 roots with **positive real parts**.
- The auxiliary equation gives **2 imaginary roots** ($s = \pm j2$).
- The total number of roots is 5, the remaining root must be real and negative: $5 - 2 - 2 = 1$ **negative real root**.

$$\boxed{\text{(C) one negative real root, two imaginary roots, and two roots with positive real parts}}$$

Q4.05.1 MCQ 2005 2M Ans: C ● ● ○

Given open-loop transfer function: $G(s)H(s) = \dfrac{K(1-s)}{(1+s)}$

The characteristic equation (C.E.) of the unity feedback system is defined as:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K(1-s)}{1+s} = 0 \implies s(1-K) + (K+1) = 0$$

Solving for the closed-loop pole s:

$$s = -\frac{(K+1)}{1-K} = \frac{K+1}{K-1}$$

The system becomes stable only when the root lies in the left-half of the s-plane (i.e., s **must be negative**).

Case 1:

$$(K+1) > 0 \quad \& \quad (K-1) < 0$$

$$K > -1 \quad \& \quad K < 1$$

$$\implies |K| < 1$$

Case 2:

$$(K+1) < 0 \quad \& \quad (K-1) > 0$$

$$K < -1 \quad \& \quad K > 1$$

$$\implies \text{Not possible}$$

$$\boxed{(C)\ \ |K| < 1}$$

Q4.04.1 MCQ | 2004 | 2M | Ans: B | ● ○ ○

Given characteristic equation:

$$s^3 - 4s^2 + s + 6 = 0$$

To find the number of roots in the left-half of the s-plane, we construct the Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & -4 & 6 \\ s^1 & \dfrac{(-4)(1)-(6)(1)}{-4} = 2.5 & \\ s^0 & 6 & \end{array}$$

Since there are **two sign changes** in the first column of the Routh array, there are exactly 2 roots lying in the right-half of the s-plane (RHS).

Since the highest power of s is 3, the total number of roots is 3. Therefore, the remaining number of roots in the left-half of the s-plane (LHS) is:

$$\text{Roots in LHS} = \text{Total roots} - \text{Roots in RHS} = 3 - 2 = 1$$

$$\boxed{\text{(B) one}}$$

Q4.03.1 MCQ | 2003 | 2M | Ans: C | ● ○ ○

The characteristic equation of the closed-loop system is given by:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+2)(s+4)} = 0 \implies s^3 + 6s^2 + 8s + K = 0$$

$$\begin{array}{c|cc} s^3 & 1 & 8 \\ s^2 & 6 & K \\ s^1 & \dfrac{48-K}{6} & \\ s^0 & K & \end{array}$$

For the system to remain stable, there must be no sign changes in the first column of the Routh array

$$\frac{48-K}{6} > 0 \implies K < 48$$

$$K > 0$$

The range of stability is:

$$0 < K < 48$$

The system just becomes unstable (marginally stable) at the critical boundary value: $K = 48$

$$\boxed{(C)\ \ \text{K} = 48}$$

Key Points

Shortcut: For a third-order system $as^3 + bs^2 + cs + d = 0$, the system becomes marginally unstable when the inner product equals the outer product ($bc = ad$), provided all coefficients are positive : $6 \times 8 = 1 \times K \implies K_{mar} = 48$.

Q4.00.1 MCQ 2000 2M Ans: C ● ○ ○

Given the characteristic equation: $2s^4 + s^3 + 3s^2 + 5s + 10 = 0$

$$\begin{array}{c|ccc} s^4 & 2 & 3 & 10 \\ s^3 & 1 & 5 & \\ s^2 & \frac{(1)(3)-(5)(2)}{1} = -7 & \frac{(1)(10)-(2)(0)}{1} = 10 & \\ s^1 & \frac{(-7)(5)-(10)(1)}{-7} = \frac{45}{7} & & \\ s^0 & 10 & & \end{array}$$

There are two sign changes in the first column. Therefore, the number of roots in the right half of the s-plane is 2.

$\boxed{(C)\ 2}$

Q4.99.1 MCQ 1999 1M Ans: B ● ○ ○

Even if there are no poles on the right half of the s-plane, we cannot say that the system must be stable. This is because we cannot rule out the possibility of repeated poles on the imaginary axis.

Repeated Poles on $j\omega$-axis lead to an unstable system where the output grows unboundedly with time ($t\sin(\omega t)$ or $t\cos(\omega t)$ terms).

$\boxed{\text{(B) could be unbounded}}$

Q4.97.1 NAT 1997 2M Ans: 2 ● ○ ○

The Characteristic Equation (C.E.) is: $s^4 + 6s^3 - 39s^2 + 19s + 84 = 0$
Constructing the Routh-Hurwitz array:

$$\begin{array}{c|ccc} s^4 & 1 & -39 & 84 \\ s^3 & 6 & 19 & 0 \\ s^2 & \frac{(6)(-39)-19}{6} = -42 & 84 & \\ s^1 & \frac{(-42)(19)-(84)(6)}{-42} = +11 & 0 & \\ s^0 & 84 & & \end{array}$$

The number of sign changes in the first column is 2. Therefore, there are 2 poles present in the right-half of the s-plane.

$\boxed{2}$

Key Points

Default Assumption: Whenever it is written just "transfer function" and the nature of feedback is not mentioned, by default assume it to be a closed-loop transfer function.

Q4.97.2 NAT 1997 5M Ans: Stable ● ● ○

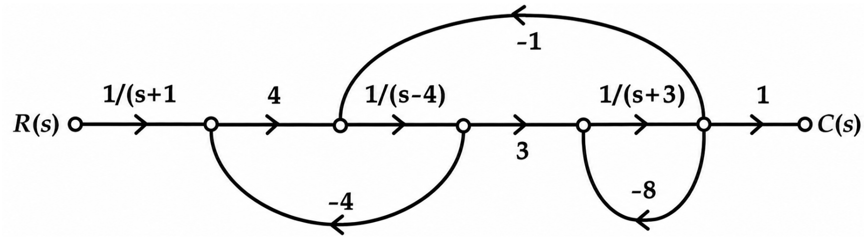


Fig. Solution Q4.97.2

- **Forward path gain:** $P_1 = \dfrac{12}{(s+1)(s+3)(s-4)}, \quad \Delta_1 = 1$
- **Individual loop gains:** $L_1 = \dfrac{-16}{s-4}, \quad L_2 = \dfrac{-3}{(s+3)(s-4)}, \quad L_3 = \dfrac{-8}{s+3}$
- **Non-touching loops:** $L_1 L_3 = \dfrac{128}{(s-4)(s+3)}$
- **System Determinant (Δ):**

$$\Delta = 1 - (L_1 + L_2 + L_3) + L_1 L_3$$

$$\Delta = 1 + \frac{16}{s-4} + \frac{8}{s+3} + \frac{3}{(s+3)(s-4)} + \frac{128}{(s+3)(s-4)}$$

$$\text{T.F} = \frac{P_1 \Delta_1}{\Delta} = \frac{12}{(s+1)\left[s^2 + 23s + 135\right]}$$

Stability Check:

- $s + 1 = 0 \implies s = -1$ (LHP)
- For $s^2 + 23s + 135 = 0$, all coefficients are positive, so its roots lie strictly in the LHP.

$\boxed{\text{Stable}}$

Q4.95.1 NAT 1995 2M Ans: zeros ● ● ○

The characteristic equation of a closed-loop system is : $1 + G(s)H(s) = 0$

The closed-loop transfer function is expressed as: $T(s) = \dfrac{G(s)}{1 + G(s)H(s)}$

- The roots of the denominator of $T(s)$ determine the closed-loop poles, which are identical to the roots of the equation $1 + G(s)H(s) = 0$. Correspondingly, these poles are the **zeros** of the function $1 + G(s)H(s)$.
- For a closed-loop system to be stable, all of its closed-loop poles (and thus the zeros of $1 + G(s)H(s)$) must lie strictly in the left half of the s-plane.

$\boxed{\text{zeros}}$

Q4.95.2 NAT | 1995 | 5M | Ans: 0<k<1.49

RH Array:

$$\begin{array}{c|ccc} s^4 & 1 & 15 & K \\ s^3 & 20 & 2 & \\ s^2 & 14.9 & K & \\ s^1 & \dfrac{29.8-20K}{14.9} & & \\ s^0 & K & & \end{array}$$

Stability Analysis:

- From s^0 row: $K > 0$
- From s^1 row: $\dfrac{29.8-20K}{14.9} > 0$

$$29.8 - 20K > 0 \implies K < 1.49$$

$$\boxed{0 < K < 1.49}$$

Q4.95.3 NAT | 1995 | 5M | Ans: 0.316 rad/sec

The system will sustain continuous oscillations at the marginal stability limit, which occurs when the row of s^1 becomes zero:

$$29.8 - 20K = 0 \implies K_{\text{mar}} = 1.49$$

Forming the auxiliary equation $A(s) = 0$ from the row just above it (s^2 row):

$$14.9s^2 + K_{\text{mar}} = 0 \implies s^2 = -\frac{1.49}{14.9} = -0.1$$

$$s = \pm j\sqrt{0.1} \approx \pm j0.316 \,\text{rad/sec}$$

$$\boxed{\omega = 0.316 \,\text{rad/sec}}$$

Q4.95.4 NAT | 1995 | 5M | Ans: 2

For $K = 5$, substituting into the first column elements of the Routh array:

- s^1 term : $\frac{29.8-20(5)}{14.9} = \frac{29.8-100}{14.9} = -4.71 < 0$ (Sign change 1)
- s^0 term : $5 > 0$ (Sign change 2)

There are exactly 2 sign changes. Hence 2 rrots lie in right half of the s-plane.

$$\boxed{2}$$

Q4.94.1 NAT | 1994 | 1M | Ans: 0

$$s^3 - 2s + 2 = 0$$

$$s = -1.77, \quad 0.88 \pm j0.59$$

Therefore, there are no positive real roots (roots that are strictly real and greater than zero).

$$\boxed{0}$$

Q4.94.2 NAT 1994 2M Ans: True ● ○ ○

- The total input-to-output dynamic characteristics of the given negative feedback loop are completely represented by its closed-loop transfer function: $T(s) = \dfrac{C(s)}{R(s)} = \dfrac{G(s)}{1 + G(s)H(s)}$
- Therefore, the stability of the entire closed-loop control system is directly equivalent to the stability of its overall transfer function $T(s)$. For $T(s)$ to be stable, all of its poles (the roots of the characteristic equation $1 + G(s)H(s) = 0$) must lie strictly in the left half of the s-plane.

$$\boxed{\text{True}}$$

Q4.92.1 NAT 1992 2M Ans: $0 \le k < \frac{4}{5}$ ● ○ ○

The characteristic equation (C.E.) of the given negative feedback system is: $1 + G(s)H(s) = 0$

$$1 + \left(\frac{s-5}{s+4}\right) K = 0 \implies s(1+K) + 4 - 5K = 0 \implies s = \frac{5K-4}{1+K}$$

For asymptotic stability, the root of the characteristic equation must lie strictly in the left half of the s-plane :

$$\frac{5K-4}{1+K} < 0$$

Given that $K \ge 0$, the denominator $(1+K)$ is always positive. Therefore, the numerator must be negative:

$$5K - 4 < 0 \implies K < \frac{4}{5}$$

$$\boxed{0 \le K < \frac{4}{5}}$$

Key Points

Asymptotic stability means that the system's response decays to zero as time goes to infinity, requiring all poles to have strictly negative real parts.

DEBUG INFO

Chapter IV

Routh's Stability Criterion

Total Questions : 34

MCQ : 22

MSQ : 0

NAT : 12

SOLUTIONS

V

Root Locus

Topics

- Basics of Root Locus
- Rules for Sketching Root Locus
- Angle of Departure and Arrival
- Complementary Root Locus

Q5.25.1 MSQ 2025 1M Ans: A,C

Given the open-loop transfer function:

$$G(s)H(s) = \frac{Ks(s+2)}{(s+5)(s+7)}$$

The open-loop poles and zeros are identified as follows:

- Poles (P): $s = -5, -7$
- Zeros (Z): $s = 0, -2$

A point on the real axis lies on the root locus if the total number of real open-loop poles and zeros to its right is odd. -1 and -6 are on the root locus.

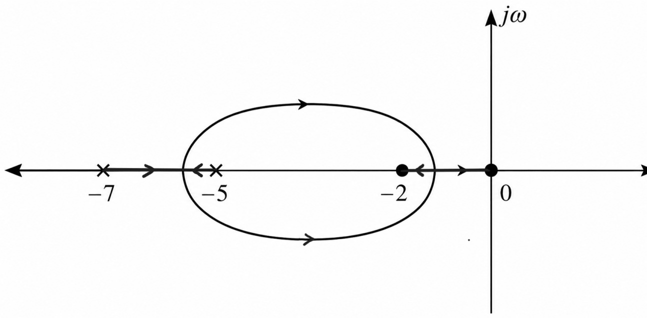


Fig. Solution Q5.25.1

$\boxed{(A)\quad -1}$ $\boxed{(C)\quad -6}$

Q5.24.1 NAT 2024 2M Ans: 4 to 8 CLICK FOR VIDEO SOLUTION

Method 1

Sum of Individual Angles:
The angle of departure (ϕ_D) from the complex open-loop pole $p_1 = -1 + j2$ is given by:

$$\phi_D = 180^\circ + \sum \phi_z - \sum \phi_p$$

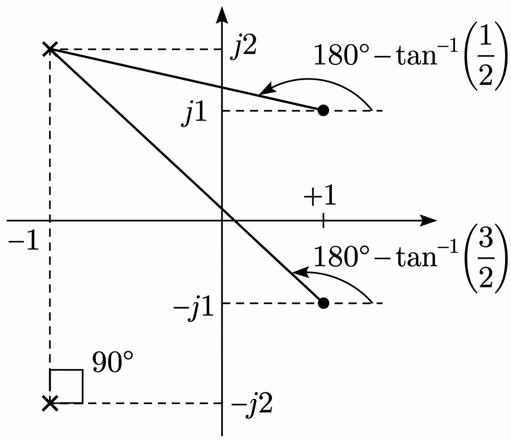


Fig. Solution Q5.24.1

Angle Contributions:

- From pole $p = -1 - j2$: $\phi_p = 90^\circ$
- From zero $z_1 = 1 + j1$:

$$\phi_{z_1} = 180^\circ - \tan^{-1}\left(\frac{2-1}{1-(-1)}\right) = 180^\circ - \tan^{-1}\left(\frac{1}{2}\right) = 153.43^\circ$$

- From zero $z_2 = 1 - j1$:

$$\phi_{z_2} = 180^\circ - \tan^{-1}\left(\frac{2-(-1)}{1-(-1)}\right) = 180^\circ - \tan^{-1}\left(\frac{3}{2}\right) = 123.69^\circ$$

$$\phi_D = 180^\circ + (153.43^\circ + 123.69^\circ) - 90^\circ = 367.12^\circ \equiv 7.12^\circ$$

Method 2

Phasor Angle Method:

$$\phi_D = 180^\circ + \angle G(s)H(s)_{s=-1+j2}$$

$$\angle G(s)H(s) = \frac{\angle(s-(1+j1)) + \angle(s-(1-j1))}{\angle(s-(-1-j2)) + \angle(s-(-1+j2))}$$

Substitute $s = -1 + j2$:

$$\angle G(s)H(s) = \frac{\angle(-2+j1) + \angle(-2+j3)}{\angle j4 + \angle 0} = 180^\circ - \tan^{-1}\left(\frac{1}{2}\right) + 180^\circ - \tan^{-1}\left(\frac{3}{2}\right) - 90^\circ$$

$$= 153.43° + 123.69° - 90° = 187.12°$$

Angle of departure : $\phi_D = 180° + 187.12° = 367.12° \equiv 7.12°$

$$\boxed{7.12°}$$

Key Points

Angle Criterion Rule:

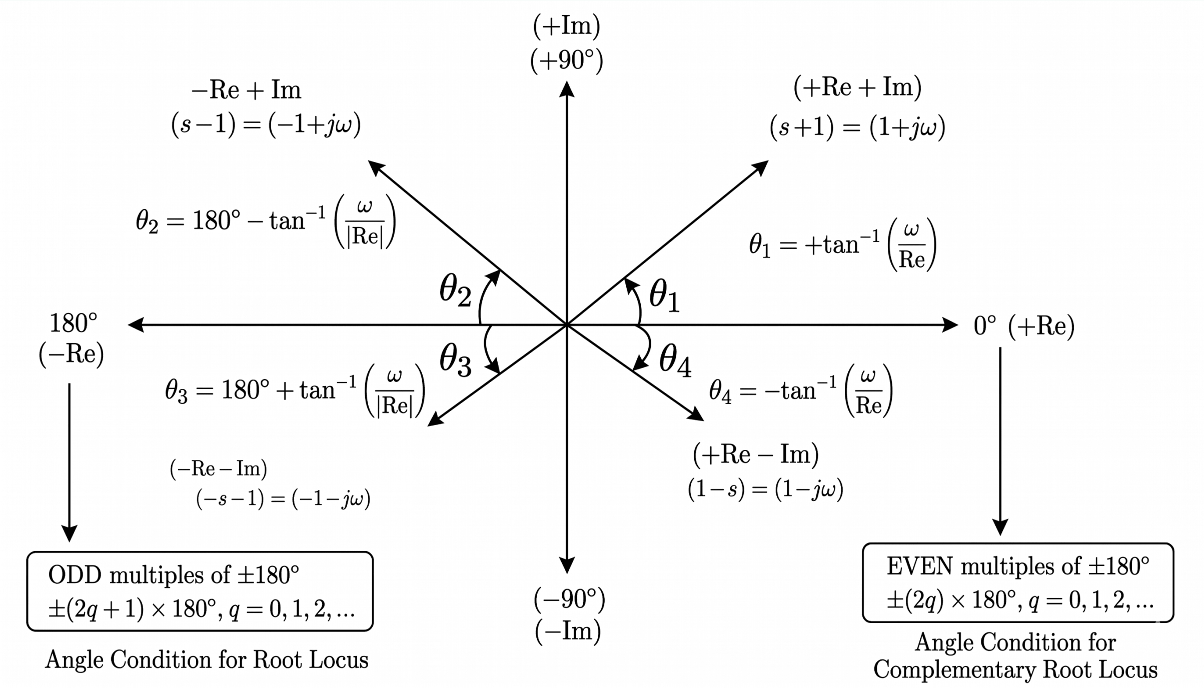


Fig. Solution Q5.24.1

Q5.17.1 MCQ 2017 2M Ans: B

The characteristic equation of the feedback control system is given by:

$$s^2 + 6Ks + 2s + 5 = 0$$

Rearranging in the form $1 + G(s)H(s) = 0$:

$$1 + \frac{6Ks}{s^2 + 2s + 5} = 0 \implies K = -\frac{s^2 + 2s + 5}{6s}$$

To find the breakaway or break-in points: $\dfrac{dK}{ds} = 0$

$$\frac{dK}{ds} = -\frac{(2s + 2)(6s) - (s^2 + 2s + 5)(6)}{(6s)^2} = 0$$

$$6(2s^2 + 2s) - 6(s^2 + 2s + 5) = 0$$

$$s^2 - 5 = 0 \implies s = \pm\sqrt{5}$$

The valid break-in point on the real axis must be negative for $K > 0$: $s = -\sqrt{5}$

$$\boxed{(B)\ \ s = -\sqrt{5}}$$

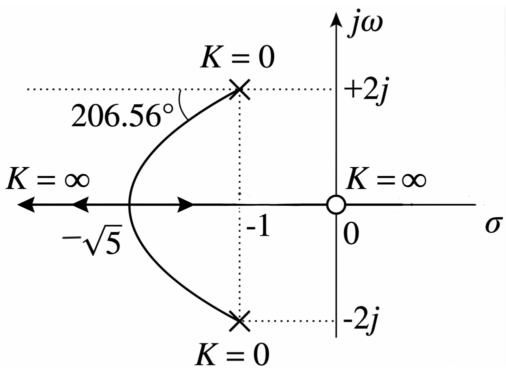


Fig. Solution Q5.17.1

Key Points

Breakaway points typically occur between two adjacent poles on the real axis if that segment is part of the root locus. Break-in points typically occur between two adjacent zeros on the real axis if that segment is part of the root locus.

Q5.16.1 MCQ 2016 2M Ans: A ● ○ ○

The characteristic equation of a unity feedback system is:

$$1 + G(s)H(s) = 0$$

Given $G(s) = \dfrac{Ks}{(s-1)(s-4)}$ and $H(s) = 1$:

$$1 + \frac{Ks}{(s-1)(s-4)} = 0$$

$$K = -\frac{(s-1)(s-4)}{s} = -s + 5 - \frac{4}{s}$$

For the breakaway point, setting $\dfrac{dK}{ds} = 0$:

$$\frac{dK}{ds} = -1 + \frac{4}{s^2} = 0 \implies s = \pm 2$$

Evaluating the gain K at $s = 2$:

$$K = \left|\frac{(2-1)(2-4)}{2}\right| = 1$$

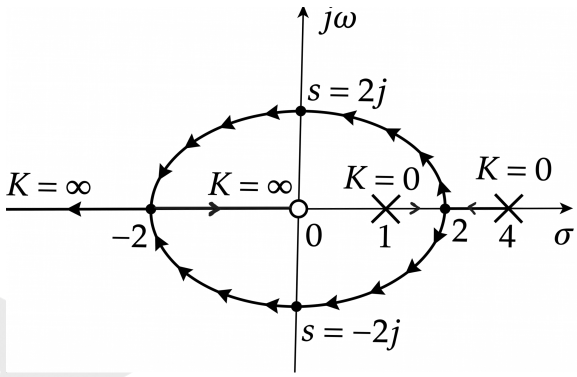


Fig. Solution Q5.16.1

$$\boxed{(A)\ 1}$$

Key Points

The gain K at any point on the root locus is given by the magnitude condition: $|G(s)H(s)| = 1$.

Q5.15.1 MCQ 2015 2M Ans: D ● ● ○

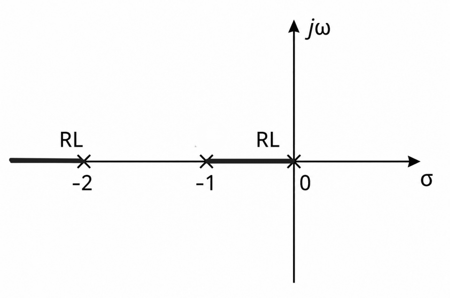
Case I: Original System

$$G(s)H(s) = \frac{K_1}{s(s+1)(s+2)} = \frac{K_1}{s^3+3s^2+2s}$$

The characteristic equation is $s^3 + 3s^2 + 2s + K_1 = 0$.

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 3 & K_1 \\ s^1 & \dfrac{6-K_1}{3} & 0 \\ s^0 & K_1 & 0 \end{array}$$

For stability, $\frac{6-K_1}{3} > 0 \implies K_1 < 6$.
At $K_1 = 6$, the system crosses the $j\omega$-axis and transits to the unstable region.



Case II: Modified System (Adding a Zero at -3)

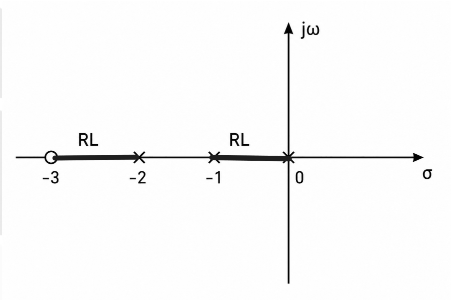
$$G(s)H(s) = \frac{K_1(s+3)}{s(s+1)(s+2)}$$

The modified characteristic equation is:

$$s^3 + 3s^2 + (2+K_1)s + 3K_1 = 0$$

$$\begin{array}{c|cc} s^3 & 1 & 2+K_1 \\ s^2 & 3 & 3K_1 \\ s^1 & 2 & \\ s^0 & 3K_1 & \end{array}$$

No sign changes in the 1st column. Thus, the root locus of the modified system never enters the right-half of the s-plane.



(D) Root locus of modified system never transits to unstable region

Key Points

Adding an open-loop zero pulls the root locus branches towards the left-half of the s-plane (LHP), which generally increases the relative stability of the system.

Q5.15.2 MCQ | 2015 | 2M | Ans: A

Given the open-loop transfer function:

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

The characteristic equation for the unity feedback system is:

$$1 + G(s) = 0$$

$$s(s^2 + 3s + 2) + K = 0$$

$$-K = s^3 + 3s^2 + 2s$$

Differentiating both sides with respect to s to find the breakaway points:

$$\frac{dK}{ds} = 0 \implies 3s^2 + 6s + 2 = 0$$

Solving the quadratic equation:

$$s = \frac{-6 \pm \sqrt{36 - 24}}{6} = -1 \pm \frac{\sqrt{3}}{3}$$

$$s = -0.42 \quad \text{or} \quad s = -1.58$$

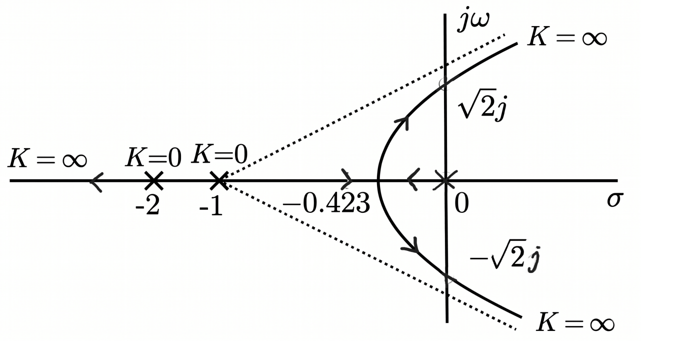


Fig. Solution Q5.15.2

$s = -0.42$ lies between 0 and -1, so it is a valid breakaway point where $K > 0$.

$s = -1.58$ does not lie on the root locus, corresponding to a negative value of K.

$$\boxed{(A) \quad -0.42}$$

Key Points

Valid breakaway points must satisfy $\frac{dK}{ds} = 0$ and exist on the root locus path.

Q5.14.1 MCQ 2014 1M Ans: C ● ○ ○

From the given root locus diagram, the sections on the real axis that belong to the root locus are:

$$-\infty < \sigma < -2 \quad \text{and} \quad -1 < \sigma < \infty$$

On the right side of these sections, the total number of open-loop poles and zeros is even. Therefore, this is a **complementary root locus (CRL)**, which corresponds to a system with **positive feedback**.
For a positive feedback system, the closed-loop transfer function is given by:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

Given a unity feedback system ($H(s) = 1$) with open-loop transfer function:

$$G(s) = \frac{K}{(s+1)(s+2)}$$

Substituting $G(s)$ and $H(s)$ into the closed-loop transfer function:

$$\frac{C(s)}{R(s)} = \frac{\frac{K}{(s+1)(s+2)}}{1 - \frac{K}{(s+1)(s+2)}} = \frac{K}{(s+1)(s+2) - K}$$

$$\boxed{(C) \quad \frac{C(s)}{R(s)} = \frac{K}{(s+1)(s+2) - K}}$$

Mistakes to be avoided

Standard Formula Reflex: Do not blindly apply the standard negative feedback formula $\frac{G(s)}{1+G(s)H(s)}$ without analyzing the real-axis segments of the given root locus.

Q5.11.1 MCQ 2011 2M Ans: A ● ● ○

The characteristic equation is defined by $1 + G(s)H(s) = 0$:

$$1 + \frac{K\left(s+\frac{2}{3}\right)}{s^2(s+2)} = 0 \implies s^3 + 2s^2 + Ks + \frac{2K}{3} = 0$$

RH Array:

$$\begin{array}{c|cc}
s^3 & 1 & K \\
s^2 & 2 & \frac{2K}{3} \\
s^1 & \frac{2K - \frac{2K}{3}}{2} = \frac{2}{3}K & \\
s^0 & \frac{2K}{3} &
\end{array}$$

For any positive value of the open-loop gain ($K > 0$):

- There are no sign changes in the first column of the Routh array.
- Thus, the closed-loop system remains completely stable for all $K > 0$.
- Since no row can become entirely zero for $K \neq 0$, the root loci never cross the $j\omega$-axis into the right-half plane.

Root Locus Construction:

- Poles and Zeros: Number of poles $P = 3$ (at $s = 0, 0, -2$), and number of zeros $Z = 1$ (at $s = -\frac{2}{3}$).
- Asymptotic Branches: $P - Z = 3 - 1 = 2$ branches terminate at infinity.
- Angle of Asymptotes:

$$\theta = \frac{(2m+1)\times 180^\circ}{P - Z} = 90^\circ \text{ and } 270^\circ$$

- Centroid (σ):

$$\sigma_g = \frac{\sum \text{Poles} - \sum \text{Zeros}}{P - Z} = \frac{(0+0-2) - \left(-\frac{2}{3}\right)}{2} = -\frac{2}{3}$$

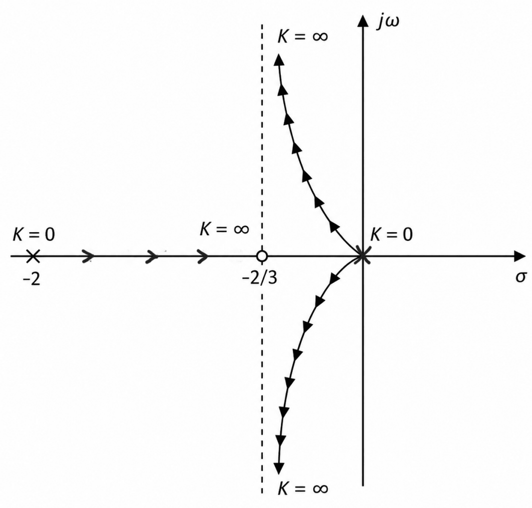


Fig. Solution Q5.11.1

As $K \to \infty$, the two branches heading to infinity approach asymptotes centered at $\mathrm{Re}(s) = -\frac{2}{3}$, while the third branch terminates at the real zero $s = -\frac{2}{3}$. Thus, all three roots share a nearly equal real part of $-\frac{2}{3}$.

$\boxed{\text{(A) three roots with nearly equal real parts exist on the left half of the s-plane}}$

Key Points

The centroid σ dictates the intersection point of the asymptotes along the real axis of the s-plane.

Q5.10.1 MCQ 2010 2M Ans: C ● ○ ○

The characteristic equation is given by:

$$s(s+1)(s+3) + k(s+2) = 0$$

Rearranging into the standard $1 + G(s)H(s) = 0$ form:

$$1 + \frac{k(s+2)}{s(s+1)(s+3)} = 0$$

The open-loop transfer function has:

- Number of zeros, $Z = 1$ at $s = -2$
- Number of poles, $P = 3$ at $s = 0, -1, -3$

Number of root locus branches tending to infinity:

$$P - Z = 3 - 1 = 2$$

The angle of asymptotes for these branches:

$$\theta = \frac{(2K+1) \times 180^\circ}{P - Z} = \frac{(2K+1) \times 180^\circ}{2} = 90^\circ, 270^\circ$$

The centroid (σ_c) where the asymptotes intersect:

$$\sigma_c = \frac{\sum \text{Poles} - \sum \text{Zeros}}{P - Z}$$

$$\sigma_c = \frac{(0 - 1 - 3) - (-2)}{2} = \frac{-4 + 2}{2} = -1$$

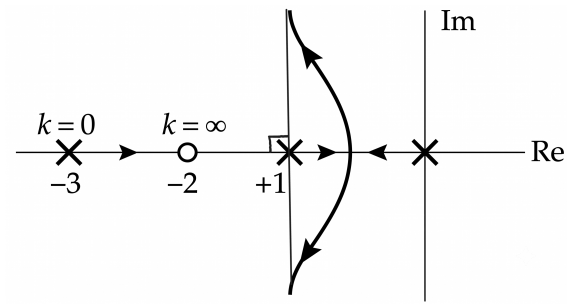


Fig. Solution Q5.10.1

Analyzing Options:

- **(A) Incorrect:** For higher values of k, two branches leave the real axis and become complex.
- **(B) Incorrect:** A breakaway point exists between the poles $s = 0$ and $s = -1$, i.e., $-1 < \text{Re}[s] < 0$.
- **(C) Correct:** The asymptotes intersect at $\sigma_c = -1$ with angles 90° and 270°.
- **(D) Incorrect:** As the branches go to infinity along $\text{Re}[s] = -1$, they never enter the right half of the s-plane for $k > 0$.

$\boxed{\text{(C) Two of its roots tend to infinity along the asymptotes Re}[s] = -1}$

Q5.06.1 MCQ 2006 2M Ans: B ● ○ ○

The open-loop characteristic function of the closed-loop control system is given by: $(s^2 - 4)(s+1) + K(s-1) = 0$

Rearranging this expression into the standard characteristic form $1 + G(s)H(s) = 0$ gives:

$$1 + \frac{K(s-1)}{(s^2-4)(s+1)} = 0 \implies 1 + \frac{K(s-1)}{(s-2)(s+2)(s+1)} = 0$$

Open-loop poles: $s = 2, -1, -2$ and Open-loop zero: $s = 1$

Real-Axis Root Locus Existence: A point on the real axis belongs to the root locus if the total count of open-loop poles and zeros to its right side is an odd number:

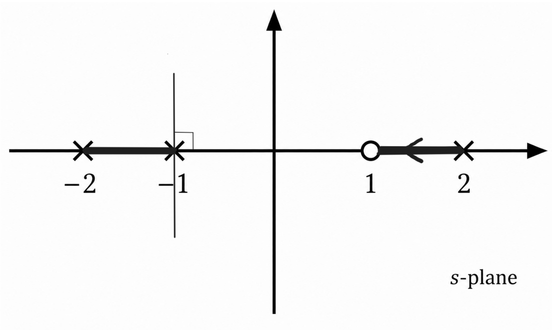


Fig. Solution Q5.06.1

Asymptotes : The number of branches terminating at infinity is:

$$P - Z = 3 - 1 = 2$$

$$\theta = \frac{(2q+1)180^\circ}{P - Z} = 90^\circ \text{ and } 270^\circ$$

Centroid :

$$\sigma_c = \frac{\sum \text{Poles} - \sum \text{Zeros}}{P - Z}$$

$$\sigma_c = \frac{(2 - 1 - 2) - (1)}{2} = \frac{-2}{2} = -1$$

The breakaway branches emerge between $s = -2$ and $s = -1$, tending toward the asymptotes centered at $\sigma_c = -1$. Meanwhile, the remaining branch on the right-half plane starts from the open-loop pole $s = 2$ and terminates directly at the zero $s = 1$.

$$\boxed{(B)}$$

Mistakes to be avoided

Direction of Travel: Remember that root locus paths always travel out from open-loop poles ($K = 0$) towards open-loop zeros ($K = \infty$).

Q5.05.1 MCQ 2005 1M Ans: A ● ○ ○

From the root locus plot, there are 3 asymptotes with angles 60°, 180°, and 300°.

- The number of asymptotes is given by $P - Z = 3$
- Since all branches terminate at infinity, the system does not have any finite zeros ($Z = 0$). Thus, the number of open-loop poles is: $P = 3$
- The centroid (intersection of asymptotes on the real axis) is located at the origin ($\sigma = 0$).
- Since all three branches originate from the origin, all three open-loop poles must be located precisely at the origin ($s = 0, 0, 0$).

Therefore, the open-loop transfer function (O.L.T.F.) is: $G(s)H(s) = \dfrac{K}{s^3}$

$$\boxed{(A)\ \frac{K}{s^3}}$$

Mistakes to be avoided

Misinterpreting Centroid Intersection: Avoid assuming that if asymptotes pass through the origin, it always means all individual poles are at zero; it simply means the average position ($\sum$ poles) is zero. However, since the branches originate directly out of the origin in all three directions, the individual pole locations themselves must be at $s = 0$.

Q5.02.1 MCQ | 2002 | 2M | Ans: B | ● ○ ○

Given open-loop transfer function:

$$G(s)H(s) = \frac{K}{s^2}$$

The system has 2 open-loop poles located at the origin ($s = 0, 0$) and no open-loop zeros ($Z = 0$).

The number of branches going to infinity is:

$$P - Z = 2 - 0 = 2$$

The centroid (σ_A) of the asymptotes is calculated as:

$$\sigma_A = \frac{\sum \text{Real part of poles} - \sum \text{Real part of zeros}}{P - Z} = \frac{0 - 0}{2} = 0$$

The angles of asymptotes (ϕ_l) are given by:

$$\phi_l = \frac{(2l + 1)180^\circ}{P - Z}, \quad \text{where } l = 0, 1$$

$$\phi_0 = \frac{180^\circ}{2} = 90^\circ, \quad \phi_1 = \frac{3 \times 180^\circ}{2} = 270^\circ$$

Thus, the two root locus branches emerge from the origin and travel vertically along the imaginary axis ($j\omega$-axis) towards infinity as K increases from 0 to ∞.

$\boxed{\text{(B)}}$

Q5.02.2 NAT | 2002 | 5M | Ans: * | ● ● ●

Given:

$$G(s) = \frac{2(s + \alpha)}{s(s + 2)(s + 10)}, \qquad H(s) = 1$$

Since the root locus is plotted with respect to the parameter α:

$$1 + G(s)H(s) = 0 \implies 1 + \frac{2(s + \alpha)}{s(s + 2)(s + 10)} = 0 \implies s^3 + 12s^2 + 22s + 2\alpha = 0 \implies 1 + \frac{2\alpha}{s^3 + 12s^2 + 22s} = 0$$

Hence, the modified open-loop transfer function is:

$$G'(s)H'(s) = \frac{2\alpha}{s(s^2 + 12s + 22)}$$

Poles, Zeros, and Asymptotes:
The modified system poles are:

$$s = 0, \quad s = \frac{-12 \pm \sqrt{144 - 88}}{2}$$

$$s = 0, \ -2.26, \ -9.74$$

Number of asymptotes $A = P - Z = 3$.

$$\theta_A = \frac{(2q+1)180^\circ}{3} = 60^\circ, \ 180^\circ, \ 300^\circ$$

Centroid:

$$\sigma_A = \frac{\sum p - \sum z}{P - Z} = \frac{-9.74 - 2.26}{3} = -4$$

Breakaway Point:
Isolating α and setting $\frac{d\alpha}{ds} = 0$:

$$\alpha = -\frac{s^3 + 12s^2 + 22s}{2} \implies 3s^2 + 24s + 22 = 0$$

$$s = \frac{-24 \pm 17.66}{6} \implies s = -1.05, \ -6.94$$

Only $s = -1.05$ lies on a valid root locus segment.

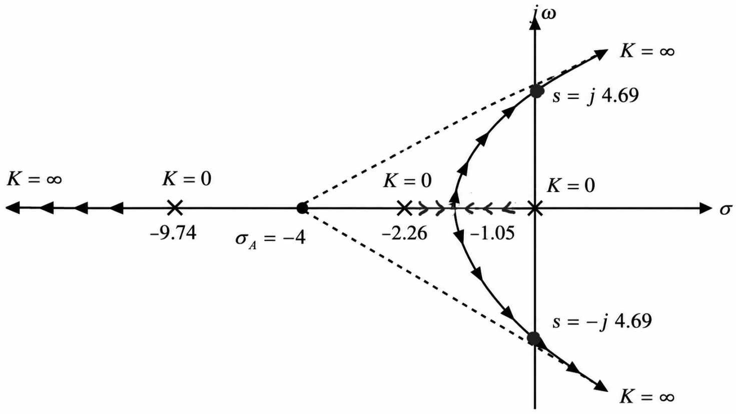
Imaginary axis crossing point:

$$\begin{array}{c|cc} s^3 & 1 & 22 \\ s^2 & 12 & 2\alpha \\ s^1 & \dfrac{264 - 2\alpha}{12} & 0 \\ s^0 & 2\alpha & 0 \end{array}$$

From the Routh array, for marginal stability:

$$\frac{264 - 2\alpha}{12} = 0 \implies \alpha = 132$$

A.E: $12s^2 + 2(132) = 0 \implies s = \pm j\,4.69.$



Mistakes to be avoided

Do not use the original zero at $s = -\alpha$ while computing asymptotes; the root locus rules must be applied to the modified open-loop transfer function where α acts as the gain.

Q5.01.1 NAT 2001 5M Ans: *

Given the characteristic equation: $s^3 + 2s^2 + Ks + K = 0$
We can rewrite it in the standard open-loop form $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s+1)}{s^3 + 2s^2} = 0 \implies 1 + \frac{K(s+1)}{s^2(s+2)} = 0$$

$$G(s)H(s) = \frac{K(s+1)}{s^2(s+2)}$$

Here, the number of open-loop poles (P) is 3 (at $s = 0, 0, -2$), and the number of open-loop zeroes (Z) is 1 (at $s = -1$). The number of asymptotes is $P - Z = 3 - 1 = 2$.

1. Angle of Asymptotes and Real Axis Intercept

$$\theta = \frac{(2k+1)\times 180^\circ}{P-Z}, \quad \text{for } k = 0, 1$$
$$= 90^\circ, 270^\circ$$

The real axis intercept (centroid, σ_A) is:

$$\sigma_A = \frac{\sum \text{Re(poles)} - \sum \text{Re(zeroes)}}{P-Z}$$
$$= \frac{(0+0-2)-(-1)}{2} = -0.5$$

2. Break-away / Break-in Points

From the characteristic equation, express K in terms of s:

$$K = -\frac{s^2(s+2)}{s+1} = -\frac{s^3+2s^2}{s+1}$$

Differentiating K with respect to s and setting it to zero ($\frac{dK}{ds} = 0$):

$$\frac{dK}{ds} = -\left[\frac{(3s^2+4s)(s+1)-(s^3+2s^2)}{(s+1)^2}\right] = 0$$

$$(3s^3+3s^2+4s^2+4s)-(s^3+2s^2) = 0$$
$$2s^3+5s^2+4s = 0 \implies s(2s^2+5s+4) = 0$$
$$s = 0, s = -1.25 \pm j0.66$$

Only $s = 0$ is a valid breakaway point because it lies on the root locus path between the two open-loop poles located at the origin.

3. Imaginary Axis Crossing Points

Using the RH Array : $s^3 + 2s^2 + Ks + K = 0$:

$$\begin{array}{c|cc} s^3 & 1 & K \\ s^2 & 2 & K \\ s^1 & \frac{K}{2} & \\ s^0 & K & \end{array}$$

For the system to sustain oscillations (crossing the imaginary axis), the s^1 row must be zero:

$$\frac{K}{2} = 0 \implies K = 0$$

Since $K = 0$ corresponds to the starting points of the root locus (at the open-loop poles $s = 0$), there is **no imaginary axis intersection point** for any $K > 0$.

4. Root Locus Sketch

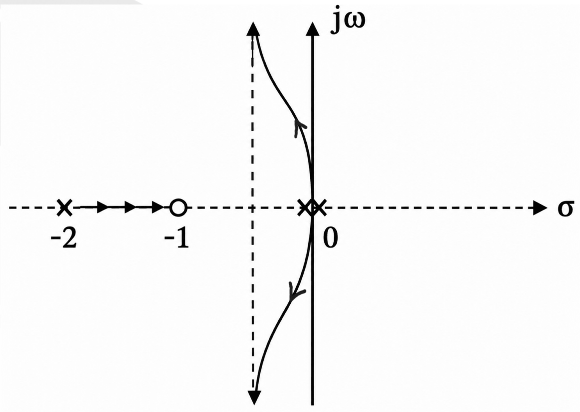


Fig. Solution Q5.01.1

Q5.00.1 | NAT | 2000 | 5M | Ans: * |

Since complex roots lie along a circle, the root locus is sketched by identifying key open-loop features:

- Open-loop poles: $s = 0$ and $s = -2$.
- Open-loop zero: $s = -5$.

The two poles break away from the real axis between 0 and -2, form a circular path in the complex plane, and break back into the real axis to the left of the zero at -5.

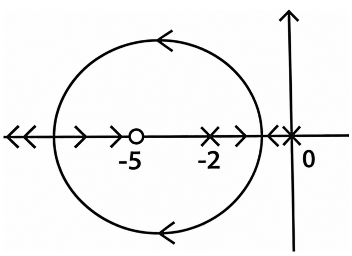


Fig. Solution Q5.00.1

Q5.00.2 | NAT | 2000 | 5M | Ans: * |

As K increases, the branches of the root locus move leftward in the complex s-plane (away from the imaginary axis and deeper into the left-half plane). This increases the negative real part of the closed-loop roots.

Therefore, the system becomes **more stable**.

Q5.00.3 NAT 2000 5M Ans: * ● ● ○

The closed-loop transfer function is given by:

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{\frac{K(s+5)}{s(s+2)}}{1+\frac{K(s+5)}{s(s+2)}} = \frac{K(s+5)}{s^2+(2+K)s+5K}$$

The characteristic equation is: $s^2 + (2+K)s + 5K = 0$
Comparing with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \sqrt{5K}; \quad 2\xi\omega_n = 2+K \implies \xi = \frac{2+K}{2\sqrt{5K}}$$

Given the desired damping ratio $\xi = 0.3$:

$$\frac{2+K}{2\sqrt{5K}} = 0.3 \implies \frac{K^2+4K+4}{20K} = 0.09$$

$$K^2 + 2.2K + 4 = 0$$

Checking the discriminant ($\Delta = b^2 - 4ac$): $\Delta = (2.2)^2 - 4(1)(4) = -11.16 < 0$
Since the roots are complex in nature, **no real value of K exists**.

Q5.92.1 MSQ 1992 1M Ans: A ● ○ ○

- **Symmetry Property:** Since the characteristic equation has real coefficients, the root locus must be perfectly symmetric about the real axis (σ-axis). Figures (a), (c), and (d) appear symmetric, while (b) is asymmetric and therefore invalid.
- **Real Axis Locus and Breakaway/Break-in Behavior:** For a branch to break away or break into the real axis and form a complex loop (like a circle), there must be a valid combination of open-loop poles and zeros

Thus, only figure (A) represents a valid root locus configuration.

$$\boxed{(A)}$$

Q5.91.1 MCQ 1991 1M Ans: A ● ○ ○

Given the open-loop transfer function: $G(s) = \dfrac{K(s+a)}{s^2(s+b)}; \quad b > a$

Step 1: Identify Poles and Zeros

- Number of poles $(P) = 3$ at $s = 0, 0, -b$.
- Number of zeros $(Z) = 1$ at $s = -a$.
- Number of asymptotes $= P - Z = 3 - 1 = 2$.

Step 2: Angles of Asymptotes

$$\theta = \frac{(2k+1)180^\circ}{P-Z} = \frac{(2k+1)180^\circ}{2}$$

For $k = 0, 1$:

$$\theta = 90^\circ, 270^\circ$$

Step 3: Centroid (σ_A)

$$\sigma_A = \frac{\sum \text{Poles} - \sum \text{Zeros}}{P-Z}$$

$$\sigma_A = \frac{(0+0-b)-(-a)}{2} = \frac{a-b}{2}$$

Since $b > a$, the centroid is **negative** ($\sigma_A < 0$).

Step 4: Real Axis Locus Test
The locus exists on the real axis only between $-a$ and $-b$, with branches breaking away from the origin into the complex plane toward $\pm 90^\circ$. This matches option (a).

$$\boxed{(A)}$$

Q5.91.2 NAT 1991 5M Ans: K>2 ● ○ ○

The characteristic equation is given by $1 + G(s) = 0$:

$$1 + \frac{K(s+2)^2}{s^2(s-1)} = 0 \implies s^3 - s^2 + K(s^2 + 4s + 4) = 0$$

$$s^3 + (K-1)s^2 + 4Ks + 4K = 0$$

RH Array:

$$\begin{array}{c|cc} s^3 & 1 & 4K \\ s^2 & K-1 & 4K \\ s^1 & \dfrac{4K(K-1)-4K}{K-1} = \dfrac{4K^2-8K}{K-1} & 0 \\ s^0 & 4K & \end{array}$$

For stable operation, all elements in the first column must be strictly positive:

- From s^2 row: $K - 1 > 0 \implies K > 1$
- From s^1 row: $4K^2 - 8K > 0 \implies 4K(K-2) > 0 \implies K > 2$

$$\boxed{K > 2}$$

Q5.91.3 NAT 1991 5M Ans: $\pm j2\sqrt{2}$ ● ○ ○

The imaginary axis crossover points correspond to the value of K that causes a row of zeros in the Routh array, which happens at the margin of stability when the s^1 row becomes zero:

$$4K^2 - 8K = 0 \implies K = 2 \quad (\text{since } K > 1)$$

Forming the auxiliary equation $A(s) = 0$ from the s^2 row just above it:

$$(K-1)s^2 + 4K = 0$$

Substituting $K = 2$:

$$(2-1)s^2 + 4(2) = 0 \implies s^2 + 8 = 0$$

$$s = \pm j2\sqrt{2}$$

$$\boxed{\pm j2\sqrt{2}}$$

Q5.91.4 NAT 1991 5M Ans: * ● ○ ○

Root Locus Analysis:

- **Poles:** $s = 0, 0$ (double pole) and $s = 1$.
- **Zeros:** $s = -2, -2$ (double zero).
- **Real Axis Test:**
  - Region $(0, 1)$ has 1 pole to its right (odd $\rightarrow$ **On RL**).
  - Region $(-2, 0)$ has 3 poles to its right (odd $\rightarrow$ **On RL**).

Since the segment between 0 and 1 lies on the root locus, the branch originating from $s = 1$ moves left towards the origin. One of the poles at $s = 0$ must break away into the complex plane, crossing the imaginary axis at $\pm j2\sqrt{2}$, and finally breaking back into the real axis to terminate at one of the double zero at $s = -2$.

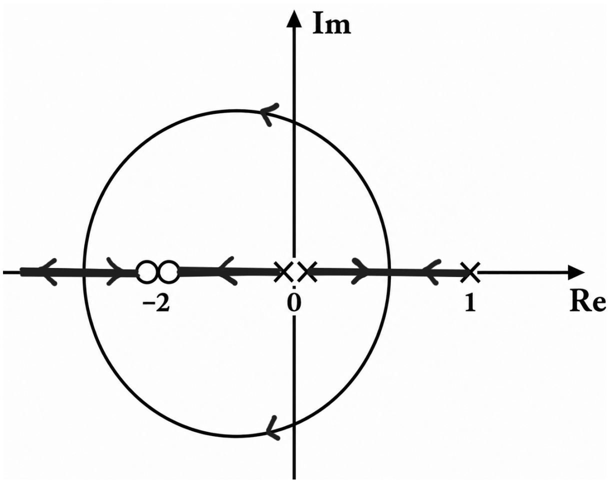


Fig. Solution Q5.91.4

DEBUG INFO

Chapter V

Root Locus

Total Questions : 22

MCQ : 11

MSQ : 2

NAT : 9

SOLUTIONS

VI

Polar Plot and Frequency Response of Second Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

Q6.24.1 NAT | 2024 | 2M | Ans: 1.11 to 1.13 | ● ● ○ | CLICK FOR VIDEO SOLUTION

From the block diagram,

$$L(s) = \frac{K_I}{s}G(s)$$

The phase of $L(j\omega)$ is

$$\angle L(j\omega) = \angle G(j\omega) - 90^\circ$$

The phase margin is evaluated at the gain crossover frequency.
For phase margin = 50°,

$$PM = 180^\circ + \angle L(j\omega_{gc})$$

$$50^\circ = 180^\circ + \angle G(j\omega_{gc}) - 90^\circ$$

$$\angle G(j\omega_{gc}) = -40^\circ$$

From the given table, $\angle G(j\omega) = -40^\circ$ at $\omega_{gc} = 0.5$ rad/s
From the table, $|G(j0.5)|_{dB} = -7$ dB
In numerical, $|G(j0.5)| = 10^{-7/20} = 0.4466$
Now, at gain crossover frequency the magnitude to be 1. Thus,

$$|L(j\omega_{gc})| = 1$$

$$\frac{K_I}{\omega_{gc}}|G(j\omega_{gc})| = 1$$

$$\frac{K_I}{0.5} \times 0.4466 = 1$$

$$K_I \approx 1.12$$

$$\boxed{1.12}$$

Q6.22.1 MCQ | 2022 | 2M | Ans: - | ● ○ ○

Using superposition, consider the DC and sinusoidal inputs separately.
For the DC input $r(t) = 1$,

$$a = G(0) \times 1$$

$$G(0) = \frac{100}{0 + 0 + 10} = 10$$

$$a = 10$$

For the sinusoidal input $0.1\sin(10t)$, the output amplitude is

$$b = 0.1\,|G(j10)|$$

$$|G(j10)| = \left|\frac{100}{(j10)^2 + 0.1(j10) + 10}\right| = \frac{100}{|-90 + j|} \approx 1.11$$

$$b = 0.1 \times 1.11 \approx 0.11$$

No option matches

$$\boxed{a = 10,\ b = 0.1}$$

Key Points

- For an LTI system, the steady-state response to a sinusoidal input remains sinusoidal at the same frequency. Only the amplitude and phase change.
- If the input is
$$x(t) = A\sin(\omega_0 t + \phi),$$
then the steady-state output is
$$y(t) = A|G(j\omega_0)|\sin(\omega_0 t + \phi + \angle G(j\omega_0))\,.$$
- For inputs containing both DC and sinusoidal components, use the principle of superposition and evaluate the response of each component separately.

Q6.17.1 MCQ | 2017 | 1M | Ans: D | ● ○ ○

The frequency response of the system is
$$G(j\omega) = \frac{1 - j\omega}{1 + j\omega}$$
Hence,
$$|G(j\omega)| = \frac{\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}} = 1$$
Therefore, only the phase changes.
$$\angle G(j\omega) = \angle(1 - j\omega) - \angle(1 + j\omega)$$
$$\phi = -\tan^{-1}\omega - \tan^{-1}\omega$$
$$\phi = -2\tan^{-1}\omega$$
Since $\omega \in [0, \infty)$,
$$\text{For } \omega = 0 \text{ rad/sec}, \ \phi = -2\tan^{-1}(0) = 0$$
$$\text{For } \omega = \infty \text{ rad/sec}, \ \phi = -2\tan^{-1}(\infty) = -\pi$$
Therefore,
$$\phi_{\min} = -\pi, \qquad \phi_{\max} = 0$$
$$\boxed{(D) \quad -\pi,\ 0}$$

Q6.17.2 NAT | 2017 | 1M | Ans: 1.01 to 1.06 | ● ● ○

The open-loop transfer function is
$$G(s) = \frac{Ke^{-s}}{s}$$
Hence,
$$G(j\omega) = \frac{Ke^{-j\omega}}{j\omega}$$
The phase of the open-loop transfer function is
$$\angle G(j\omega) = -\omega - \frac{\pi}{2}$$
For phase margin = 30°,
$$PM = 180^\circ + \angle G(j\omega_{gc})$$
$$30^\circ = 180^\circ - \omega_{gc} - 90^\circ$$
$$\omega_{gc} = 60^\circ = \frac{\pi}{3} \text{ rad/s}$$

At gain crossover frequency,

$$|G(j\omega_{gc})| = 1$$

$$\left|\frac{Ke^{-j\omega_{gc}}}{j\omega_{gc}}\right| = 1$$

$$\frac{K}{\omega_{gc}} = 1$$

$$K = \omega_{gc} = \frac{\pi}{3}$$

$$K \approx 1.047$$

$$\boxed{1.05}$$

Key Points

- A pure time delay e^{-sT} contributes only phase and does not affect magnitude:

$$|e^{-j\omega T}| = 1, \qquad \angle e^{-j\omega T} = -\omega T \text{ radians}$$

Q6.16.1 MCQ 2016 1M Ans: B ● ○ ○

The frequency response of the system is

$$G(j\omega) = \frac{j\omega}{j\omega + 2}$$

For the input $\cos(2t)$, substitute $\omega = 2$:

$$G(j2) = \frac{j2}{2 + j2}$$

The output amplitude is

$$A = |G(j2)| = \frac{|j2|}{|2 + j2|} = \frac{2}{\sqrt{2^2 + 2^2}} = \frac{1}{\sqrt{2}}$$

The phase shift is

$$\phi = \angle G(j2) = \angle(j2) - \angle(2 + j2) = 90^\circ - 45^\circ = 45^\circ$$

Therefore,

$$A = \frac{1}{\sqrt{2}}, \qquad \phi = 45^\circ$$

$$\boxed{(B)\ \ \frac{1}{\sqrt{2}},\ +45^\circ}$$

Q6.16.2 MCQ 2016 1M Ans: A ● ○ ○

The phase of the transfer function is

$$\angle G(j\omega) = -3\tan^{-1}(\omega)$$

At the phase cross-over frequency,

$$\angle G(j\omega_{pc}) = -180^\circ$$

$$-3\tan^{-1}(\omega_{pc}) = -180^\circ$$

$$\tan^{-1}(\omega_{pc}) = 60^\circ$$

$$\omega_{pc} = \tan 60^\circ$$

$$\omega_{pc} = \sqrt{3} \text{ rad/s}$$

$$\boxed{(A) \quad \sqrt{3} \text{ rad/s}}$$

Q6.15.1 MCQ 2015 2M Ans: D ● ○ ○

Since the system is a second-order real system and has a pole at $s = 2 - j3$, its complex conjugate must also be a pole. Hence, the poles are

$$s = 2 \pm j3$$

A perfectly flat magnitude response of unity implies that

$$|G(j\omega)| = 1, \qquad \forall\, \omega$$

Therefore, the system is an all-pass system. For a continuous-time all-pass system, every pole at $s = p$ has a corresponding zero at $s = -p^*$. Hence, corresponding to the poles at $s = 2 \pm j3$, the zeros are located at

$$s = -2 \pm j3$$

Therefore,

$$\boxed{\text{(D) Poles at } (2 \pm j3), \text{ zeros at } (-2 \pm j3)}$$

Key Points

- A real system must have complex poles and zeros in conjugate pairs.
- In a continuous-time all-pass system, zeros are obtained by reflecting the poles about the imaginary axis.

Q6.14.1 MCQ 2014 1M Ans: B ● ○ ○

For the input $x(t) = \cos(3t)$, the steady-state output amplitude is

$$A = |H(j3)|$$

Given,

$$H(s) = \frac{1}{s(s+4)}$$

$$H(j3) = \frac{1}{(j3)(4+j3)}$$

$$|H(j3)| = \frac{1}{|j3|\,|4+j3|} = \frac{1}{3 \times \sqrt{4^2+3^2}} = \frac{1}{3 \times 5} = \frac{1}{15}$$

$$\boxed{\text{(B) } \frac{1}{15}}$$

Q6.12.1 MCQ 2012 1M Ans: C ● ○ ○

The steady-state output becomes zero when

$$|G(j\omega)| = 0$$

Since

$$G(s) = \frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$$

$$G(j\omega) = \frac{(9-\omega^2)(j\omega+2)}{(j\omega+1)(j\omega+3)(j\omega+4)}$$

The denominator is finite for all the given values of ω. Hence, $G(j\omega) = 0$ only when Numerator = 0

$$9 - \omega^2 = 0$$

$$\omega^2 = 9$$

$$\omega = 3 \text{ rad/s}$$

$$\boxed{\text{(C) } \omega = 3 \text{ rad/s}}$$

Q6.11.1 MCQ 2011 1M Ans: A ● ○ ○

Phase crossover frequency occurs when

$$\angle G(j\omega) = -180^\circ$$

From the table,

$$|G(j\omega_{pc})| = 0.5$$

Therefore, the gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{1}{0.5} = 2$$

In dB,

$$GM_{dB} = 20\log_{10}(2) \approx 6 \text{ dB}$$

Gain crossover frequency occurs when

$$|G(j\omega)| = 1$$

From the table,

$$\angle G(j\omega_{gc}) = -150^\circ$$

Therefore, the phase margin is

$$PM = 180^\circ + \angle G(j\omega_{gc}) = 180^\circ - 150^\circ = 30^\circ$$

$$\boxed{\text{(A) 6 dB and } 30^\circ}$$

Key Points

- Gain crossover frequency (ω_{gc}) is the frequency at which

$$|G(j\omega)| = 1$$

- Phase crossover frequency (ω_{pc}) is the frequency at which

$$\angle G(j\omega) = -180^\circ$$

- The gain margin and phase margin are given by

$$GM = \frac{1}{|G(j\omega_{pc})|}, \qquad PM = 180^\circ + \angle G(j\omega_{gc})$$

Q6.10.1 MCQ 2010 2M Ans: A ● ● ○

Given,

$$G(s) = \frac{1}{s(s+1)(s+2)}$$

Putting $s = j\omega$,

$$G(j\omega) = \frac{1}{j\omega(j\omega+1)(j\omega+2)}$$

$$G(j\omega) = \frac{1}{j\omega\left[(2-\omega^2) + j3\omega\right]}$$

$$G(j\omega) = \frac{1}{-3\omega^2 + j\omega(2-\omega^2)}$$

After rationalizing,

$$G(j\omega) = \frac{-3\omega^2 - j\omega(2-\omega^2)}{9\omega^4 + \omega^2(2-\omega^2)^2} = \frac{-3\omega^2}{9\omega^4 + \omega^2(2-\omega^2)^2} + j\frac{\omega(\omega^2-2)}{9\omega^4 + \omega^2(2-\omega^2)^2}$$

Hence,

$$\Re\{G(j\omega)\} = \frac{-3\omega^2}{9\omega^4 + \omega^2(2-\omega^2)^2}$$

$$\Im\{G(j\omega)\} = \frac{\omega(\omega^2-2)}{9\omega^4 + \omega^2(2-\omega^2)^2}$$

Real part is negative for all $\omega > 0$.
Imaginary part changes sign when

$$\omega^2 - 2 = 0$$

$$\omega = \sqrt{2}$$

Therefore, the Polar plot crosses the real axis at $\omega = \sqrt{2}$
Substituting $\omega = \sqrt{2}$ in G($j\omega$),

$$G(j\sqrt{2}) = -\frac{1}{6}$$

Therefore, the Polar plot cuts the real axis at $\left(-\frac{1}{6}, 0\right)$

For $0 < \omega < \sqrt{2}$

$$\omega^2 - 2 < 0$$

Hence, imaginary part is negative.
Therefore, the Polar plot lies in Quadrant III.
For $\omega > \sqrt{2}$

$$\omega^2 - 2 > 0$$

Hence, imaginary part is positive.
Therefore, the Polar plot lies in Quadrant II.
Now,

$$|G(j\omega)| = \frac{1}{\omega\sqrt{(1+\omega^2)(4+\omega^2)}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\omega - \tan^{-1}\left(\frac{\omega}{2}\right)$$

$$G(j0) = -\frac{3}{4} - j\infty \qquad \angle G(j0) = -90^\circ$$

$$G(j\infty) = 0 \qquad \angle G(j\infty) = -270^\circ$$

Therefore, the correct plot is option (A).

$$\boxed{(A)}$$

Q6.09.1 MCQ 2009 2M Ans: D ● ● ○

Given,

$$G(s) = \frac{e^{-0.1s}}{s}$$

The phase of the frequency response is

$$\angle G(j\omega) = -90^\circ - 0.1\omega\left(\frac{180^\circ}{\pi}\right)$$

At phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180^\circ$$

$$-90^\circ - 0.1\omega_{pc}\left(\frac{180^\circ}{\pi}\right) = -180^\circ$$

$$0.1\omega_{pc}\left(\frac{180^\circ}{\pi}\right) = 90^\circ$$

$$\omega_{pc} = \frac{\pi}{0.2} = 5\pi \text{ rad/s}$$

The magnitude at phase crossover frequency is

$$|G(j\omega_{pc})| = \frac{1}{\omega_{pc}} = \frac{1}{5\pi}$$

Hence, the gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|} = 5\pi$$

In dB,

$$GM_{dB} = 20\log_{10}(5\pi) \approx 23.9 \text{ dB}$$

$$\boxed{\text{(D) } 23.9 \text{ dB}}$$

Mistakes to be avoided

- Do not substitute -0.1ω directly as degrees. The exponential term $e^{-j\omega T}$ produces a phase of $-\omega T$ radians which must be converted to degrees as $-\omega T\left(\frac{180^\circ}{\pi}\right)$ before equating it to -180°.

Q6.05.1 MCQ 2005 1M Ans: A ● ○ ○

Given: The open loop transfer function is,

$$G(s) = \frac{s+1}{s^2}$$

$$G(j\omega) = \frac{1+j\omega}{(j\omega)^2} = \frac{\sqrt{1+\omega^2}}{\omega^2} \angle -180^\circ + \tan^{-1}\omega$$

ω	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0	∞	-180°
∞	0	-90°

From the above table, $\omega_{pc} = 0$ rad/s and $|G(j\omega_{pc})| = \infty$
Gain margin is given by,

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{1}{\infty} = 0$$

Therefore, the correct option is,

$$\boxed{\text{(A) } 0}$$

Q6.04.1 MCQ 2004 2M Ans: B ● ○ ○

Given,

$$G(s) = \frac{s}{s+1} \quad \text{and} \quad x(t) = \sin t$$

For a sinusoidal input, the steady-state output is

$$y_{ss}(t) = |G(j\omega)| \sin(\omega t + \angle G(j\omega))$$

Here, $\omega = 1$ rad/s. Therefore,

$$G(j) = \frac{j}{1+j} = \frac{1}{\sqrt{2}} \angle 45^\circ$$

Hence, the steady-state response is

$$y(t) = \frac{1}{\sqrt{2}} \sin(t + 45^\circ)$$

$$\boxed{\text{(B) } \frac{1}{\sqrt{2}} \sin(t + 45^\circ)}$$

Q6.04.2 MCQ 2004 2M Ans: C ● ● ○

Given,

$$G(s) = \frac{as+1}{s^2}$$

$$G(j\omega) = \frac{1 + ja\omega}{(j\omega)^2} = \frac{\sqrt{1 + a^2\omega^2}}{\omega^2} \angle -180^\circ + \tan^{-1}(a\omega)$$

At gain crossover frequency,

$$|G(j\omega_{gc})| = 1$$

$$\frac{\sqrt{1 + a^2\omega_{gc}^2}}{\omega_{gc}^2} = 1$$

$$1 + a^2\omega_{gc}^2 = \omega_{gc}^4$$

Since the phase margin is 45°,

$$PM = 180^\circ + \angle G(j\omega_{gc})$$

$$45^\circ = 180^\circ + \left[-180^\circ + \tan^{-1}(a\omega_{gc})\right]$$

$$\tan^{-1}(a\omega_{gc}) = 45^\circ$$

$$a\omega_{gc} = 1$$

$$a = \frac{1}{\omega_{gc}}$$

Substituting $a\omega_{gc} = 1$ in $1 + a^2\omega_{gc}^2 = \omega_{gc}^4$,

$$1 + a^2\omega_{gc}^2 = \omega_{gc}^4$$
$$1 + 1 = \omega_{gc}^4$$
$$\omega_{gc} = 2^{1/4}$$
$$a = \frac{1}{2^{1/4}} \approx 0.841$$

$$\boxed{\text{(C) } 0.841}$$

Q6.01.1 NAT | 2001 | 0M | Ans: 0 | ● ● ○

Given,

$$G(s) = \frac{10000}{s(s+10)^2}$$
$$G(j\omega) = \frac{10000}{j\omega(j\omega+10)^2}$$

At $\omega = 20$ rad/s,

$$|G(j20)| = \frac{10000}{20\left(\sqrt{10^2+20^2}\right)^2} = \frac{10000}{20(500)} = 1$$

Therefore,

$$20\log_{10}|G(j20)| = 20\log_{10}(1) = 0\,\text{dB}$$

$$\boxed{0}$$

Q6.01.2 NAT | 2001 | 0M | Ans: -36.87 | ● ● ○

Gain crossover frequency is obtained from

$$|G(j\omega_{gc})| = 1$$

From Q6.01.1, $\omega_{gc} = 20$ rad/s.
Phase angle of the open loop transfer function is

$$\angle G(j\omega) = -90^\circ - 2\tan^{-1}\left(\frac{\omega}{10}\right)$$

Therefore,

$$\angle G(j20) = -90^\circ - 2\tan^{-1}(2) = -90^\circ - 2(63.435^\circ) = -216.87^\circ$$

Hence, the phase margin is

$$PM = 180^\circ + \angle G(j20) = 180^\circ - 216.87^\circ = -36.87^\circ$$

$$\boxed{-36.87}$$

Q6.01.3 NAT 2001 0M Ans: -13.98 ● ● ○

Phase crossover frequency is obtained from

$$\angle G(j\omega_{pc}) = -180^\circ$$
$$-90^\circ - 2\tan^{-1}\left(\frac{\omega_{pc}}{10}\right) = -180^\circ$$
$$2\tan^{-1}\left(\frac{\omega_{pc}}{10}\right) = 90^\circ$$
$$\tan^{-1}\left(\frac{\omega_{pc}}{10}\right) = 45^\circ$$
$$\omega_{pc} = 10 \text{ rad/s}$$

Magnitude at phase crossover frequency is

$$|G(j10)| = \frac{10000}{10\left(\sqrt{10^2+10^2}\right)^2} = \frac{10000}{10(200)} = 5$$

Hence,

$$GM = \frac{1}{|G(j10)|} = \frac{1}{5}$$

Therefore, gain margin in dB is

$$GM_{dB} = 20\log_{10}\left(\frac{1}{5}\right) = -13.98 \text{ dB}$$

$$\boxed{-13.98}$$

Q6.01.4 NAT 2001 0M Ans: Unstable ● ○ ○

From Q6.01.2,

$$PM = -36.87^\circ < 0$$

Also, from Q6.01.3,

$$GM = -13.98 \text{ dB} < 0$$

Since both the phase margin and gain margin are negative, the closed-loop system is unstable.

$$\boxed{\text{Unstable}}$$

Q6.00.1 NAT 2000 1M Ans: 51.91° ● ○ ○

For $\tau_D = 0$, the open-loop transfer function becomes

$$G_1(s) = \frac{1}{s(s+1)(s+2)}$$

It is given that $|G_1(j\omega)| = 1$ when $\omega_{gc} = 0.466$ rad/s
The phase angle at gain crossover frequency is

$$\angle G_1(j\omega_{gc}) = -90^\circ - \tan^{-1}(\omega_{gc}) - \tan^{-1}\left(\frac{\omega_{gc}}{2}\right)$$
$$\angle G_1(j0.466) = -90^\circ - \tan^{-1}(0.466) - \tan^{-1}(0.233) = -90^\circ - 24.98^\circ - 13.11^\circ = -128.09^\circ$$

Therefore, the phase margin is

$$PM = 180^\circ + \angle G_1(j\omega_{gc}) = 180^\circ - 128.09^\circ = 51.91^\circ$$

$$\boxed{51.91}$$

Q6.00.2 NAT 2000 Ans: * ● ● ○

The delay term in the open-loop transfer function is $e^{-s\tau_D} = e^{-j\omega\tau_D}$
Its magnitude is $\left|e^{-j\omega\tau_D}\right| = 1$, so the gain of the system remains unchanged. However, it introduces an additional phase lag of $\angle e^{-j\omega\tau_D} = -\omega\tau_D$ rad.
Hence, the phase margin in the presence of dead time becomes

$$\text{PM}_{\text{new}} = \text{PM}_{\text{old}} - \omega_{gc}\tau_D.$$

Therefore, as the dead time τ_D increases, the additional phase lag increases, causing the phase margin to decrease. Consequently, the stability of the closed-loop system decreases.
Hence,

$$\boxed{\text{As dead time increases, the stability decreases}}$$

Q6.00.3 NAT 2000 Ans: 1.94 ● ● ○

From Q6.00.1, the phase margin without dead time is

$$\text{PM}_{\text{old}} = 51.91^\circ = \frac{51.91\pi}{180} \text{ rad} = 0.906 \text{ rad}$$

From Q6.00.2,

$$\text{PM}_{\text{new}} = \text{PM}_{\text{old}} - \omega_{gc}\tau_D$$

The maximum allowable dead time corresponds to the stability limit, where the phase margin becomes zero.

$$0 = \text{PM}_{\text{old}} - \omega_{gc}\tau_{D,\max}$$

$$\omega_{gc}\tau_{D,\max} = \text{PM}_{\text{old}}$$

$$\tau_{D,\max} = \frac{\text{PM}_{\text{old}}}{\omega_{gc}}$$

Substituting $\omega_{gc} = 0.466$ rad/s,

$$\tau_{D,\max} = \frac{0.906}{0.466} = 1.944 \text{ s}$$

Hence,

$$\boxed{\tau_{D,\max} \approx 1.94 \text{ s}}$$

Mistakes to be avoided

The characteristic equation of the closed-loop system is $s(s+1)(s+2) + e^{-s\tau_D} = 0$. Using the first-order approximation, $e^{-s\tau_D} \approx 1 - s\tau_D$, the characteristic equation becomes

$$s^3 + 3s^2 + (2 - \tau_D)s + 1 = 0$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 2-\tau_D \\ s^2 & 3 & 1 \\ s^1 & \dfrac{5-3\tau_D}{3} & 0 \\ s^0 & 1 & \end{array}$$

For stability,

$$\frac{5-3\tau_D}{3} > 0$$

$$\tau_D < \frac{5}{3}$$

Hence, the maximum permissible dead time is

$$\tau_{D,\max} = \frac{5}{3} \text{ s} = 1.67 \text{ s}$$

The above result is obtained using the first-order approximation $e^{-s\tau_D} \approx 1 - s\tau_D$, which is valid only when $|s\tau_D| \ll 1$. In this problem, $\omega_{gc}\tau_{D,\max} \approx 0.9$ rad, which is not sufficiently small. Hence, the approximation introduces a noticeable error, giving $\tau_{D,\max} \approx 1.67$ s, whereas the more accurate value is $\tau_{D,\max} \approx 1.94$ s, obtained directly from the phase margin relation. Since the gain crossover frequency is already given, the phase margin method is the preferred approach here.

Q6.97.1 NAT | 1997 | 1M | Ans: 33.62 | ● ● ○

Given,

$$G(s) = \frac{1}{s(s+2)(s+4)}$$

$$G(j\omega) = \frac{1}{j\omega(j\omega+2)(j\omega+4)}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{4}\right)$$

At phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180^\circ$$

$$-90^\circ - \tan^{-1}\left(\frac{\omega_{pc}}{2}\right) - \tan^{-1}\left(\frac{\omega_{pc}}{4}\right) = -180^\circ$$

$$\tan^{-1}\left(\frac{\omega_{pc}}{2}\right) + \tan^{-1}\left(\frac{\omega_{pc}}{4}\right) = 90^\circ$$

Using identity, $\tan^{-1} A + \tan^{-1} B = 90^\circ \Rightarrow$ AB=1,

$$\frac{\omega_{pc}}{2} \times \frac{\omega_{pc}}{4} = 1$$

$$\omega_{pc}^2 = 8$$

$$\omega_{pc} = 2\sqrt{2} \text{ rad/s}$$

Magnitude at phase crossover frequency is

$$|G(j\omega_{pc})| = \frac{1}{\omega_{pc}\sqrt{\omega_{pc}^2 + 2^2}\sqrt{\omega_{pc}^2 + 4^2}}$$

$$|G(j2\sqrt{2})| = \frac{1}{(2\sqrt{2})(2\sqrt{3})(2\sqrt{6})} = \frac{1}{48}$$

Hence,

$$GM = \frac{1}{|G(j\omega_{pc})|} = 48$$

Therefore, gain margin in dB is

$$GM_{dB} = 20\log_{10}(48) = 33.62 \text{ dB}$$

$$\boxed{33.62}$$

Q6.96.1 MCQ 1996 1M Ans: D

Given,

$$\frac{C(s)}{R(s)} = \frac{1}{1+s}$$

For the input $r(t) = \sin t$, the excitation frequency is $\omega = 1$ rad/s. Therefore,

$$T(j) = \frac{1}{1+j}$$

Magnitude is

$$|T(j)| = \frac{1}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}}$$

Phase is

$$\angle T(j) = -\tan^{-1}(1) = -45^\circ = -\frac{\pi}{4}$$

Hence, the steady-state response is

$$c(t) = \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)$$

$$\boxed{\text{(D) } \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)}$$

Q6.94.1 MCQ 1994 4M Ans: B

(a) Given,

$$G(s) = \frac{s}{(s+1)(s+2)}$$

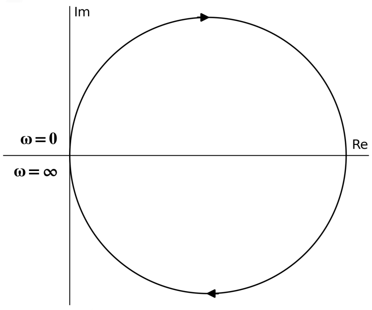
Put $s = j\omega$,

$$G(j\omega) = \frac{j\omega}{(1+j\omega)(2+j\omega)}$$

$$= \frac{\omega}{\sqrt{1+\omega^2}\sqrt{4+\omega^2}}\angle\left(90^\circ - \tan^{-1}\omega - \tan^{-1}\left(\frac{\omega}{2}\right)\right)$$

$$G(j0) = 0\angle 90^\circ$$

$$G(j\infty) = 0\angle -90^\circ$$



Therefore, the polar plot is T.

(b) Given,

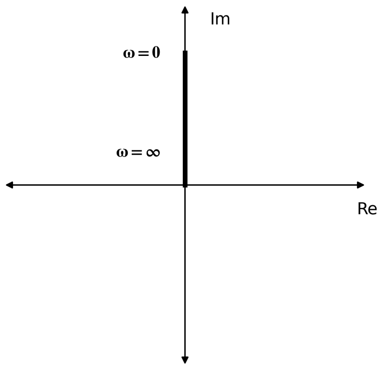
$$G(s) = \frac{s^2+1}{s^3}$$

Put $s = j\omega$,

$$G(j\omega) = \frac{1-\omega^2}{(j\omega)^3} = \frac{|1-\omega^2|}{\omega^3}\angle -270^\circ$$

$$G(j0) = \infty\angle -270^\circ$$

$$G(j\infty) = 0\angle -270^\circ$$



Therefore, the polar plot is R.

(c) Given,

$$G(s) = \frac{s^2 - 1}{s^2 + 1}$$

Put $s = j\omega$,

$$G(j\omega) = \frac{-\omega^2 - 1}{1 - \omega^2}$$

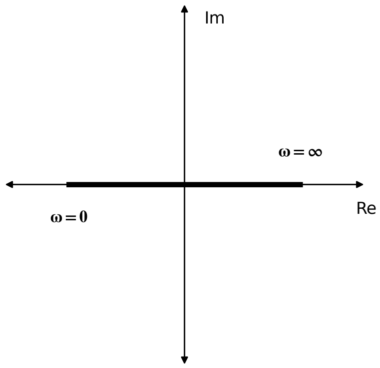
$$|G(j\omega)| = \frac{\omega^2 + 1}{|1 - \omega^2|}$$

$$\angle G(j\omega) = 180^\circ, \qquad 0 < \omega < 1$$

$$\angle G(j\omega) = 0^\circ, \qquad \omega > 1$$

$$G(j0) = 1\angle 180^\circ$$

$$G(j\infty) = 1\angle 0^\circ$$



Therefore, the polar plot is Q.

(d) Given,

$$G(s) = \frac{1}{s^2 + 10}$$

Put $s = j\omega$,

$$G(j\omega) = \frac{1}{10 - \omega^2}$$

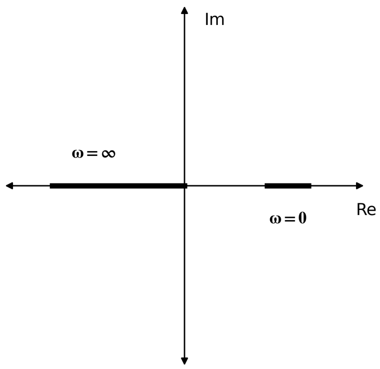
$$|G(j\omega)| = \frac{1}{|10 - \omega^2|}$$

$$\angle G(j\omega) = 0^\circ, \qquad 0 < \omega < \sqrt{10}$$

$$\angle G(j\omega) = 180^\circ, \qquad \omega > \sqrt{10}$$

$$G(j0) = 0.1\angle 0^\circ$$

$$G(j\infty) = 0\angle 180^\circ$$



Therefore, the polar plot is U.

$$\boxed{\text{(B) a-T, b-R, c-Q, d-U}}$$

DEBUG INFO

Chapter VI

Polar Plot and Frequency Response of Second Order Systems

Total Questions : 25

MCQ : 15

MSQ : 0

NAT : 10

SOLUTIONS

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

Q7.25.1 MCQ | 2025 | 1M | Ans: D | ● ● ○

Method 1

For the given unity feedback system, the Nyquist criterion is

$$N = P - Z$$

Since $G(s)$ is strictly stable,

$$P = 0$$

The Nyquist plot encircles the critical point $(-1, 0)$ once in the anti-clockwise direction.
Assuming Nyquist Contour in clockwise direction,

$$N = 1$$

$$Z = P - N = 0 - 1 = -1$$

This is an absurd answer. So the Nyquist contour is not in clockwise direction. Hence, the nyquist contour is in anticlockwise direction.

$$N = -1$$

$$Z = P - N = 0 + 1 = 1$$

Thus, the closed-loop characteristic equation has one right-half-plane pole.
Therefore, the closed-loop system is unstable.

Method 2

As the open-loop system is stable,

$$P = 0$$

The Nyquist plot encircles the critical point $(-1, 0)$ once.

$$N \neq 0$$

From the Nyquist criterion,

$$N = P - Z$$

Since $P = 0$ and $N \neq 0$,

$$Z \neq 0$$

Hence, the closed-loop system has right-half-plane pole(s) and is unstable.

$$\boxed{(D)\ \text{The system is unstable}}$$

Key Points

- In the Nyquist criterion,

$$N = P - Z$$

  N is the number of encirclements of the $(0, 0)$ point by the $1 + G(s)H(s)$ plot in the $1 + GH$ plane (OR) N is the number of encirclements of the $(-1, j0)$ point by the $G(s)H(s)$ plot in the GH plane.

  P is the total number of poles of $1 + G(s)H(s)$ lying in the right-half of the s-plane, which are also the poles of $G(s)H(s)$.

  Z is the total number of zeros of $1 + G(s)H(s)$ lying in the right-half of the s-plane, which are also the poles of the closed-loop transfer function.

- For the closed-loop system to be stable,

$$Z = 0.$$

Q7.23.1 MCQ | 2023 | 1M | Ans: C | ● ○ ○

The infinite semicircular arc of the Nyquist contour corresponds to

$$s = Re^{j\theta}, \qquad R \to \infty$$

Given,

$$G(s)H(s) = \frac{3s+5}{s-1}$$

Put $s = Re^{j\theta}$

$$G(Re^{j\theta})H(Re^{j\theta}) = \frac{3(Re^{j\theta})+5}{(Re^{j\theta})-1}$$

As, $R \to \infty$, the highest-order terms dominate.

$$G(Re^{j\theta})H(Re^{j\theta}) \approx \frac{3(Re^{j\theta})}{(Re^{j\theta})}$$

$$G(Re^{j\theta})H(Re^{j\theta}) \approx 3$$

Hence, the infinite semicircular arc in the s-plane is mapped to the point $G(s)H(s) = 3$

$$\boxed{(C)\ \ G(s)H(s) = 3}$$

Q7.22.1 MCQ | 2022 | 2M | Ans: B | ● ● ○

We need the encirclements of $(-1, j0)$ by the Nyquist plot of $G(s)$.
This is equivalent to finding the encirclements of the origin by the plot of $1 + G(s) = 0$
According to the Nyquist Stability Criterion,

$$N = P - Z$$

The poles of $1 + G(s)$ are the same as the poles of $G(s)$.

$$G(s) = \frac{1}{s(s+1)}$$

Since the Nyquist contour encloses the entire right-half plane and a small neighbourhood around the pole at the origin, the contour encloses one pole of $G(s)$.

$$P = 1$$

For Z,

$$1 + G(s) = 0$$

$$1 + \frac{1}{s(s+1)} = 0$$

$$s^2 + s + 1 = 0$$

$$s = \frac{-1 \pm j\sqrt{3}}{2}$$

Both roots lie in the left-half plane.

$$Z = 0$$

Therefore,

$$N = P - Z = 1 - 0 = 1$$

$$\boxed{(B)\ \ 1}$$

Q7.20.1 NAT 2020 1M Ans: 2 to 2 ● ● ○

The given transfer function is

$$G(s) = \frac{K}{(s+a)(s-b)(s+c)}$$

Since $a, b, c > 0$, $G(s)$ has one right-half-plane pole at

$$s = b$$

Therefore,

$$P = 1$$

The Nyquist plot of $1 + G(s)$ encircles the origin once in the clockwise direction.
According to the Nyquist Stability Criterion,

$$N = P - Z$$

Taking clockwise encirclement as negative,

$$N = -1$$

$$-1 = 1 - Z$$

$$Z = 2$$

Here, Z is the number of right-half-plane zeros of $1 + G(s)$, which are also the right-half-plane poles of the closed loop transfer function, i.e.,

$$\frac{G(s)}{1 + G(s)}$$

Hence, the number of poles of closed loop transfer function are lying in the open right-half plane are 2

$$\boxed{2}$$

Q7.19.1 MCQ 2019 1M Ans: A ● ○ ○

The Nyquist plot crosses the negative real axis when its phase is $-180°$.
Given,

$$G(s) = \frac{\pi e^{-0.25s}}{s}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{\pi e^{-j0.25\omega}}{j\omega}$$

Hence,

$$\angle G(j\omega) = -0.25\omega - \frac{\pi}{2}$$

For crossing the negative real axis,

$$-0.25\omega - \frac{\pi}{2} = -\pi$$

$$0.25\omega = \frac{\pi}{2}$$

$$\omega = 2\pi$$

At $\omega = 2\pi$,

$$|G(j\omega)| = \frac{\pi}{\omega} = \frac{\pi}{2\pi} = \frac{1}{2}$$

Therefore, the Nyquist plot passes through $(-0.5, j0)$

$$\boxed{(A)\ \ (-0.5, j0)}$$

Q7.16.1 MCQ 2016 2M Ans: A ● ● ○

For the given loop transfer function,

$$G(s)H(s) = \frac{s+3}{s^2(s-3)}$$

The poles of $G(s)H(s)$ are at

$$s = 0,\ 0,\ 3$$

Hence,

$$P = 1$$

For Z,

$$1 + G(s)H(s) = 0$$

$$1 + \frac{s+3}{s^2(s-3)} = 0$$

$$s^3 - 3s^2 + s + 3 = 0$$

Using Routh array,

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & -3 & 3 \\ s^1 & 2 & 0 \\ s^0 & 3 & \end{array}$$

There are two sign changes in the first column.

$$Z = 2$$

Using the Nyquist Stability Criterion,

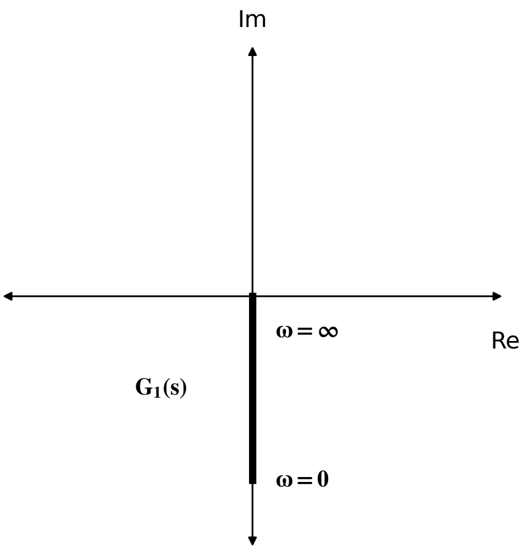
$$N = P - Z = 1 - 2 = -1$$

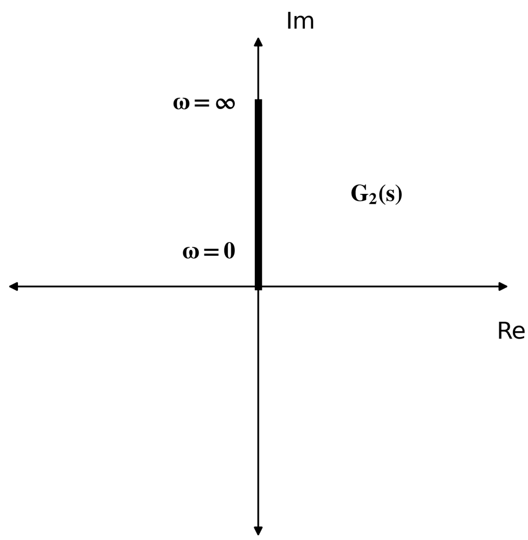
Therefore, the Nyquist plot encircle the point $(-1, j0)$ once in clockwise direction.

$$\boxed{\text{(A) once in clockwise direction}}$$

Q7.15.1 MCQ 2015 1M Ans: B ● ○ ○

From the given Nyquist plots,





Since the locus lies entirely on the negative imaginary axis,

$$G_1(j\omega) = K_1 e^{-j90^\circ} = -jK_1, \qquad K_1 > 0$$

Since the locus lies entirely on the positive imaginary axis,

$$G_2(j\omega) = K_2 e^{j90^\circ} = jK_2, \qquad K_2 > 0$$

Therefore,

$$G_1(j\omega)G_2(j\omega) = (-jK_1)(jK_2) = K_1K_2$$

Since $K_1 > 0$ and $K_2 > 0$,

$$G_1(j\omega)G_2(j\omega) > 0$$

Hence, the Nyquist plot of the product is a constant with phase 0°.
Therefore, the correct Nyquist plot is option (B).

$$\boxed{(B)}$$

Key Points

- Any point on the positive real axis can be represented as K, where $K > 0$.
- Any point on the negative real axis can be represented as $-K$, where $K > 0$.
- Any point on the positive imaginary axis can be represented as jK, where $K > 0$.
- Any point on the negative imaginary axis can be represented as $-jK$, where $K > 0$.
- Here, K denotes the magnitude (distance from the origin) and is not necessarily a constant gain.

Q7.09.1 MCQ 2009 1M Ans: D ● ● ○

From the polar plot, the locus crosses the negative real axis at $(-1.42, j0)$.
Now if we draw the complete nyquist plot, it would be:

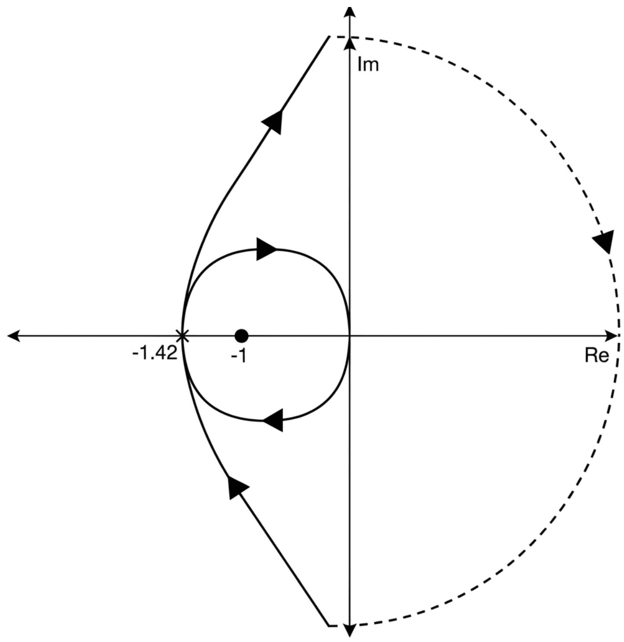


Fig. Solution Q7.09.1

The plot encircles the critical point $(-1, j0)$ once in the clockwise direction. Assuming the Nyquist Contour is in clock wise direction, hence, $N = -2$

It is given that the open-loop system is stable. Hence, $P = 0$

By Nyquist criterion,

$$N = P - Z$$

$$-2 = 0 - Z$$

$$Z = 2$$

Therefore, the closed-loop system has two poles in the right half plane and is unstable.

$$\boxed{\text{(D) Unstable with two poles on the RH s-plane}}$$

Key Points

- Nyquist criterion:

$$N = P - Z$$

  - N = encirclements of $(-1, j0)$ by $G(s)$
  - P = number of right half plane poles of $G(s)$
  - Z = number of right half plane zeros of $1 + G(s)$
- If $Z > 0$, the closed-loop system is unstable.

Q7.07.1 MCQ 2007 2M Ans: B ● ● ○

Given,

$$G(s) = \frac{1}{s(s+1)(s+2)}$$

Put $s = j\omega$,

$$G(j\omega) = \frac{1}{j\omega(j\omega+1)(j\omega+2)} = -\frac{3}{(1+\omega^2)(4+\omega^2)} + j\frac{\omega^2 - 2}{\omega(1+\omega^2)(4+\omega^2)}$$

$$\text{Let, } G(j\omega) = x + jy$$

Comparing real and imaginary parts,

$$x = -\frac{3}{(1+\omega^2)(4+\omega^2)} \qquad y = \frac{\omega^2 - 2}{\omega(1+\omega^2)(4+\omega^2)}$$

As $\omega \to 0^+$,

$$x \to -\frac{3}{4}, \qquad y \to -\infty$$

Therefore, the tangent at $\omega \to 0^+$ is vertical and the Nyquist plot is asymptotic to $x = -\frac{3}{4}$

$$\boxed{(B)\ x = -\frac{3}{4}}$$

Q7.06.1 MCQ | 2006 | 2M | Ans: A | ● ○ ○

For a stable open-loop system, the Nyquist criterion states that the closed-loop system is stable if the Nyquist plot does not encircles (encloses when polar plot is given) the critical point $(-1, j0)$.
From the given plots,

Plot	Encirclement of $(-1, j0)$	Closed-loop Stability
1	No	Stable
2	Yes	Unstable
3	Yes	Unstable
4	Yes	Unstable

Therefore, only plot (1) represents a stable closed-loop system.

$$\boxed{(A)\ (1) \text{ only}}$$

Key Points

- For an open-loop stable system ($P = 0$), closed-loop stability requires $N = 0$, i.e., no encirclement of the critical point.
- The direction of traversal is irrelevant here since only the existence of encirclement is required.

Q7.05.1 MCQ | 2005 | 2M | Ans: B | ● ● ○

Given,

$$G(s)H(s) = \frac{\pi e^{-0.25s}}{s}$$

Put $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{\pi e^{-j0.25\omega}}{j\omega} = \frac{\pi}{\omega} e^{-j\left(90^\circ + \frac{\omega}{4}\right)}$$

Therefore,

$$|G(j\omega)H(j\omega)| = \frac{\pi}{\omega}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ - \frac{\omega}{4}$$

The Nyquist plot crosses the negative real axis when

$$\angle G(j\omega)H(j\omega) = -180^\circ$$

$$-90^\circ - \frac{\omega}{4} = -180^\circ$$
$$\frac{\omega}{4} = \frac{\pi}{2}$$
$$\omega = 2\pi$$

At $\omega = 2\pi$,

$$|G(j\omega)H(j\omega)| = \frac{\pi}{2\pi} = \frac{1}{2} \qquad \text{and} \qquad \angle G(j\omega)H(j\omega) = -90^\circ - \frac{2\pi}{4} = -\frac{\pi}{2} - \frac{\pi}{2} = -\pi$$

Hence,

$$G(j\omega)H(j\omega) = \frac{1}{2}\angle -\pi = -\frac{1}{2}$$

Therefore, the Nyquist plot passes through $\left(-\frac{1}{2}, j0\right)$

$$\boxed{(B)\ \left(-\frac{1}{2}, j0\right)}$$

Q7.04.1 MCQ 2004 1M Ans: A ● ○ ○

Phase margin is defined as

$$\text{PM} = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc})$$

The Nyquist plot passes through the point $(-1, j0)$. Therefore,

$$|G(j\omega_{gc})H(j\omega_{gc})| = 1 \qquad \text{and} \qquad \angle G(j\omega_{gc})H(j\omega_{gc}) = -180^\circ$$

Hence,

$$\text{PM} = 180^\circ + (-180^\circ) = 0^\circ$$

$$\boxed{(A)\ 0^\circ}$$

Key Points

- If the Nyquist plot passes through $(-1, j0)$, then the phase at the gain crossover frequency is -180°. Hence, the phase margin is zero and the closed-loop system is on the verge of instability.

Q7.95.1 NAT 1995 Ans: * ● ● ●

The open-loop transfer function is

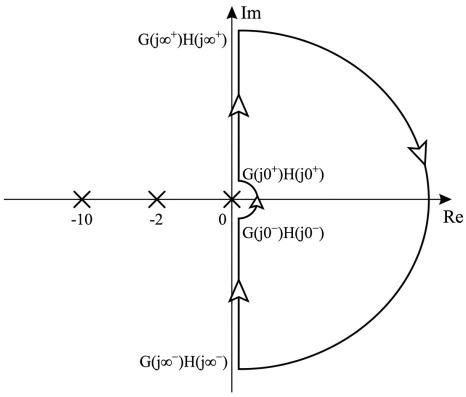
$$G(s)H(s) = \frac{10K}{s(1+0.1s)(1+0.5s)}$$

Nyquist contour poles are at:

$$s = 0, \qquad s = -2, \qquad s = -10$$

Hence, the Nyquist contour contains an indentation around the pole at the origin.

Segment	Mapping of $G(j\omega)H(j\omega)$
Segment–I	$G(j0^+)H(j0^+) \to G(j\infty^+)H(j\infty^+)$
Segment–II	$G(j\infty^+)H(j\infty^+) \to G(j\infty^-)H(j\infty^-)$
Segment–III	$G(j\infty^-)H(j\infty^-) \to G(j0^-)H(j0^-)$
Segment–IV	$G(j0^-)H(j0^-) \to G(j0^+)H(j0^+)$



Segment–I:
Put $s = j\omega$ in the given transfer function,

$$G(j\omega)H(j\omega) = \frac{10K}{j\omega(1 + j0.1\omega)(1 + j0.5\omega)} = \frac{10K}{\omega\sqrt{1 + 0.01\omega^2}\sqrt{1 + 0.25\omega^2}}\angle\left(-90^\circ - \tan^{-1}(0.1\omega) - \tan^{-1}(0.5\omega)\right)$$

$$G(j0^+)H(j0^+) = \infty\angle(-90^\circ)$$

$$G(j\infty^+)H(j\infty^+) = 0\angle(-270^\circ)$$

Again,

$$G(j\omega)H(j\omega) = \frac{10K}{j\omega(1 + j0.6\omega - 0.05\omega^2)} = \frac{10K}{-0.6\omega^2 + j\omega(1 - 0.05\omega^2)}$$

Multiplying numerator and denominator by the conjugate,

$$G(j\omega)H(j\omega) = \frac{10K\left[-0.6\omega^2 - j\omega(1 - 0.05\omega^2)\right]}{0.36\omega^4 + \omega^2(1 - 0.05\omega^2)^2} = \frac{-6K}{0.36\omega^2 + (1 - 0.05\omega^2)^2} - j\frac{10K(1 - 0.05\omega^2)}{\omega\left[0.36\omega^2 + (1 - 0.05\omega^2)^2\right]}$$

Real part is negative for all $\omega > 0$.
Imaginary part changes sign when

$$1 - 0.05\omega^2 = 0$$

$$\omega = \sqrt{20}$$

Therefore, the Nyquist plot crosses the real axis at $\omega = \sqrt{20}$
Substituting $\omega = \sqrt{20}$ in OLTF,

$$G(j\sqrt{20})H(j\sqrt{20}) = -\frac{5K}{6}$$

Therefore, the Nyquist plot cuts the real axis at $\left(-\frac{5K}{6}, 0\right)$

For $0 < \omega < \sqrt{20}$

$$1 - 0.05\omega^2 > 0$$

Hence imaginary part is negative.
Therefore, the Nyquist plot lies in Quadrant–III.
For $\omega > \sqrt{20}$

$$1 - 0.05\omega^2 < 0$$

Hence imaginary part is positive.
Therefore, the Nyquist plot lies in Quadrant–II.
Segment–II:
Put

$$s = Re^{j\theta}, \qquad R \to \infty, \qquad \theta : 90^\circ \to -90^\circ \ \text{(CW)}$$

$$G(Re^{j\theta})H(Re^{j\theta}) = \lim_{R\to\infty} \frac{10K}{Re^{j\theta}(1 + 0.1Re^{j\theta})(1 + 0.5Re^{j\theta})}$$

$$\approx \lim_{R\to\infty} \frac{200K}{R^3 e^{j3\theta}} = 0e^{-j3\theta}$$

Hence the mapping is from $-270^\circ \to 270^\circ$ with approximately zero radius in anticlockwise direction.
Segment–III:
Segment–III is the mirror image of Segment–I.
Segment–IV:
Put

$$s = re^{j\theta}, \qquad r \to 0, \qquad \theta : -90^\circ \to 90^\circ \ \text{(ACW)}$$

$$G(re^{j\theta})H(re^{j\theta}) = \lim_{r\to 0} \frac{10K}{re^{j\theta}(1+0.1re^{j\theta})(1+0.5re^{j\theta})}$$

$$\approx \lim_{r\to 0} \frac{10K}{re^{j\theta}} = \infty e^{-j\theta}$$

Hence the mapping is from 90° → −90° with infinite radius in clockwise direction.
Therefore, the complete Nyquist plot is

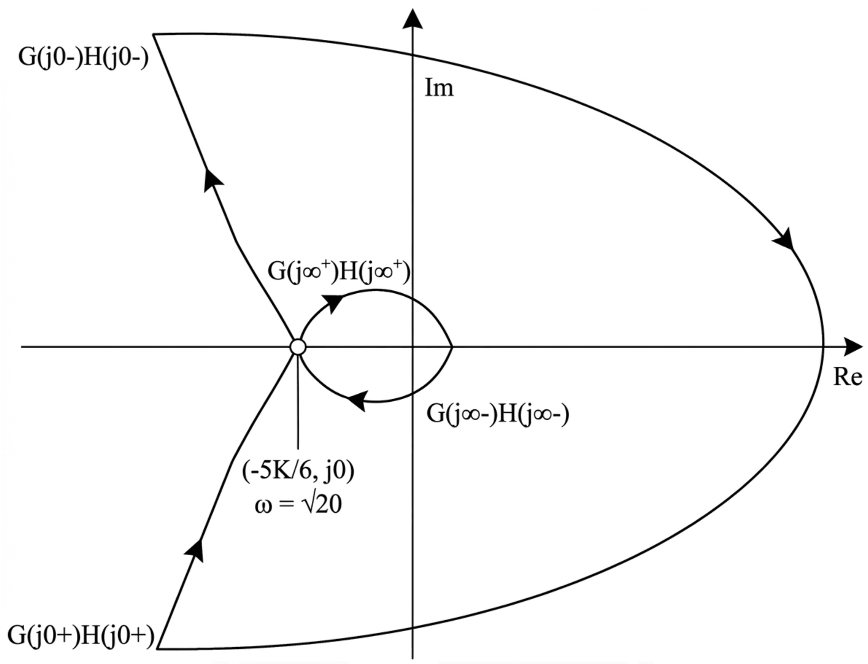


Fig. Solution Q7.95.1

Q7.95.2 NAT 1995 Ans: *

From Q7.95.1, the Nyquist plot crosses the negative real axis when $\omega_{pc} = \sqrt{20}$ rad/s.
At $\omega = \sqrt{20}$ rad/s,

$$G(j\sqrt{20})H(j\sqrt{20}) = -\frac{5K}{6}$$

Hence, the phase crossover point is $\left(-\frac{5K}{6},\, j0\right)$

Gain margin is defined as

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{5K/6} = \frac{6}{5K}$$

Therefore,

$$\boxed{\omega_{pc} = \sqrt{20}\text{ rad/s}} \quad \text{and} \quad \boxed{GM = \frac{6}{5K}}$$

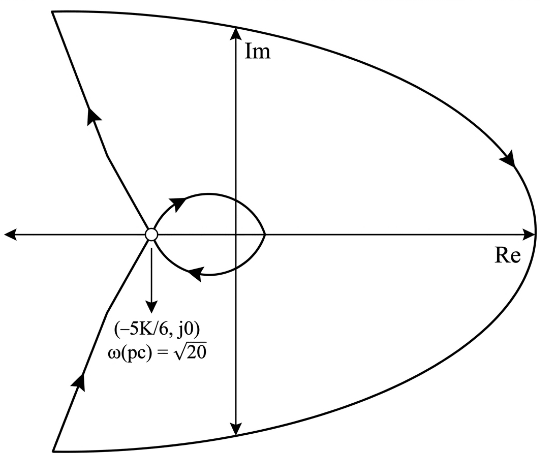


Fig. Solution Q7.95.2

Q7.95.3 | NAT | 1995 | Ans: 0° | ● ○ ○

For $K = 1.2$,

$$G(j\omega)H(j\omega) = -\frac{5(1.2)}{6} = -1$$

Since the phase crossover point lies exactly on the unit circle, hence, $\omega_{gc} = \sqrt{20}$ rad/sec.

The phase of the open-loop transfer function at gain crossover frequency is,

$$\angle G(j\sqrt{20})H(j\sqrt{20}) = -180^\circ$$

Therefore,

$$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc}) = 180^\circ - 180^\circ = 0^\circ$$

$$\boxed{PM = 0^\circ}$$

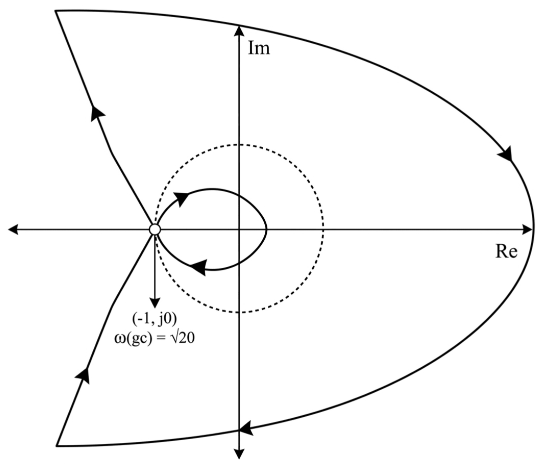


Fig. Solution Q7.95.3

Q7.92.1 | MCQ | 1992 | 1M | Ans: B | ● ○ ○

The open-loop transfer function is

$$G(s) = \frac{1}{(s-1)(s+2)(s+3)}$$

The open-loop transfer function has one right-half-plane pole at $s = 1$. Therefore,

$$P = 1$$

Since the system has unity feedback, the characteristic equation will be,

$$1 + G(s) = 0$$

For closed-loop stability,

$$Z = 0$$

Using the Nyquist stability criterion (assuming nyquist contour is clockwise),

$$N = P - Z = 1 - 0 = 1$$

Thus, the Nyquist plot must make one anti-clockwise encirclement of the origin.

$$\boxed{(B)\ \text{Once}}$$

Q7.91.1 | MCQ | 1991 | 2M | Ans: B | ● ● ○

From the given Nyquist plot,

- The plot starts at -180°
- The plot ends at -360°
- The plot does not intersect the real axis

Therefore, the open-loop transfer function is a Type '2' and Order '4' system. And it has no zeros.
Hence, only option (B) satisfies the above conditions.

$$\boxed{(B)\ \frac{K}{s^2(s+2)(s+5)}}$$

DEBUG INFO

Chapter VII

Nyquist Plot

Total Questions : 17

MCQ : 13

MSQ : 0

NAT : 4

SOLUTIONS

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd Order Systems

Q8.26.1 MCQ 2026 1M Ans: A ● ○ ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope
$\omega < \omega_0$	-20 dB/dec
$\omega > \omega_0$	0 dB/dec

A slope of -20 dB/dec indicates one pole at the origin.
A slope increase of $+20$ dB/dec at $\omega = \omega_0$ indicates one zero.
Therefore, the transfer function can be written as

$$G(s) = K\frac{\left(1 + \frac{s}{\omega_0}\right)}{s}$$

Put $s = j\omega$,

$$G(j\omega) = K\frac{(1 + \frac{j\omega}{\omega_0})}{(j\omega)}$$

For $\omega >> \omega_0$, both zero and pole are activated.

$$G(j\omega) \approx K\frac{(\frac{j\omega}{\omega_0})}{(j\omega)}$$

Let, $\omega = \omega_0$,

$$G(j\omega_0) \approx K\frac{(\frac{j\omega_0}{\omega_0})}{(j\omega_0)}$$

$$G(j\omega_0) \approx K\frac{1}{\omega_0}$$

Consider magnitude in dB,

$$20\log G(j\omega) \approx 20\log\frac{K}{\omega_0}$$

From the plot, $\omega > \omega_0$ the high-frequency asymptote is 0 dB.

$$20\log\left(\frac{K}{\omega_0}\right) = 0$$

$$\frac{K}{\omega_0} = 1$$

$$K = \omega_0$$

Hence,

$$G(s) = \omega_0\frac{\left(1 + \frac{s}{\omega_0}\right)}{s} = \frac{1 + \frac{s}{\omega_0}}{\frac{s}{\omega_0}}$$

$$\boxed{\text{(A) } \frac{1 + \frac{s}{\omega_0}}{\frac{s}{\omega_0}}}$$

Q8.24.2 NAT 2024 2M Ans: 0.09 to 0.11 ● ● ○ CLICK FOR VIDEO SOLUTION

The asymptotic phase plot remains constant at -90° till at least $\omega = 10$ rad/s.
Therefore, it has one pole at the origin and hence the open-loop transfer function behaves as an integrator:

$$G(s) = \frac{K}{s}$$

The gain crossover frequency is given as

$$\omega_{gc} = 10 \text{ rad/s}$$

At the gain crossover frequency,

$$|G(j\omega_{gc})| = 1$$

Hence,

$$\left|\frac{K}{j10}\right| = 1$$

$$\frac{K}{10} = 1$$

$$K = 10$$

For a unity feedback system, the velocity error constant is

$$K_v = \lim_{s\to 0} sG(s)$$

$$K_v = \lim_{s\to 0} s\left(\frac{10}{s}\right)$$

$$K_v = 10$$

The steady-state error for a unit ramp input is

$$e_{ss} = \frac{1}{K_v} = \frac{1}{10} = 0.1$$

$$\boxed{0.10}$$

Q8.23.1 MCQ 2023 2M Ans: D ● ● ○

By looking at the available data, it is evident that the transfer function will be in the form:

$$G(s) = K\,e^{-Ts}$$

From the magnitude plot, the gain is constant at 8 dB.

$$20\log_{10} K = 8$$

$$K = 10^{8/20} \approx 2.51$$

Therefore,

$$G(s) = 2.51\,e^{-Ts}$$

The phase plot is a straight line passing through $(\omega, \phi) = (1, -60^\circ)$. Hence for a pure time delay,

$$\phi(\omega) = -\omega T$$

Converting 60° into radians,

$$60^\circ = \frac{\pi}{3} = 1.047 \text{ rad}$$

At $\omega = 1$ rad/s,

$$-\omega T = -1.047$$
$$-(1)T = -1.047$$
$$T = 1.047 \text{ s}$$

Hence,

$$G(s) = 2.51e^{-1.047s}$$

$$\boxed{(D)\ 2.51e^{-1.047s}}$$

Q8.22.1 MCQ 2022 1M Ans: A ● ○ ○

The magnitude plot is constant for all frequencies. Hence, the system is a first-order all-pass system.
For a first-order all-pass system, the possible transfer functions are

$$T_1(s) = \frac{s-1}{s+1} \quad \text{and} \quad T_2(s) = \frac{1-s}{1+s}$$

The phase plot must be used to identify the correct transfer function.

For

$$T_1(s) = \frac{s-1}{s+1}$$

putting $s = j\omega$,

$$T_1(j\omega) = \frac{j\omega - 1}{j\omega + 1}$$
$$\angle T_1(j\omega) = 180^\circ - \tan^{-1}\omega - \tan^{-1}\omega$$
$$\angle T_1(j\omega) = 180^\circ - 2\tan^{-1}\omega$$

As $\omega \to \infty$,

$$\angle T_1(j\omega) = 180^\circ - 2 \times 90^\circ$$
$$\angle T_1(j\omega) = 0^\circ$$

This does not agrees with the given phase plot.

For

$$T_2(s) = \frac{1-s}{1+s}$$

putting $s = j\omega$,

$$T_2(j\omega) = \frac{1-j\omega}{1+j\omega}$$
$$\angle T_2(j\omega) = -\tan^{-1}\omega - \tan^{-1}\omega$$
$$\angle T_2(j\omega) = -2\tan^{-1}\omega$$

As $\omega \to \infty$,

$$\angle T_2(j\omega) = -2 \times 90^\circ$$
$$\angle T_2(j\omega) = -180^\circ$$

This agrees with the given phase plot.

Therefore,

$$T(s) = \frac{1-s}{1+s}$$

Hence, the system has one LHP pole and one RHP zero at the same frequency.

$$\boxed{(A)\ \text{One LHP pole and one RHP zero at the same frequency}}$$

Q8.21.1 NAT 2021 1M Ans: 0.09 to 0.11 ● ● ○

Transforming the given circuit to Laplace Domain,

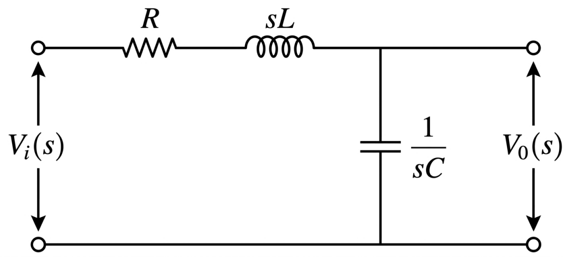


Fig. Solution Q8.21.1

The output is taken across the capacitor.
Therefore,

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{1}{LCs^2 + RCs + 1}$$

Comparing with the standard second-order low-pass transfer function

$$H(s) = \frac{1}{\frac{s^2}{\omega_n^2} + \frac{2\zeta s}{\omega_n} + 1}$$

we obtain

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(1 \times 10^{-3})(250 \times 10^{-6})}}$$

$$\omega_n = 2000 \text{ rad/s}$$

which matches the frequency marked in the Bode plot.
The resonant peak is 26 dB.

$$M_r = 10^{26/20}$$

$$M_r \approx 20$$

For a lightly damped second-order system,

$$M_r \approx \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$20 = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$\zeta = 0.025$$

From the standard form,

$$2\zeta\omega_n = \frac{R}{L}$$

$$R = 2\zeta\omega_n L = 2(0.025)(2000)(1 \times 10^{-3}) = 0.1\,\Omega$$

$$\boxed{0.10}$$

Q8.19.1 MCQ 2019 2M Ans: B

From the asymptotic Bode magnitude plot,

Frequency Range	Slope	Inference
$\omega < 1$	-20 dB/dec	One pole at the origin
$\omega = 1$	$-20 \to -40$	Pole at $\omega = 1$
$\omega = 20$	$-40 \to -60$	Pole at $\omega = 20$

Therefore,

$$G(s) = \frac{K}{s\left(1+\frac{s}{1}\right)\left(1+\frac{s}{20}\right)}$$

Hence, the transfer function has

$$\text{3 poles and 0 zeros}$$

Therefore, Statement I is false.
Since $G(s)$ is a minimum-phase transfer function, every pole contributes a phase lag of $-90°$ at very high frequencies.

$$\angle G(j\omega) = (-90°) + (-90°) + (-90°) = -270° = -\frac{3\pi}{2}$$

Therefore, Statement II is true.

$$\boxed{(B)\ \text{Statement I is false and Statement II is true}}$$

Key Points

- A pole decreases the Bode magnitude slope by 20 dB/decade.
- A zero increases the Bode magnitude slope by 20 dB/decade.
- For a minimum-phase system, each pole contributes $-90°$ and each zero contributes $+90°$ at very high frequencies.

Q8.16.1 MCQ 2016 2M Ans: A

From the given asymptotic Bode magnitude plot,

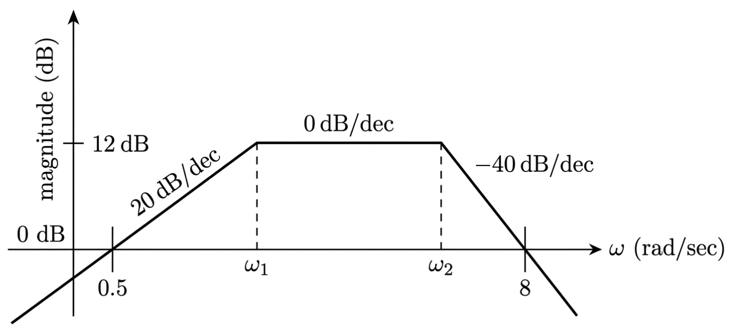


Fig. Solution Q8.16.1

The slope changes can be summarized as:

Frequency Range	Slope	Inference
$\omega < \omega_1$	$+20$ dB/dec	Zero at the origin
$\omega_1 \to \omega_2$	0 dB/dec	Pole at $\omega = \omega_1$
$\omega > \omega_2$	-40 dB/dec	Two poles at $\omega = \omega_2$

From the plot,
For ω_1:

$$\frac{12-0}{\log_{10}\omega_1 - \log_{10}0.5} = 20\,\text{dB/dec}$$
$$\omega_1 \approx 2 \text{ rad/sec}$$

For ω_2:

$$\frac{12-0}{\log_{10}\omega_2 - \log_{10}8} = -40\,\text{dB/dec}$$
$$\omega_2 \approx 4 \text{ rad/sec}$$

Therefore,

$$G(s) = K\frac{s}{\left(1+\frac{s}{2}\right)\left(1+\frac{s}{4}\right)^2} = K\frac{s}{(1+0.5s)(1+0.25s)^2}$$

For $\omega < 2$ rad/s,

$$G(s) \approx Ks$$

Putting $s = j\omega$,

$$G(j\omega) \approx Kj\omega$$

Magnitude,

$$|G(j\omega)| = K\omega$$

In dB,

$$20\log|G(j\omega)| = 20\log(K\omega)$$

From the Bode plot, the low-frequency asymptote passes through $(\omega, M) = (0.5,\ 0\,\text{dB})$. Hence,

$$20\log|G(j0.5)| = 0$$
$$\log(0.5K) = 0$$
$$0.5K = 1$$
$$K = 2$$

Therefore,

$$G(s) = \frac{2s}{(1+0.5s)(1+0.25s)^2}$$

$$\boxed{(A)\quad \frac{2s}{(1+0.5s)(1+0.25s)^2}}$$

Q8.15.1 MCQ 2015 1M Ans: D ● ○ ○

From the given plot,

Frequency Range	Slope Calculation
$0.1 \to 10$	$\frac{20-20}{\log_{10}10 - \log_{10}0.1} = 0\,\text{dB/dec}$
$10 \to 1000$	$\frac{-20-20}{\log_{10}1000 - \log_{10}10} = -20\,\text{dB/dec}$
$1000 \to 100000$	$\frac{-20-(-20)}{\log_{10}100000 - \log_{10}1000} = 0\,\text{dB/dec}$

A slope decrease of 20 dB/dec at $\omega = 10$ rad/sec indicates one pole.
A slope increase of 20 dB/dec at $\omega = 1000$ rad/sec indicates one zero.
Therefore, transfer function can be written as,

$$G(s) = K\frac{\left(1 + \frac{s}{1000}\right)}{\left(1 + \frac{s}{10}\right)}$$

For $\omega < 10$ rad/sec, neither zero nor pole is activated.

$$\therefore \; G(j\omega) \approx K$$

At $\omega = 1$ rad/sec, magnitude from the plot is 20 dB.

$$20 \log |G(j1)| = 20$$

$$20 \log K = 20$$

$$K = 10$$

Hence,

$$G(s) = 10\frac{\left(1 + \frac{s}{1000}\right)}{\left(1 + \frac{s}{10}\right)} = \frac{s + 1000}{10(s + 10)}$$

$$\boxed{(D) \; \frac{s + 1000}{10(s + 10)}}$$

Q8.14.1 NAT | 2014 | 2M | Ans: 0.7 to 0.8 | ● ● ○

From the given Bode plot,
It is staring with a slope of −6 dB/octave. Hence, one pole at origin
At $\omega = 2$ rad/sec, the slope changes from −6 dB/octave to 0 dB/octave. Hence, there is one zero at $\omega_z = 2$ rad/sec
At $\omega = 4$ rad/sec, the slope changes from 0 dB/octave to +6 dB/octave. Hence, there is one zero at $\omega_z = 4$ rad/sec
At $\omega = 8$ rad/sec, the slope changes from +6 dB/octave to 0 dB/octave. Hence, there is one pole at $\omega_p = 8$ rad/sec
At $\omega = 24$ rad/sec, the slope changes from 0 dB/octave to −6 dB/octave. Hence, there is one pole at $\omega_p = 24$ rad/sec
At $\omega = 36$ rad/sec, the slope changes from −6 dB/octave to −12 dB/octave. Hence, there is one pole at $\omega_p = 36$ rad/sec
From the above information, the transfer function would be,

$$G(s) = \frac{K(s + 2)(s + 4)}{s\,(s + 8)\,(s + 24)\,(s + 36)}$$

Given Transfer function is,

$$G(s) = \frac{K(1 + 0.5s)(1 + as)}{s\left(1 + \frac{s}{8}\right)(1 + bs)\left(1 + \frac{s}{36}\right)} = \frac{144Ka}{b}\,\frac{(s + 2)\left(s + \frac{1}{a}\right)}{s(s + 8)\left(s + \frac{1}{b}\right)(s + 36)}$$

Comparing the above two transfer functions, we get

$$\frac{1}{a} = 4 \qquad \Rightarrow \qquad a = \frac{1}{4},$$

$$\frac{1}{b} = 24 \qquad \Rightarrow \qquad b = \frac{1}{24}.$$

A system with initial slope $-20n$ dB/dec or $-6n$ dB/oct cuts the frequency axis at $\omega = K^{1/n}$, where n is the number of poles at origin. Hence,

$$8 = K^{1/n}$$

Since, $n = 1$,

$$K = 8$$

Therefore,

$$\frac{a}{bK} = \frac{\frac{1}{4}}{\left(\frac{1}{24}\right)(8)} = \frac{3}{4} = 0.75$$

$$\boxed{0.75}$$

Q8.14.2 MCQ | 2014 | 2M | Ans: A | ● ● ○

The given circuit is an active low pass filter. Hence, the output is,

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{1 + sRC}$$

The cut off frequency is given by,

$$f_0 = \frac{1}{2\pi RC}$$

From the circuit, $R = 1k\ \Omega$ and $C = 1\mu\ F$

$$f_0 = \frac{1}{2\pi(1k)(1\mu)} = \frac{1000}{2\pi}$$

$$\omega = 2\pi f_0 = 10^3 \text{ rad/sec}$$

Thus, the magnitude plot remains at 0 dB for $\omega < 10^3$ rad/sec and decreases with a slope of -20 dB/dec for $\omega > 10^3$ rad/sec. Also, the phase angle is

$$\phi = -\tan^{-1}(\omega RC)$$

$$\phi = -\tan^{-1}\left(\frac{\omega}{10^3}\right)$$

This changes from 0 to $-\pi/2$ as ω increases.
Therefore, the correct Bode plots are shown in Option A.

$$\boxed{(A)}$$

Q8.14.3 MCQ | 2014 | 2M | Ans: A | ● ○ ○

The given transfer function is

$$G(s) = \frac{5(s+4)}{s(s+0.25)(s^2+4s+25)}.$$

Writing the transfer function in time-constant form,

$$G(s) = \frac{5(4)\left(1+\frac{s}{4}\right)}{s\,(0.25)\left(1+\frac{s}{0.25}\right)25\left[\left(\frac{s}{5}\right)^2+\frac{4}{5}\left(\frac{s}{5}\right)+1\right]}$$

$$G(s) = 3.2\frac{\left(1+\frac{s}{4}\right)}{s\left(1+\frac{s}{0.25}\right)\left[\left(\frac{s}{5}\right)^2+\frac{4}{5}\left(\frac{s}{5}\right)+1\right]}$$

Therefore, the constant gain term is

$$K = 3.2.$$

The corner frequencies are

$$\omega = 0.25\,\text{rad/sec}, \qquad \omega = 4\,\text{rad/sec}, \qquad \omega_n = \sqrt{25} = 5\,\text{rad/sec}.$$

Hence, the highest corner frequency is

$$\omega_h = 5\,\text{rad/sec}.$$

$$\boxed{(A)\ 3.2,\ 5.0}$$

Q8.13.1 MCQ | 2013 | 1M | Ans: B | ● ○ ○

From the given Bode magnitude plot,

$$\text{Slope} = \frac{-8-32}{\log_{10}10-\log_{10}1} = -40\,\text{dB/dec}$$

A slope of -40 dB/dec indicates two poles at the origin. Also, the phase is negative for all ω. Hence, the system contains only poles. Therefore, the transfer function can be written as,

$$G(s) = \frac{K}{s^2}$$

For $\omega < 10$ rad/sec, only the two poles at the origin are effective.

$$G(s) \approx \frac{K}{s^2}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{K}{(j\omega)^2}$$
$$G(j\omega) = \frac{-K}{\omega^2}$$

Therefore, magnitude is,

$$|G(j\omega)| = \frac{K}{\omega^2}$$

In dB,

$$20\log(|G(j\omega)|) = 20\log\left(\frac{K}{\omega^2}\right)$$

At $\omega = 1$ rad/sec, the magnitude from the plot is 32 dB. Hence,

$$32 = 20\log\left(\frac{K}{1^2}\right)$$
$$20\log K = 32$$
$$K = 10^{1.6} = 39.8$$
$$\therefore G(s) = \frac{39.8}{s^2}$$

$$\boxed{(B)\ \frac{39.8}{s^2}}$$

Q8.09.1 MCQ 2009 1M Ans: B ● ● ○

Frequency Range	Slope	Inference
$0.1 \rightarrow 2$	-40 dB/dec	2 Poles at origin
$2 \rightarrow 5$	-60 dB/dec	1 Pole at $\omega = 2$ rad/sec
$5 \rightarrow 25$	-40 dB/dec	1 Zero at $\omega = 5$ rad/sec
$\omega > 25$	-60 dB/dec	1 Pole at $\omega = 25$ rad/sec

Therefore, the transfer function can be written as

$$G(s) = K\frac{\left(1+\frac{s}{5}\right)}{s^2\left(1+\frac{s}{2}\right)\left(1+\frac{s}{25}\right)}$$

For $\omega < 2$ rad/sec, none of the finite poles and zeros are activated. Therefore,

$$G(j\omega) \approx \frac{K}{(j\omega)^2}.$$

At $\omega = 0.1$ rad/sec, the magnitude from the plot is 80 dB.

$$20\log\left|\frac{K}{(j0.1)^2}\right| = 80$$

$$20\log\left(\frac{K}{0.01}\right) = 80$$

$$K = 100.$$

Hence,

$$G(s) = 100\frac{\left(1+\frac{s}{5}\right)}{s^2\left(1+\frac{s}{2}\right)\left(1+\frac{s}{25}\right)} = \frac{1000(s+5)}{s^2(s+2)(s+25)}$$

$$\boxed{(B)\ \frac{1000(s+5)}{s^2(s+2)(s+25)}}$$

Q8.08.1 MCQ 2008 2M Ans: C ● ○ ○

From the given Bode plot, the initial slope is -40 dB/dec. Hence, there are two poles at the origin.
At $\omega = 0.1$ rad/sec, the slope changes from -40 dB/dec to -20 dB/dec. Hence, there is one zero at $\omega_z = 0.1$ rad/sec.
At higher frequencies, the slope changes from -20 dB/dec to 0 dB/dec. Hence, there is one more zero.
Therefore, the transfer function has two poles and two zeros.

$\boxed{(C)\ \text{Two poles and two zeros}}$

Q8.06.1 MCQ | 2006 | 2M | Ans: A | ● ● ○

The given transfer function is

$$H(j\omega) = \frac{10^4(1+j\omega)}{(10+j\omega)(100+j\omega)^2}.$$

Writing it in time-constant form,

$$H(j\omega) = \frac{10^4(1+j\omega)}{10\left(1+\frac{j\omega}{10}\right)100^2\left(1+\frac{j\omega}{100}\right)^2}$$

$$H(j\omega) = \frac{1+j\omega}{10\left(1+\frac{j\omega}{10}\right)\left(1+\frac{j\omega}{100}\right)^2}.$$

Therefore,

- One zero at $\omega = 1$ rad/sec,
- One pole at $\omega = 10$ rad/sec,
- Two poles at $\omega = 100$ rad/sec.

Also,

$$|H(0)| = 0.1$$

$$20\log|H(0)| = -20\,\text{dB}.$$

Hence, the asymptotic Bode magnitude plot starts at -20 dB.
At $\omega = 1$ rad/sec, the slope increases by 20 dB/dec.
At $\omega = 10$ rad/sec, the slope decreases by 20 dB/dec.
At $\omega = 100$ rad/sec, the slope decreases by 40 dB/dec.
Therefore, the slopes are

$$\omega < 1 \Rightarrow -20\,\text{dB/dec}$$

$$1 < \omega < 10 \Rightarrow +20\,\text{dB/dec}$$

$$10 < \omega < 100 \Rightarrow 0\,\text{dB/dec}$$

$$\omega > 100 \Rightarrow -40\,\text{dB/dec}.$$

Hence, the correct Bode magnitude plot is shown in Option A.

$$\boxed{(A)}$$

Key Points

- Since the horizontal axis is $\log(\omega)$, the markings represent logarithmic values of frequency.
- $\log(\omega) = 0 \Rightarrow \omega = 1$ rad/sec.
- $\log(\omega) = 1 \Rightarrow \omega = 10$ rad/sec.
- $\log(\omega) = 2 \Rightarrow \omega = 100$ rad/sec.
- $\log(\omega) = 3 \Rightarrow \omega = 1000$ rad/sec.

Mistakes to be avoided

Always check whether the frequency axis is plotted as ω or $\log(\omega)$. Directly interpreting $\log(\omega) = 1, 2, 3$ as $\omega = 1, 2, 3$ rad/sec leads to incorrect corner frequencies and an incorrect Bode plot.

Q8.03.1 MCQ | 2003 | 2M | Ans: A | ● ● ○

The given transfer function is

$$G(s) = \frac{K}{1 + \frac{s}{a}}.$$

$$G(j\omega) = \frac{K}{1 + j\frac{\omega}{a}}.$$

$$G(j0.5a) = \frac{K}{1 + j0.5}.$$

The actual phase angle is

$$\phi_{\text{actual}} = -\tan^{-1}(0.5) = -26.57°.$$

Since $0.1a < 0.5a < a < 10a$, the point $\omega = 0.5a$ lies on the straight-line portion of the asymptotic phase plot. Therefore,

$$\phi_{\text{approx}} = (-45°/\text{decade})\log_{10}\left(\frac{0.5a}{0.1a}\right) = -45°\log_{10}(5) = -45°(0.699) = -31.46°$$

Hence, the error in phase angle is

$$\left|\phi_{\text{approx}} - \phi_{\text{actual}}\right| = |-31.46 + 26.57| = 4.89°.$$

The actual magnitude is

$$\begin{aligned} 20\log|G(j0.5a)| &= 20\log K - 20\log\sqrt{1 + (0.5)^2} \\ &= 20\log K - 20\log\sqrt{1.25} \\ &= 20\log K - 0.97\,\text{dB}. \end{aligned}$$

Since $\omega = 0.5a < a$, the asymptotic magnitude is

$$20\log|G(j0.5a)|_{\text{approx}} = 20\log K.$$

Therefore, the error in dB gain is 0.97 dB.

$$\boxed{(A)\ 4.9°,\ 0.97\,\text{dB}}$$

Q8.01.1 MCQ | 2001 | 2M | Ans: D | ● ● ○

Frequency Range	Slope	Inference
$0.1 \to 2$	-40 dB/dec	2 Poles at origin
$2 \to 5$	-60 dB/dec	1 Pole at $\omega = 2$ rad/sec
$5 \to 25$	-40 dB/dec	1 Zero at $\omega = 5$ rad/sec
$\omega > 25$	-60 dB/dec	1 Pole at $\omega = 25$ rad/sec

Therefore, the transfer function can be written as

$$G(s) = K\frac{\left(1 + \frac{s}{5}\right)}{s^2\left(1 + \frac{s}{2}\right)\left(1 + \frac{s}{25}\right)}$$

For $\omega < 2$ rad/sec, none of the finite poles and zeros are activated. Therefore,

$$G(j\omega) \approx \frac{K}{(j\omega)^2}.$$

At $\omega = 0.1$ rad/sec, the magnitude from the plot is 54 dB.

$$20\log\left|\frac{K}{(j0.1)^2}\right| = 54$$

$$20\log\left(\frac{K}{0.01}\right) = 54$$

$$K = 5.01$$

Hence,

$$G(s) = 5.01\frac{\left(1+\frac{s}{5}\right)}{s^2\left(1+\frac{s}{2}\right)\left(1+\frac{s}{25}\right)} = \frac{50.1(s+5)}{s^2(s+2)(s+25)} \approx \frac{50(s+5)}{s^2(s+2)(s+25)}$$

$$\boxed{(B)\ \frac{50(s+5)}{s^2(s+2)(s+25)}}$$

Q8.99.1 MCQ 1999 2M Ans: D ● ○ ○

From the given Bode plot, the magnitude is 0 dB for $f < f_1$. Hence, there is no pole or zero at the origin.
At $f = f_1$, the slope increases by 6 dB/octave (20 dB/dec). Hence, there is one zero at $f = f_1$.
Therefore, the transfer function is of the form

$$A = K\left(1 + j\frac{f}{f_1}\right)$$

Since the low-frequency gain is 0 dB,

$$20\log K = 0$$

$$K = 1$$

Hence,

$$A = 1 + j\frac{f}{f_1}$$

$$\boxed{(D)\ 1 + j\frac{f}{f_1}}$$

Q8.91.1 MSQ 1991 2M Ans: B, D ● ● ○

From the given Bode plot, the initial slope is −20 dB/dec. Hence, there is one pole at the origin.
At $\omega = 0.05$ rad/sec, the slope changes from −20 dB/dec to −40 dB/dec. Hence, there is one pole at $\omega_p = 0.05$ rad/sec.
At $\omega = 0.1$ rad/sec, the slope changes from −40 dB/dec to −20 dB/dec. Hence, there is one zero at $\omega_z = 0.1$ rad/sec.
Therefore, the transfer function can be written as

$$G(s) = K\frac{\left(1+\frac{s}{0.1}\right)}{s\left(1+\frac{s}{0.05}\right)}$$

For $\omega < 0.05$ rad/sec, neither finite pole nor zero is activated. Therefore,

$$G(j\omega) \approx \frac{K}{j\omega}$$

At $\omega = 0.01$ rad/sec, magnitude from the plot is 60 dB.

$$20\log\left|\frac{K}{j0.01}\right| = 60$$

$$20\log(100K) = 60$$

$$K = 10$$

Hence,

$$G(s) = 10\frac{\left(1+\frac{s}{0.1}\right)}{s\left(1+\frac{s}{0.05}\right)} = \frac{5(s+0.1)}{s(s+0.05)} = \frac{10(1+10s)}{s(1+20s)}$$

Therefore, the correct options are

$$\boxed{(B)\ \frac{10(1+10s)}{s(1+20s)}}$$

$$\boxed{(D)\ \frac{5(s+0.1)}{s(s+0.05)}}$$

DEBUG INFO

Chapter VIII

Bode Plot

Total Questions : 19

MCQ : 15

MSQ : 1

NAT : 3

SOLUTIONS

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Q9.26.1 MCQ 2026 2M Ans: D ● ○ ○

The state equation is

$$\dot{x} = \begin{bmatrix} a & b \\ -a & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

The poles of the transfer function are the eigenvalues of the state matrix

$$A = \begin{bmatrix} a & b \\ -a & 0 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$

$$\begin{vmatrix} s - a & -b \\ a & s \end{vmatrix} = 0$$

$$s(s - a) - (-ab) = 0$$

$$s^2 - as + ab = 0$$

The poles are given as $-2 + j3$ and $-2 - j3$. Therefore,

$$(s + 2 - j3)(s + 2 + j3) = 0$$

$$(s + 2)^2 + 3^2 = 0$$

$$s^2 + 4s + 13 = 0$$

Comparing the above equation with $s^2 - as + ab = 0$ gives,

$$-a = 4$$

$$a = -4$$

and

$$ab = 13$$

$$(-4)b = 13$$

$$b = -\frac{13}{4}$$

$$b = -3.25$$

$$\boxed{(D) \;\; a = -4, \; b = -3.25}$$

Q9.26.2 MCQ 2026 2M Ans: C ● ● ○

Given

$$A = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

The new state vector is

$$\mathbf{z} = T\mathbf{x}$$

where

$$T = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

Therefore,

$$\mathbf{x} = T^{-1}\mathbf{z}$$

Since

$$|T| = (2)(0) - (1)(-1) = 1$$

$$T^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Put} \quad \mathbf{x} = T^{-1}\mathbf{z} \ \text{ in } \ \dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \ \text{ and } \ y = C\mathbf{x} + D\mathbf{u}$$

$$\frac{d}{dt}\left[T^{-1}\mathbf{z}\right] = A\left[T^{-1}\mathbf{z}\right] + B\mathbf{u}$$

$$T^{-1}\dot{\mathbf{z}} = AT^{-1}\mathbf{z} + B\mathbf{u} \qquad \text{and} \qquad y = CT^{-1}\mathbf{z} + D\mathbf{u}$$

$$\dot{\mathbf{z}} = TAT^{-1}\mathbf{z} + TB\mathbf{u}$$

Let,

$$A' = TAT^{-1}$$

$$B' = TB$$

$$C' = CT^{-1}$$

$$D' = D$$

First,

$$A' = TAT^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix}$$

Second,

$$B' = TB = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

Third,

$$C' = CT^{-1} = \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix}$$

Hence,

$$\dot{\mathbf{z}} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}\mathbf{u}$$

$$y = \begin{bmatrix} 2 & 3 \end{bmatrix}\mathbf{z}$$

$$\boxed{(C)\ \ \dot{\mathbf{z}}\begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}\mathbf{u},\ y = \begin{bmatrix} 2 & 3 \end{bmatrix}\mathbf{z}}$$

Q9.25.3 | NAT | 2025 | 2M | Ans: 3 to 3 | ● ○ ○

For a state-space system,

$$\dot{x}(t) = Ax(t) + Br(t)$$

the poles are the eigenvalues of the state matrix A.
Given,

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = 0$$

$$s(s+3) - (-1)(2) = 0$$

$$s^2 + 3s + 2 = 0$$

$$(s+1)(s+2) = 0$$

$$s = -1, -2$$

The sum of the magnitudes of the poles is

$$|-1| + |-2| = 1 + 2 = 3$$

$$\boxed{3}$$

Q9.23.1 NAT 2023 2M Ans: 2 to 2 ● ○ ○

The transfer function of the state-space system is

$$G(s) = C(sI - A)^{-1}B + D$$

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & -2 \end{bmatrix}, \quad D = 1$$

First, compute

$$sI - A = \begin{bmatrix} s & -1 \\ 1 & s+2 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 2s + 1} \begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}$$

$$(sI - A)^{-1}B = \frac{1}{(s+1)^2} \begin{bmatrix} 1 \\ s \end{bmatrix}$$

$$C(sI - A)^{-1}B = \frac{1}{(s+1)^2} \begin{bmatrix} 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} = \frac{3 - 2s}{(s+1)^2}$$

Therefore,

$$G(s) = \frac{3 - 2s}{(s+1)^2} + 1 = \frac{s^2 + 4}{(s+1)^2}$$

For an input $\sin(\omega t)$, the steady-state output is zero when

$$|G(j\omega)| = 0$$

Hence,

$$G(j\omega) = 0$$

$$(j\omega)^2 + 4 = 0$$

$$-\omega^2 + 4 = 0$$

$$\omega^2 = 4$$

$$\omega = 2 \text{ rad/s}$$

$$\boxed{2}$$

Q9.21.1 NAT 2021 2M Ans: 1 to 1

The given system is

$$\dot{x} = -x + u$$

$$y = x$$

The control law is

$$u = -Kx$$

Substituting into the state equation,

$$\dot{x} = -x - Kx$$

$$\dot{x} = -(1 + K)x$$

Therefore, the closed-loop state matrix is

$$A_{cl} = -(1 + K)$$

Hence, the closed-loop pole is

$$s = -(1 + K)$$

Given that the closed-loop pole is located at

$$s = -2$$

Therefore,

$$-(1 + K) = -2$$

$$1 + K = 2$$

$$K = 1$$

$$\boxed{1}$$

Q9.19.1 MCQ 2019 2M Ans: A

The state equation is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\alpha & -2\beta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha \end{bmatrix} r$$

The characteristic equation is obtained from

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ \alpha & s + 2\beta \end{vmatrix} = 0$$

$$s(s + 2\beta) + \alpha = 0$$

$$s^2 + 2\beta s + \alpha = 0$$

Comparing with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$, we obtain

$$\omega_n^2 = \alpha$$

$$\therefore \quad \omega_n = \sqrt{\alpha}$$

$$2\xi\omega_n = 2\beta$$

$$\xi\omega_n = \beta$$

$$\xi = \frac{\beta}{\sqrt{\alpha}}$$

$$\boxed{(A) \ \ \xi = \frac{\beta}{\sqrt{\alpha}}, \ \omega_n = \sqrt{\alpha}}$$

Q9.18.1 MCQ 2018 2M Ans: C ● ● ●

Given :

$$\frac{dx_1(t)}{dt} = x_2(t) - x_1(t)$$
$$\frac{dx_2(t)}{dt} = x_1(t) - x_2(t)$$

The state-space model of the above equation is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$[sI - A] = \begin{bmatrix} s+1 & -1 \\ -1 & s+1 \end{bmatrix}$$

$$\phi(s) = [sI - A]^{-1} = \frac{\text{Adj}[sI - A]}{|sI - A|}$$

$$\text{Adj}[sI - A] = \begin{bmatrix} s+1 & 1 \\ 1 & s+1 \end{bmatrix}$$

$$|sI - A| = (s+1)^2 - 1 = s^2 + 2s$$

$$[sI - A]^{-1} = \frac{1}{s^2 + 2s} \begin{bmatrix} s+1 & 1 \\ 1 & s+1 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{s+1}{s(s+2)} & \frac{1}{s(s+2)} \\ \frac{1}{s(s+2)} & \frac{s+1}{s(s+2)} \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{1}{2s} + \frac{1}{2(s+2)} & \frac{1}{2s} - \frac{1}{2(s+2)} \\ \frac{1}{2s} - \frac{1}{2(s+2)} & \frac{1}{2s} + \frac{1}{2(s+2)} \end{bmatrix}$$

Taking inverse Laplace transform,

$$\phi(t) = \frac{1}{2} \begin{bmatrix} 1 + e^{-2t} & 1 - e^{-2t} \\ 1 - e^{-2t} & 1 + e^{-2t} \end{bmatrix}$$

$$x(t) = \phi(t)x(0) = \frac{1}{2} \begin{bmatrix} 1 + e^{-2t} & 1 - e^{-2t} \\ 1 - e^{-2t} & 1 + e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

$$x(t) = \frac{1}{2} \begin{bmatrix} x_1(0) + x_2(0) + e^{-2t}[x_1(0) - x_2(0)] \\ x_1(0) + x_2(0) + e^{-2t}[x_2(0) - x_1(0)] \end{bmatrix}$$

$$x_f = \lim_{t \to \infty} x(t) = \frac{1}{2} \begin{bmatrix} x_1(0) + x_2(0) \\ x_1(0) + x_2(0) \end{bmatrix}$$

$$x_{1f} = x_{2f} < \infty$$

$$\boxed{(C) \;\; x_{1f} = x_{2f} < \infty}$$

Q9.17.1 MCQ 2017 2M Ans: D ● ○ ○

Given,

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The transfer function is

$$G(s) = C(sI - A)^{-1}B$$

First,

$$sI - A = \begin{bmatrix} s-1 & -2 \\ -2 & s \end{bmatrix}$$

$$|sI - A| = \begin{vmatrix} s-1 & -2 \\ -2 & s \end{vmatrix} = s(s-1) - 4 = s^2 - s - 4$$

$$(sI - A)^{-1} = \frac{1}{s^2 - s - 4}\begin{bmatrix} s & 2 \\ 2 & s-1 \end{bmatrix}$$

Now,

$$(sI - A)^{-1}B = \frac{1}{s^2 - s - 4}\begin{bmatrix} s & 2 \\ 2 & s-1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{s^2 - s - 4}\begin{bmatrix} s+4 \\ 2s \end{bmatrix}$$

Hence,

$$G(s) = \begin{bmatrix} 1 & 0 \end{bmatrix}\frac{1}{s^2 - s - 4}\begin{bmatrix} s+4 \\ 2s \end{bmatrix} = \frac{s+4}{s^2 - s - 4}$$

$$\boxed{(D) \quad \frac{s+4}{s^2 - s - 4}}$$

Q9.17.2 NAT 2017 2M Ans: 1.280 to 1.287 ● ● ○

Given state-space model,

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix}x(t)$$

with

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and unit step input $u(t) = 1$. Taking Laplace transform of the state equation, we get

$$X(s) = [sI - A]^{-1}x(0) + [sI - A]^{-1}BU(s)$$

First compute,

$$sI - A = \begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}$$

$$|sI - A| = s(s+2)$$

Therefore,

$$[sI - A]^{-1} = \frac{1}{s(s+2)}\begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

Since $U(s) = \dfrac{1}{s}$,

The zero-input response is

$$[sI-A]^{-1}x(0) = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{s} \\ 0 \end{bmatrix}$$

The zero-state response is

$$[sI-A]^{-1}BU(s) = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s} = \begin{bmatrix} \frac{1}{s^2(s+2)} \\ \frac{1}{s(s+2)} \end{bmatrix}$$

Hence,

$$X(s) = \begin{bmatrix} \frac{1}{s} + \frac{1}{s^2(s+2)} \\ \frac{1}{s(s+2)} \end{bmatrix}$$

Since $y(t) = x_1(t)$, we need only the first state:

$$X_1(s) = \frac{1}{s} + \frac{1}{s^2(s+2)}$$

Using partial fractions,

$$X_1(s) = \frac{3}{4s} + \frac{1}{2s^2} + \frac{1}{4(s+2)}$$

Taking inverse Laplace transform,

$$x_1(t) = \frac{3}{4} + \frac{t}{2} + \frac{1}{4}e^{-2t}$$

Thus,

$$y(t) = \frac{3}{4} + \frac{t}{2} + \frac{1}{4}e^{-2t}$$

At $t = 1$,

$$y(1) = \frac{3}{4} + \frac{1}{2} + \frac{1}{4}e^{-2} = 1.25 + 0.25(0.1353) = 1.2838$$

$$\boxed{1.284}$$

Q9.16.1 NAT | 2016 | 2M | Ans: 5.9 to 6.1 | ● ○ ○

Given state equation,

$$\dot{x}(t) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} x(t) \qquad y(t) = c^T x(t) \qquad c = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

with initial condition,

$$x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Taking Laplace transform,

$$X(s) = [sI-A]^{-1}x(0)$$

First compute,

$$sI - A = \begin{bmatrix} s-1 & 0 \\ 0 & s-2 \end{bmatrix}$$

Therefore,

$$[sI - A]^{-1} = \begin{bmatrix} \frac{1}{s-1} & 0 \\ 0 & \frac{1}{s-2} \end{bmatrix}$$

Hence,

$$X(s) = \begin{bmatrix} \frac{1}{s-1} & 0 \\ 0 & \frac{1}{s-2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s-1} \\ \frac{1}{s-2} \end{bmatrix}$$

Taking inverse Laplace transform,

$$x(t) = \begin{bmatrix} e^t \\ e^{2t} \end{bmatrix}$$

Therefore,

$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} e^t \\ e^{2t} \end{bmatrix} = e^t + e^{2t}$$

At $t = \log_e 2$

$$y(\log_e 2) = e^{\log_e 2} + e^{2\log_e 2} = 2 + 4 = 6$$

$$\boxed{6}$$

Q9.15.1 MCQ 2015 2M Ans: B ● ● ○

Let the state vector be

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

From the signal flow graph,

$$\dot{x}_1 = 5x_1 - 2x_2 + u_1$$

$$\dot{x}_2 = 2x_1 + x_2 + u_1 + u_2$$

$$y_1 = x_1, \qquad y_2 = x_1 - x_2$$

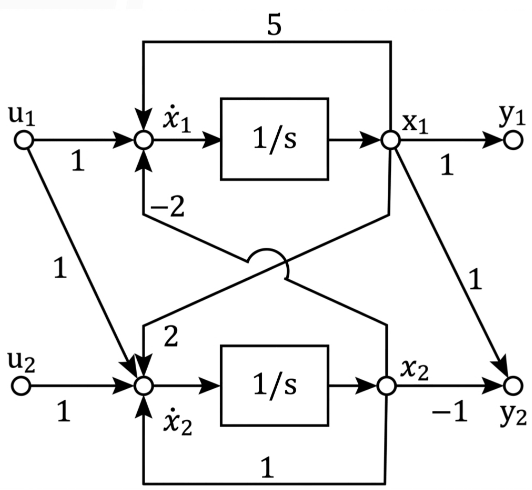


Fig. Solution Q9.15.1

In matrix form,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 5 & -2 \\ 2 & 1 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} 1 & -1 \end{bmatrix}$$

For input u_1,

$$C_1 = \begin{bmatrix} B_1 & AB_1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}$$

$$\text{rank}(C_1) = 1$$

Hence, the system is not controllable with u_1.

For input u_2,

$$C_2 = \begin{bmatrix} B_2 & AB_2 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}$$

$$\text{rank}(C_2) = 2$$

Hence, the system is controllable with u_2.

For output y_1,

$$O_1 = \begin{bmatrix} C_1 \\ C_1 A \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & -2 \end{bmatrix}$$

$$\text{rank}(O_1) = 2$$

Hence, the system is observable from y_1.

For output y_2,

$$O_2 = \begin{bmatrix} C_2 \\ C_2 A \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix}$$

$$\text{rank}(O_2) = 1$$

Hence, the system is not observable from y_2.

Therefore, the SISO system is controllable and observable only when u_2 is the only input and y_1 is the only output. Therefore, the correct option is,

$$\boxed{(B) \text{ When } u_2 \text{ is the only input and } y_1 \text{ is the only output}}$$

Q9.15.2 MCQ 2015 2M Ans: A ● ○ ○

Given,

$$\dot{x} = \begin{bmatrix} 2 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \qquad y = \begin{bmatrix} 3 & 0 \end{bmatrix} x$$

The transfer function is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

Here,

$$sI - A = \begin{bmatrix} s-2 & -1 \\ 2 & s \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 - 2s + 2} \begin{bmatrix} s & 1 \\ -2 & s-2 \end{bmatrix}$$

Therefore,

$$\frac{Y(s)}{U(s)} = \begin{bmatrix} 3 & 0 \end{bmatrix} \frac{1}{s^2 - 2s + 2} \begin{bmatrix} s & 1 \\ -2 & s-2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{s^2 - 2s + 2} \begin{bmatrix} 3 & 0 \end{bmatrix} \begin{bmatrix} s+1 \\ s-4 \end{bmatrix} = \frac{3(s+1)}{s^2 - 2s + 2}$$

$$\therefore \frac{Y(s)}{U(s)} = \frac{3(s+1)}{s^2 - 2s + 2}$$

$$\boxed{(A) \ \frac{3(s+1)}{s^2 - 2s + 2}}$$

Q9.14.1 MCQ 2014 1M Ans: C ● ○ ○

Method 1

The state matrix is

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Since,

$$A = I + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = I + N$$

Here,

$$N = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad N^2 = 0$$

Therefore,

$$e^{At} = e^{(I+N)t} = e^t e^{Nt} = e^t (I + tN)$$

Hence,

$$e^{At} = e^t \begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix} = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$$

Method 2

The state transition matrix is $\phi(t) = e^{At}$

Since, $e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots$

First, compute the powers of A:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

Therefore,

$$e^{At} = I + \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} t + \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \frac{t^2}{2!} + \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \frac{t^3}{3!} + \cdots$$

$$= \begin{bmatrix} 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots & 0 \\ t + \frac{2t^2}{2!} + \frac{3t^3}{3!} + \cdots & 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$$

$$\boxed{(C) \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}}$$

Q9.14.2 MCQ 2014 2M Ans: A ● ○ ○

For $a = 1$,

$$A = \begin{bmatrix} a & a-1 \\ a+1 & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

The poles are the eigenvalues of A.

$$|sI - A| = 0$$

$$\begin{vmatrix} s-1 & 0 \\ -2 & s-1 \end{vmatrix} = 0$$

$$(s-1)^2 = 0$$

$$s_1 = s_2 = 1$$

Therefore, the pole locations are

$$\boxed{(A)\ 1 \pm j0}$$

Q9.14.3 MCQ 2014 2M Ans: C ● ○ ○

Given,

$$P = \begin{bmatrix} -1 & 1 \\ 0 & -3 \end{bmatrix}, \qquad Q = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

The controllability matrix is

$$C = [Q \quad PQ] = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

$$|C| = -1 \neq 0$$

Hence, the system is controllable.

The observability matrix is

$$O = \begin{bmatrix} R \\ RP \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$$

$$|O| = 0$$

Hence, the system is not observable.

$\boxed{(C) \text{ Controllable but not observable}}$

Q9.14.4 MCQ 2014 2M Ans: A ● ● ○

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The transfer function is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B$$

$$sI - A = \begin{bmatrix} s & -1 \\ 1 & s+1 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + s + 1}\begin{bmatrix} s+1 & 1 \\ -1 & s \end{bmatrix}$$

$$\frac{Y(s)}{U(s)} = \begin{bmatrix} 1 & 0 \end{bmatrix}\frac{1}{s^2 + s + 1}\begin{bmatrix} s+1 & 1 \\ -1 & s \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2 + s + 1}$$

For a unit step input,

$$U(s) = \frac{1}{s}$$

Hence,

$$Y(s) = \frac{1}{s\left(s^2 + s + 1\right)}$$

Now,

$$E(s) = U(s) - Y(s) = \frac{1}{s} - \frac{1}{s\left(s^2 + s + 1\right)}$$

Therefore,

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0}\left[1 - \frac{1}{s^2 + s + 1}\right] = 1 - \frac{1}{1} = 0$$

$\boxed{(A)\ 0}$

Q9.13.1 MCQ 2013 2M Ans: A ● ○ ○

Given,

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The controllability matrix is $= \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$

$$\text{Determinant} = 1 \neq 0$$

Hence, the system is controllable.

The observability matrix is $= \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$

$$\text{Determinant} = 0$$

Hence, the system is not observable.

$$\boxed{(A)\ \text{Controllable but not observable}}$$

Q9.13.2 MCQ 2013 2M Ans: A ● ○ ○

Given,

$$\dot{x}_1 = -2x_1 + u$$

$$\dot{x}_2 = -x_2 + u$$

$$y = x_1$$

Taking Laplace transform with zero initial conditions,

$$sX_1(s) = -2X_1(s) + U(s)$$

$$(s+2)X_1(s) = U(s)$$

$$X_1(s) = \frac{U(s)}{s+2}$$

Since $y = x_1$,

$$Y(s) = X_1(s) = \frac{U(s)}{s+2}$$

For a unit step input,

$$U(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s+2)}$$

$$= \frac{1}{2s} - \frac{1}{2(s+2)}$$

$$y(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

$$\boxed{(A)\ \frac{1}{2} - \frac{1}{2}e^{-2t}}$$

Q9.12.1 MCQ 2012 2M Ans: D ● ● ○

Given state space model,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The controllability matrix is

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

First, compute AB:

$$AB = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix}$$

Next, compute A^2B:

$$A^2B = A(AB)$$

$$= \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1a_2 \\ 0 \\ 0 \end{bmatrix}$$

Hence,

$$Q_c = \begin{bmatrix} 0 & 0 & a_1a_2 \\ 0 & a_2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Its determinant is

$$|Q_c| = -a_1a_2^2$$

For complete controllability,

$$|Q_c| \neq 0$$

$$-a_1a_2^2 \neq 0$$

$$\therefore \quad a_1 \neq 0, \quad a_2 \neq 0$$

Observe that a_3 does not appear in the controllability determinant. Therefore, controllability is independent of a_3.

$$\boxed{(D)\ a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0}$$

Q9.10.1 MCQ 2010 2M Ans: C ● ○ ○

Given,

$$A = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$

$$\begin{vmatrix} s+1 & -2 \\ 0 & s-2 \end{vmatrix} = 0$$

$$(s+1)(s-2) = 0$$

$$s = -1, \quad s = 2$$

Since one eigenvalue lies in the right-half plane, the system is unstable.

The controllability matrix is

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$|Q_c| = -2 \neq 0$$

Hence, the system is completely controllable.

$$\boxed{(C)\ \text{Unstable but controllable}}$$

Q9.09.1 MCQ 2009 2M Ans: C ● ● ○

Given,

$$\dot{x}_1 = -3x_1 + x_2 + 2u$$
$$\dot{x}_2 = -2x_2 + u$$
$$y = x_1$$

Method 1

Taking Laplace transform with zero initial conditions,

$$sX_1(s) = -3X_1(s) + X_2(s) + 2U(s)$$
$$(s+3)X_1(s) = X_2(s) + 2U(s)$$
$$sX_2(s) = -2X_2(s) + U(s)$$
$$(s+2)X_2(s) = U(s)$$
$$X_2(s) = \frac{U(s)}{s+2}$$

Substituting in the first equation,

$$(s+3)X_1(s) = \frac{U(s)}{s+2} + 2U(s)$$
$$X_1(s) = \frac{1+2(s+2)}{(s+2)(s+3)}U(s)$$
$$X_1(s) = \frac{2s+5}{(s+2)(s+3)}U(s)$$

Since $y = x_1$,

$$Y(s) = X_1(s)$$
$$\therefore \frac{Y(s)}{U(s)} = \frac{2s+5}{s^2+5s+6}$$

Method 2

Here,

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$
$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D = 0$$

Using

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$
$$sI - A = \begin{bmatrix} s+3 & -1 \\ 0 & s+2 \end{bmatrix}$$
$$(sI-A)^{-1} = \frac{1}{(s+2)(s+3)}\begin{bmatrix} s+2 & 1 \\ 0 & s+3 \end{bmatrix}$$
$$\frac{Y(s)}{U(s)} = \frac{1}{(s+2)(s+3)}\begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} 2(s+2)+1 \\ s+3 \end{bmatrix}$$
$$= \frac{2s+5}{s^2+5s+6}$$

$$\boxed{(C)\ \frac{2s+5}{s^2+5s+6}}$$

Q9.09.2 MCQ 2009 2M Ans: B ● ● ○

The state matrix is

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix}$$

The state transition matrix is

$$\phi(t) = e^{At}$$

Using the series expansion,

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots$$

First, compute the powers of A:

$$A^2 = \begin{bmatrix} 9 & -5 \\ 0 & 4 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} -27 & 19 \\ 0 & -8 \end{bmatrix}$$

Therefore,

$$e^{At} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix}t + \begin{bmatrix} 9 & -5 \\ 0 & 4 \end{bmatrix}\frac{t^2}{2!} + \begin{bmatrix} -27 & 19 \\ 0 & -8 \end{bmatrix}\frac{t^3}{3!} + \cdots$$

$$= \begin{bmatrix} 1 - 3t + \frac{9t^2}{2!} - \frac{27t^3}{3!} + \cdots & t - \frac{5t^2}{2!} + \frac{19t^3}{3!} + \cdots \\ 0 & 1 - 2t + \frac{4t^2}{2!} - \frac{8t^3}{3!} + \cdots \end{bmatrix}$$

The 1x1 and 2x2 entries are the expansions of e^{-3t} and e^{-2t}, respectively. The 1x2 entry simplifies to $e^{-2t} - e^{-3t}$. Hence,

$$\phi(t) = e^{At} = \begin{bmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{bmatrix}$$

$$\boxed{(B) \begin{bmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{bmatrix}}$$

Q9.08.1 MCQ 2008 2M Ans: D ● ○ ○

Given,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D = 0$$

Using the state-space formula,

$$G(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix}$$

$$G(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s(s+2)} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} = \frac{1}{s(s+2)}$$

$$\boxed{(D) \ \frac{1}{s(s+2)}}$$

Q9.08.2 MCQ 2008 2M Ans: A ● ○ ○

From Q9.08.1,

$$G(s) = \frac{1}{s(s+2)}$$

The open-loop transfer function has one pole at the origin. Therefore, System type = 1.
For a unity feedback system,

$$K_p = \lim_{s \to 0} G(s) = \lim_{s \to 0} \frac{1}{s(s+2)} = \infty$$

Hence, for a unit step input,

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = 0$$

$$\boxed{(A) \ 0}$$

Key Points

- For a Type-1 system, the steady-state error for a unit step input is zero

Q9.06.1 MCQ | 2006 | 1M | Ans: B | ● ○ ○

Given,

$$H(s) = \frac{3(s-2)}{4s^3 - 2s + 1}$$

Dividing the denominator by 4,

$$H(s) = \frac{\frac{3}{4}(s-2)}{s^3 + 0s^2 - \frac{1}{2}s + \frac{1}{4}}$$

Comparing with the standard transfer function,

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^{n-1} + b_1 s^{n-2} + \cdots + b_{n-1}}{s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_n}$$

Here,

$$a_1 = 0, \qquad a_2 = -\frac{1}{2}, \qquad a_3 = \frac{1}{4}$$

The controllable canonical state matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{4} & \frac{1}{2} & 0 \end{bmatrix}$$

Multiplying the last row by 4 does not change the structure of the canonical matrix represented in the options. Hence,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 2 & -4 \end{bmatrix}$$

$$\boxed{(B) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 2 & -4 \end{bmatrix}}$$

Q9.05.1 MCQ | 2005 | 2M | Ans: A | ● ● ○

The state matrix is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$$

The state transition matrix is

$$\phi(t) = e^{At}$$

Using the series expansion,

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots$$

First, compute the powers of A:

$$A^2 = \begin{bmatrix} 0 & -3 \\ 0 & 9 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} 0 & 9 \\ 0 & -27 \end{bmatrix}$$

Therefore,

$$e^{At} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix} t + \begin{bmatrix} 0 & -3 \\ 0 & 9 \end{bmatrix} \frac{t^2}{2!} + \begin{bmatrix} 0 & 9 \\ 0 & -27 \end{bmatrix} \frac{t^3}{3!} + \cdots$$

$$= \begin{bmatrix} 1 & t - \frac{3t^2}{2!} + \frac{9t^3}{3!} - \cdots \\ 0 & 1 - 3t + \frac{9t^2}{2!} - \frac{27t^3}{3!} + \cdots \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{1}{3}\left(1 - e^{-3t}\right) \\ 0 & e^{-3t} \end{bmatrix}$$

$$\boxed{(A)\ \begin{bmatrix} 1 & \frac{1}{3}(1 - e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix}}$$

Q9.05.2 MCQ 2005 2M Ans: C ● ● ○

From Q9.05.1,

$$\phi(t) = \begin{bmatrix} 1 & \frac{1}{3}(1 - e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix}$$

Also,

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad X(0) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \qquad u(t) = 1$$

The state response is

$$X(t) = \phi(t)X(0) + \int_0^t \phi(t-\tau)Bu(\tau)\,d\tau$$

Since $u(\tau) = 1$,

$$\phi(t)X(0) = \begin{bmatrix} 1 & \frac{1}{3}(1 - e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -e^{-3t} \\ 3e^{-3t} \end{bmatrix}$$

$$\phi(t-\tau)B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\int_0^t \phi(t-\tau)B\,d\tau = \int_0^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} d\tau = \begin{bmatrix} t \\ 0 \end{bmatrix}$$

Therefore,

$$X(t) = \begin{bmatrix} -e^{-3t} \\ 3e^{-3t} \end{bmatrix} + \begin{bmatrix} t \\ 0 \end{bmatrix} = \begin{bmatrix} t - e^{-3t} \\ 3e^{-3t} \end{bmatrix}$$

$$\boxed{(C)\ X(t) = \begin{bmatrix} t - e^{-3t} \\ 3e^{-3t} \end{bmatrix}}$$

Q9.04.1 MCQ 2004 2M Ans: A ● ○ ○

Given,

$$A = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -2 \\ -2 & s \end{vmatrix} = 0$$

$$s^2 - 4 = 0$$
$$(s-2)(s+2) = 0$$
$$s = 2, \qquad s = -2$$

$$\boxed{(A)\ \ -2 \text{ and } +2}$$

Q9.03.1 MCQ 2003 1M Ans: A ● ○ ○

Since there is no external input,

$$x(t) = \phi(t)x(0)$$

Given,

$$\phi(t) = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix}, \qquad x(0) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

At $t = 1$ s,

$$x(1) = \begin{bmatrix} e^{-2} & 0 \\ 0 & e^{-1} \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2e^{-2} \\ 3e^{-1} \end{bmatrix} = \begin{bmatrix} 0.271 \\ 1.104 \end{bmatrix}$$

$$\boxed{(A)\ \begin{bmatrix} 0.271 \\ 1.104 \end{bmatrix}}$$

Q9.03.2 MCQ 2003 2M Ans: A ● ● ○

Given,

$$\frac{d^2\omega}{dt^2} + \frac{B}{J}\frac{d\omega}{dt} + \frac{K^2}{LJ}\omega = \frac{K}{LJ}V_a$$

Choose the state variables as

$$x_1 = \frac{d\omega}{dt}, \qquad x_2 = \omega$$

Then,

$$\dot{x}_1 = \frac{d^2\omega}{dt^2}, \qquad \dot{x}_2 = \frac{d\omega}{dt} = x_1$$

From the given equation,

$$\dot{x}_1 = -\frac{B}{J}x_1 - \frac{K^2}{LJ}x_2 + \frac{K}{LJ}V_a$$
$$\dot{x}_2 = x_1$$

Therefore,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{B}{J} & -\frac{K^2}{LJ} \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{K}{LJ} \\ 0 \end{bmatrix} V_a$$

Hence,

$$P = \begin{bmatrix} -\frac{B}{J} & -\frac{K^2}{LJ} \\ 1 & 0 \end{bmatrix}$$

$$\boxed{(A)\ \begin{bmatrix} -\frac{B}{J} & -\frac{K^2}{LJ} \\ 1 & 0 \end{bmatrix}}$$

Q9.02.1 MCQ 2002 1M Ans: C ● ○ ○

For the system

$$\dot{X} = AX$$

Taking Laplace transform,

$$sX(s) - X(0) = AX(s)$$
$$(sI - A)X(s) = X(0)$$
$$X(s) = (sI - A)^{-1}X(0)$$

Taking inverse Laplace transform,

$$X(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}X(0)\right\}$$

Since,

$$X(t) = \phi(t)X(0)$$
$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

$$\boxed{(C)\ \phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}}$$

Q9.02.2 MCQ 2002 2M Ans: B ● ○ ○

Given,

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$
$$\begin{vmatrix} s-2 & -3 \\ 0 & s-5 \end{vmatrix} = 0$$
$$(s-2)(s-5) = 0$$
$$s = 2, \qquad s = 5$$

Both eigenvalues lie in the right-half plane.
Hence, the system is unstable.

The controllability matrix is

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$
$$AB = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$
$$Q_c = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$
$$|Q_c| = 0$$

Hence, the system is uncontrollable.

$$\boxed{(B)\ \text{Uncontrollable but unstable}}$$

Q9.02.3 MCQ 2002 2M Ans: B ● ● ○

Given,

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 4 & 0 \end{bmatrix}$$

Since $u(t)$ is a unit impulse input and $X(0) = 0$,

$$X(s) = (sI - A)^{-1}B$$

Here,

$$sI - A = \begin{bmatrix} s-2 & 0 \\ 0 & s-4 \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-4} \end{bmatrix}$$

$$X(s) = \begin{bmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-4} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s-2} \\ \frac{1}{s-4} \end{bmatrix}$$

Therefore,

$$Y(s) = CX(s) = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{s-2} \\ \frac{1}{s-4} \end{bmatrix} = \frac{4}{s-2}$$

$$y(t) = 4e^{2t}$$

$$\boxed{(B)\ 4e^{2t}}$$

Q9.02.4 NAT 2002 5M Ans: * ● ● ●

Given,

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = u(t)e^{-t}$$

The state variables are chosen as

$$x_1 = y, \qquad x_2 = \left(\frac{dy}{dt} - y\right)e^t$$

Since $x_1 = y$,

$$\dot{x}_1 = \frac{dy}{dt}$$

From the definition of x_2,

$$x_2 e^{-t} = \frac{dy}{dt} - y = \dot{x}_1 - x_1$$

$$\therefore \qquad \dot{x}_1 = x_1 + e^{-t}x_2$$

Differentiating x_2,

$$\dot{x}_2 = \frac{d}{dt}\left[\left(\frac{dy}{dt} - y\right)e^t\right] = e^t\left(\frac{d^2y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} - y\right) = e^t\left(\frac{d^2y}{dt^2} - y\right)$$

From the given differential equation,

$$\frac{d^2y}{dt^2} = -\frac{dy}{dt} + 2y + u(t)e^{-t}$$

$$\frac{d^2y}{dt^2} - y = -\frac{dy}{dt} + y + u(t)e^{-t}$$

$$\frac{d^2y}{dt^2} - y = -\left(\frac{dy}{dt} - y\right) + u(t)e^{-t}$$

$$\frac{d^2y}{dt^2} - y = -x_2e^{-t} + u(t)e^{-t}$$

$$\left(\frac{d^2y}{dt^2} - y\right)e^t = -x_2 + u(t)$$

$$\therefore \qquad \dot{x}_2 = -x_2 + u(t)$$

Hence, the state model is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & e^{-t} \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

For a unit step input,

$$u(t) = 1$$

$$\dot{x}_2 + x_2 = 1$$

Taking Laplace transform and using $x_2(0) = 0$,

$$(s+1)X_2(s) = \frac{1}{s}$$

$$X_2(s) = \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

$$\therefore \qquad x_2(t) = 1 - e^{-t}$$

$$\boxed{\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & e^{-t} \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)}$$

$$\boxed{x_2(t) = 1 - e^{-t}}$$

Q9.01.1 MCQ 2001 1M Ans: A ● ● ○

Given,

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix}, \qquad x(0) = \begin{bmatrix} 10 \\ -10 \end{bmatrix}$$

The state transition matrix is

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\} = \mathcal{L}^{-1}\left\{\begin{bmatrix} s+3 & -1 \\ 0 & s+2 \end{bmatrix}^{-1}\right\} = \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+3} & \frac{1}{(s+3)(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}\right\} = \begin{bmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{bmatrix}$$

Therefore,

$$x(t) = \phi(t)x(0) = \begin{bmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 10 \\ -10 \end{bmatrix}$$

As $t \to \infty$,

$$e^{-2t} \to 0, \qquad e^{-3t} \to 0$$

$$x_{ss} = \lim_{t\to\infty} x(t) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boxed{(A) \begin{bmatrix} 0 \\ 0 \end{bmatrix}}$$

Q9.00.1 NAT 2000 Ans: * ● ● ●

The state response is

$$x(t) = e^{At}x(0)$$

Given,

$$e^{At} = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

At $t = 1$,

$$\begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} = e^{-1} \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} e^{-1}\left(2x_1(0) + x_2(0)\right) \\ -e^{-1}x_1(0) \end{bmatrix}$$

$$x_1(1) = e^{-1}\left(2x_1(0) + x_2(0)\right)$$

$$x_2(1) = -e^{-1}x_1(0)$$

Since

$$x(2) = e^{2A}x(0) = e^{A}(e^{A}x(0)) = e^{A}x(1),$$

$$x_1(2) = e^{-1}\left(2x_1(1) + x_2(1)\right)$$

Substituting the values of $x_1(1)$ and $x_2(1)$,

$$x_1(2) = e^{-1}\left[2e^{-1}\left(2x_1(0) + x_2(0)\right) - e^{-1}x_1(0)\right] = e^{-2}\left[4x_1(0) + 2x_2(0) - x_1(0)\right] = e^{-2}\left[3x_1(0) + 2x_2(0)\right]$$

Given,

$$x_1(2) = 2$$

$$e^{-2}\left[3x_1(0) + 2x_2(0)\right] = 2$$

$$\therefore \quad 3x_1(0) + 2x_2(0) = 2e^2$$

Hence, all initial states satisfying $3x_1(0) + 2x_2(0) = 2e^2$, produce the set of states:

$$x_1(1) = e^{-1}\left(2x_1(0) + x_2(0)\right)$$

$$x_2(1) = -e^{-1}x_1(0)$$

Q9.00.2 NAT 2000 Ans: * ● ● ○

Given,

$$e^{At} = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}$$

Taking Laplace transform,

$$\Phi(s) = \mathcal{L}\{e^{At}\} = \begin{bmatrix} \mathcal{L}\{e^{-t}\} + \mathcal{L}\{te^{-t}\} & \mathcal{L}\{te^{-t}\} \\ -\mathcal{L}\{te^{-t}\} & \mathcal{L}\{e^{-t}\} - \mathcal{L}\{te^{-t}\} \end{bmatrix}$$

$$\Phi(s) = \begin{bmatrix} \dfrac{1}{s+1} + \dfrac{1}{(s+1)^2} & \dfrac{1}{(s+1)^2} \\ -\dfrac{1}{(s+1)^2} & \dfrac{1}{s+1} - \dfrac{1}{(s+1)^2} \end{bmatrix} = \frac{1}{(s+1)^2} \begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}$$

Hence,

$$(sI - A)^{-1} = \Phi(s) = \frac{1}{(s+1)^2} \begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}$$

$$\boxed{(sI - A)^{-1} = \frac{1}{(s+1)^2} \begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}}$$

Q9.00.3 NAT 2000 Ans: * ● ● ○

From Q9.00.2,

$$(sI - A)^{-1} = \frac{1}{(s+1)^2}\begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}$$

Taking inverse,

$$sI - A = (s+1)^2\begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}^{-1} = (s+1)^2\frac{1}{(s+1)^2}\begin{bmatrix} s & -1 \\ 1 & s+2 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 1 & s+2 \end{bmatrix}$$

Since,

$$sI - A = \begin{bmatrix} s-a_{11} & -a_{12} \\ -a_{21} & s-a_{22} \end{bmatrix}$$

Comparing the elements,

$$a_{11} = 0, \qquad a_{12} = 1, \qquad a_{21} = -1, \qquad a_{22} = -2$$

Hence,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}}$$

Q9.98.1 NAT 1998 5M Ans: ● ● ○

Given,

$$A = \begin{bmatrix} -5 & 1 \\ -6 & 0 \end{bmatrix}$$

The Laplace transform of the state transition matrix is

$$\Phi(s) = (sI - A)^{-1} = \begin{bmatrix} s+5 & -1 \\ 6 & s \end{bmatrix}^{-1} = \frac{1}{s(s+5)+6}\begin{bmatrix} s & 1 \\ -6 & s+5 \end{bmatrix} = \frac{1}{(s+2)(s+3)}\begin{bmatrix} s & 1 \\ -6 & s+5 \end{bmatrix}$$

Since

$$x(t) = e^{At}x(0) = \mathcal{L}^{-1}\{\Phi(s)\}x(0)$$

Given,

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_1(t) = \mathcal{L}^{-1}\left\{\frac{s}{(s+2)(s+3)}\right\}$$

$$x_1(t) = -2e^{-2t} + 3e^{-3t}$$

At $t = 1$,

$$x_1(1) = -2e^{-2} + 3e^{-3} = -0.121$$

$$\boxed{\Phi(s) = \frac{1}{(s+2)(s+3)}\begin{bmatrix} s & 1 \\ -6 & s+5 \end{bmatrix}}$$

$$\boxed{x_1(1) = -0.121}$$

Q9.97.1 NAT 1997 5M Ans: * ● ● ○

Given,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -40 & -44 & -14 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \qquad D = 0$$

The transfer function is

$$G(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 40 & 44 & s+14 \end{bmatrix}$$

Since $B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, only the second column of $(sI - A)^{-1}$ is required.

The cofactor corresponding to the element in the second row and second column is

$$M_{22} = \begin{vmatrix} s & 0 \\ 40 & s+14 \end{vmatrix} = s(s+14)$$

Also,

$$|sI - A| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 40 & 44 & s+14 \end{vmatrix} = s^3 + 14s^2 + 44s + 40 = (s+10)(s+2)^2$$

Therefore,

$$(sI - A)^{-1}B = \frac{1}{|sI - A|}\begin{bmatrix} * \\ s(s+14) \\ * \end{bmatrix}$$

Hence,

$$G(s) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \frac{1}{|sI - A|}\begin{bmatrix} * \\ s(s+14) \\ * \end{bmatrix} = \frac{s(s+14)}{|sI - A|} = \frac{s(s+14)}{s^3 + 14s^2 + 44s + 40} = \frac{s(s+14)}{(s+10)(s+2)^2}$$

$$\boxed{G(s) = \frac{s(s+14)}{(s+10)(s+2)^2}}$$

Q9.95.1 MCQ 1995 1M Ans: A ● ○ ○

Given,

$$A = \begin{bmatrix} -4 & -1 \\ 3 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D = 0$$

The transfer function is

$$G(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s+4 & 1 \\ -3 & s+1 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s+4)(s+1)+3}\begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix} = \frac{1}{s^2+5s+7}\begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix}$$

$$G(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s^2+5s+7}\begin{bmatrix} s+1 & -1 \\ 3 & s+4 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{s^2+5s+7}\begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} s \\ s+7 \end{bmatrix} = \frac{s}{s^2+5s+7}$$

$$\boxed{(A)\ \frac{s}{s^2+5s+7}}$$

Q9.95.2 NAT 1995 Ans: *

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

The characteristic equation is

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ 1 & s+2 \end{vmatrix} = 0$$

$$s(s+2) + 1 = 0$$

$$(s+1)^2 = 0$$

Hence, the characteristic equation is $s^2 + 2s + 1 = 0$
The state transition matrix is

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}\right\} = \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+1} + \frac{1}{(s+1)^2} & \frac{1}{(s+1)^2} \\ -\frac{1}{(s+1)^2} & \frac{1}{s+1} - \frac{1}{(s+1)^2} \end{bmatrix}\right\}$$

Therefore,

$$\phi(t) = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}$$

Hence,

$$\boxed{\text{Characteristic Equation: } s^2 + 2s + 1 = 0}$$

$$\boxed{\phi(t) = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}}$$

Q9.95.3 NAT 1995 Ans: *

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad r(t) = U(t)$$

Also,

$$x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The state response is

$$x(t) = \phi(t)x(0) + \int_0^t \phi(t-\tau)Br(\tau)\,d\tau$$

Since $x(0) = 0$ and $r(\tau) = 1$,

$$x(t) = \int_0^t \phi(t-\tau)\begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau$$

From Q9.95.2,

$$\phi(t) = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}$$

$$\phi(t-\tau)\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} (t-\tau)e^{-(t-\tau)} \\ e^{-(t-\tau)} - (t-\tau)e^{-(t-\tau)} \end{bmatrix}$$

Hence,

$$x_2(t) = \int_0^t \left[e^{-(t-\tau)} - (t-\tau)e^{-(t-\tau)}\right] d\tau$$

Let

$$u = t - \tau$$

$$x_2(t) = \int_0^t (e^{-u} - ue^{-u})\, du = [-e^{-u}]_0^t - [-(u+1)e^{-u}]_0^t = (1-e^{-t}) - \left[1-(t+1)e^{-t}\right] = te^{-t}$$

$$\boxed{x_2(t) = te^{-t}}$$

Q9.94.1 MCQ 1994 1M Ans: C ● ○ ○

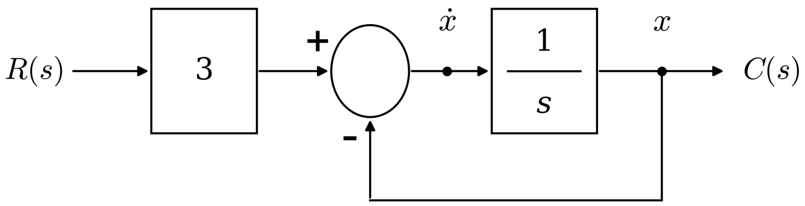


Fig. Solution Q9.94.1

Let the output of the integrator be the state variable x.

$$x = C(s)$$

From the summing junction,

$$\dot{x} = -x + 3r$$

Comparing with the standard state equation, $\dot{x} = Ax + Br$

$$A = [-1]$$

$$\boxed{(C)\ \ [-1]}$$

Q9.93.1 MCQ 1993 1M Ans: A ● ○ ○

Given the state-space representation,

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

Taking Laplace transform with zero initial conditions,

$$sX(s) = AX(s) + BU(s)$$

$$(sI - A)X(s) = BU(s)$$

$$X(s) = (sI - A)^{-1}BU(s)$$

Therefore,

$$Y(s) = CX(s) + DU(s) = C(sI-A)^{-1}BU(s) + DU(s) = \left[C(sI-A)^{-1}B + D\right]U(s)$$

Hence, the transfer function is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

$$\boxed{(A)\ C(sI - A)^{-1}B + D}$$

Q9.93.2 MCQ 1993 1M Ans: B ● ● ○

Let,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}e\\ f\end{bmatrix}u(t)$$

$$A = \begin{bmatrix}a & b\\ c & d\end{bmatrix}, \qquad B = \begin{bmatrix}e\\ f\end{bmatrix}$$

Therefore,

$$AB = \begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}e\\ f\end{bmatrix} = \begin{bmatrix}ae + bf\\ ce + df\end{bmatrix}$$

Since, $x_1(t) = x_2(t)$, we can write

$$x_1(t) = x_2(t) = x(t)$$

Therefore,

$$\dot{x}_1 = (a + b)x(t) + eu(t) \quad \text{and} \quad \dot{x}_2 = (c + d)x(t) + fu(t)$$

Since $x_1(t) = x_2(t)$ for all t, their derivatives must also be equal, i.e.,

$$\dot{x}_1(t) = \dot{x}_2(t)$$

$$(a + b)x(t) + eu(t) = (c + d)x(t) + fu(t)$$

Since the above relation must hold for all $x(t)$ and $u(t)$, the corresponding coefficients must be equal. Hence,

$$a + b = c + d, \qquad e = f$$

The controllability matrix is

$$Q_c = \begin{bmatrix}B & AB\end{bmatrix} = \begin{bmatrix}e & ae + bf\\ f & ce + df\end{bmatrix}$$

Using $e = f$ and $a + b = c + d$,

$$Q_c = \begin{bmatrix}e & e(a + b)\\ e & e(c + d)\end{bmatrix} = \begin{bmatrix}e & e(a + b)\\ e & e(a + b)\end{bmatrix}$$

$$|Q_c| = 0$$

Hence, the system is uncontrollable.

$$\boxed{(B)\ \text{Uncontrollable}}$$

Q9.88.1 MCQ 1988 1M Ans: C ● ○ ○

Given,

$$A = \begin{bmatrix}-2 & 0\\ 0 & -4\end{bmatrix}$$

The state transition matrix is

$$\phi(t) = e^{At}$$

Since A is a diagonal matrix,

$$e^{At} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-4t} \end{bmatrix}$$

$$\therefore \quad \phi(t) = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-4t} \end{bmatrix}$$

$$\boxed{(C)\ \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-4t} \end{bmatrix}}$$

DEBUG INFO

Chapter IX

State Space Analysis

Total Questions : 47

MCQ : 34

MSQ : 0

NAT : 13

SOLUTIONS

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

Q10.25.1 NAT 2025 2M Ans: 0.1 to 0.1

Given,

$$D(s) = 1 + K_D s$$

$$G(s) = \frac{1000\sqrt{2}}{s(s+10)^2}$$

The open-loop transfer function is

$$L(s) = D(s)G(s) = \frac{1000\sqrt{2}(1 + K_D s)}{s(s+10)^2}$$

Given gain crossover frequency and Phase margin are: $\omega_{gc} = 10$ rad/s and $45°$.
Therefore,

$$45° = -180° + \angle L(j10)$$

$$\angle L(j10) = -180° + 45° = -135°$$

Now,

$$\angle L(j10) = \angle(1 + j10K_D) - \angle(j10) - 2\angle(10 + j10) = \tan^{-1}(10K_D) - 90° - 90° = \tan^{-1}(10K_D) - 180°$$

Equating with the required phase,

$$\tan^{-1}(10K_D) - 180° = -135°$$

$$\tan^{-1}(10K_D) = 45°$$

$$10K_D = 1$$

$$K_D = 0.1$$

Verify the gain crossover condition:

$$|L(j10)| = \frac{1000\sqrt{2}\,|1 + j|}{10\,|10 + j10|^2} = \frac{1000\sqrt{2}\,\sqrt{2}}{10\,(10\sqrt{2})^2} = 1$$

Hence, $\omega_{gc} = 10$ rad/s is satisfied.

$$\boxed{0.1}$$

Q10.23.1 MCQ 2023 1M Ans: D

Given,

$$G(s) = \frac{1}{s-1}$$

The proportional gain and integral gain of the PI controller be K_P and K_I respectively. Thus,

$$K_P = 3, \qquad K_I = 1$$

Therefore,

$$C(s) = K_P + \frac{K_I}{s} = 3 + \frac{1}{s} = \frac{3s+1}{s}$$

Let the block diagram be,

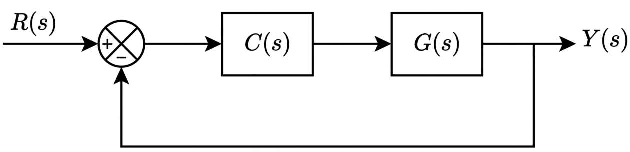


Fig. Solution Q10.23.1

The open-loop transfer function is

$$C(s)G(s) = \frac{3s+1}{s(s-1)}$$

For a unit-gain negative feedback system,

$$\frac{Y(s)}{R(s)} = \frac{C(s)G(s)}{1+C(s)G(s)} = \frac{\frac{3s+1}{s(s-1)}}{1+\frac{3s+1}{s(s-1)}} = \frac{3s+1}{(s+1)^2}$$

For a unit step input, $R(s) = \frac{1}{s}$, hence

$$Y(s) = \frac{3s+1}{s(s+1)^2}$$

Using the Final Value Theorem,

$$y(\infty) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} \frac{3s+1}{(s+1)^2} = 1$$

The controller output is

$$U(s) = C(s)(R(s) - Y(s)) = \frac{3s+1}{s(s-1)} \times \left(\frac{1}{s} - \frac{3s+1}{s(s+1)^2}\right) = \frac{(3s+1)(s-1)}{s(s+1)^2}$$

Again using the Final Value Theorem,

$$u(\infty) = \lim_{s\to 0} sU(s) = \lim_{s\to 0} \frac{(3s+1)(s-1)}{(s+1)^2} = -1$$

Therefore, the final values of controller output and plant output are

$$u(\infty) = -1, \qquad y(\infty) = 1$$

$$\boxed{\text{(D)} \ -1,\ 1}$$

Q10.23.2 MCQ 2023 2M Ans: B ● ● ○

Given lead compensator,

$$K(s) = \frac{1+\frac{s}{a}}{1+\frac{s}{\beta a}}, \qquad \beta > 1,\ a > 1$$

For a lead compensator, the frequency at which maximum phase lead occurs is

$$\omega_m = a\sqrt{\beta}$$

Given,

$$\omega_m = 4 \text{ rad/s}$$

Therefore,

$$a\sqrt{\beta} = 4$$
$$a^2\beta = 16$$

Also, the gain amplification provided by the compensator at ω_m is $20\log_{10}\left(\sqrt{\beta}\right)$
Given gain amplification is 6 dB,

$$20\log_{10}\left(\sqrt{\beta}\right) = 6$$

$$\beta \approx 4$$

Thus

$$a\sqrt{\beta} = 4,$$
$$a(2) = 4$$
$$a = 2$$

$$\boxed{\text{(B) } 2,\ 4}$$

Q10.21.1 NAT 2021 2M Ans: 10.80 to 11 ● ○ ○

Given,

$$G_p(s) = \frac{14.4}{s(1+0.1s)} \qquad G_c(s) = 1$$

The open-loop transfer function is

$$G(s) = G_p(s)G_c(s) = \frac{14.4}{s(1+0.1s)}$$

For unity feedback,

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\frac{14.4}{s(1+0.1s)}}{1+\frac{14.4}{s(1+0.1s)}} = \frac{144}{s^2+10s+144}$$

Comparing the characteristic equation with the standard second-order form, $s^2 + 2\zeta\omega_n s + \omega_n^2$

$$\omega_n^2 = 144 \qquad => \omega_n = 12 \text{ rad/s}$$

$$2\zeta\omega_n = 10$$
$$2\zeta(12) = 10$$
$$\zeta = \frac{5}{12}$$

The damped natural frequency is

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 12\sqrt{1-\left(\frac{5}{12}\right)^2} \approx 10.91 \text{ rad/s}$$

$$\boxed{10.91}$$

Q10.21.2 NAT 2021 2M Ans: 9.50 to 9.60 ● ○ ○

For disturbance response, set $R(s) = 0$. Then from the block diagram,

$$E(s) = -C(s)$$

The transfer function from disturbance input to output is

$$\frac{C(s)}{D(s)} = \frac{G_p(s)}{1+G_c(s)G_p(s)}$$

For a unit step disturbance, $D(s) = \frac{1}{s}$

Therefore, the steady-state error is, $e_{ss} = \lim_{s\to 0} sE(s) = -\lim_{s\to 0} sC(s) = -\lim_{s\to 0} \dfrac{G_p(s)}{1+G_c(s)G_p(s)}$

At $s = 0$,

$$G_p(0) = 2.2, \qquad G_c(0) = K$$

Hence,

$$|e_{ss}| = \frac{2.2}{1+2.2K}$$

Given that the steady-state error should not exceed 0.1,

$$\frac{2.2}{1+2.2K} \le 0.1$$

$$2.2 \le 0.1 + 0.22K$$

$$0.22K \ge 2.1$$

$$K \ge 9.545$$

$$\boxed{9.55}$$

Q10.19.1 MCQ 2019 2M Ans: B ● ○ ○

The transfer function of the phase lead compensator is

$$D(s) = 3\left(\frac{s+\frac{1}{3T}}{s+\frac{1}{T}}\right) = \frac{1+s(3T)}{1+sT}$$

Comparing with the standard form

$$D(s) = K_c\left(\frac{1+sT_z}{1+sT_p}\right)$$

we get

$$K_c = 1, \qquad T_z = 3T, \qquad T_p = T$$

For a phase lead compensator, the frequency at which the phase is maximum is

$$\omega_m = \frac{1}{\sqrt{T_z T_p}}$$

Therefore,

$$\omega_m = \frac{1}{\sqrt{(3T)(T)}} = \frac{1}{\sqrt{3T^2}} = \sqrt{\frac{1}{3T^2}}$$

$$\boxed{(B)\ \sqrt{\frac{1}{3T^2}}\ \text{rad/s}}$$

Key Points

- For a phase lead compensator, the frequency corresponding to maximum phase is $\omega_m = \dfrac{1}{\sqrt{T_z T_p}}$
- Always convert the compensator into the standard form $K_c \dfrac{1+sT_z}{1+sT_p}$ before applying the formula.

Q10.17.1 MCQ 2017 1M Ans: A ● ○ ○

The given compensator is

$$C(s) = \frac{\left(1 + \frac{s}{0.1}\right)\left(1 + \frac{s}{100}\right)}{(1 + s)\left(1 + \frac{s}{10}\right)}$$

The pole-zero locations are $\omega_{z1} = 0.1$, $\omega_{z2} = 100$ and $\omega_{p1} = 1$, $\omega_{p2} = 10$.

The factor $\dfrac{1 + \frac{s}{0.1}}{1 + s}$ introduces phase lead since $\omega_{z1} < \omega_{p1}$. The factor $\dfrac{1 + \frac{s}{100}}{1 + \frac{s}{10}}$ introduces phase lag since $\omega_{z2} > \omega_{p2}$

For a lead compensator, the maximum phase lead occurs in the frequency range between its zero and pole frequencies. Hence,

$$0.1 < \omega < 1$$

$$\boxed{(A)\ \ 0.1 < \omega < 1}$$

Key Points

- A compensator of the form $\dfrac{1 + sT_z}{1 + sT_p}$ introduces phase lead when $\omega_z < \omega_p$.
- The frequency corresponding to maximum phase lead lies between the zero and pole frequencies of the lead network.
- Analyse each pole-zero pair separately when multiple lead and lag networks are cascaded.

Q10.14.1 MCQ 2014 2M Ans: C ● ● ○

The magnitude plot has a slope of 20 dB/decade between $\omega = \dfrac{1}{3}$ rad/s and $\omega = 1$ rad/s. Hence, the network has

$$\omega_z = \frac{1}{3}\ \text{rad/s}, \qquad \omega_p = 1\ \text{rad/s}$$

Since $\omega_z < \omega_p$ the network is a phase lead compensator. The maximum phase lead is

$$\phi_m = \sin^{-1}\left(\frac{\omega_p - \omega_z}{\omega_p + \omega_z}\right) = \sin^{-1}\left(\frac{1 - \frac{1}{3}}{1 + \frac{1}{3}}\right) = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

The gain at $\omega = 1$ rad/s is, $G_m = 20\log_{10}\left(\dfrac{1}{1/3}\right) = 20\log_{10}(3) = 9.54$ dB.

The corresponding gain at maximum phase lead frequency is half of this value, $G_m = \dfrac{9.54}{2} = 4.77$ dB

$$\boxed{(C)\ \ +30^\circ \text{ and } 4.77\,\text{dB}}$$

Key Points

- For a phase lead compensator, the frequency corresponding to maximum phase lead is $\omega_m = \sqrt{\omega_z \omega_p}$. Hence, ω_m lies exactly at the midpoint between ω_z and ω_p on the $\log\omega$ axis.
- Between ω_z and ω_p, the Bode magnitude plot is a straight line with a slope of 20 dB/decade. Therefore, the gain at ω_m is exactly halfway between the starting and ending gain values.

- Consequently, the gain at maximum phase lead frequency is half of the total gain rise:

$$G_m = \frac{1}{2} \times 20 \log_{10}\left(\frac{\omega_p}{\omega_z}\right) = 10 \log_{10}\left(\frac{\omega_p}{\omega_z}\right).$$

Q10.12.1 MCQ 2012 2M Ans: A

Given,

$$G_c(s) = \frac{s+a}{s+b}$$

For a lead compensator, the zero must be dominant. That means the zero should be nearer to the origin than the pole.
Zero location: $s = -a$ and Pole location: $s = -b$
For lead compensation,

$$-a > -b$$

$$a < b$$

Checking the options:
Option A:

$$a = 1, \quad b = 2$$

$$1 < 2$$

It satisfies the lead compensator condition.

$$\boxed{\text{(A) a=1, b=2}}$$

Q10.12.2 MCQ 2012 2M Ans: A

From Q10.12.1, the lead compensator is

$$G_c(s) = \frac{s+1}{s+2}$$

Maximum phase occurs at the geometric mean of pole and zero locations.

$$\omega_m = \sqrt{z \times p}$$

Here, $z = -1$ and $p = -2$

$$\omega_m = \sqrt{1 \times 2}$$

$$\omega_m = \sqrt{2}\,\text{rad/s}$$

$$\boxed{(A)\ \sqrt{2}\,\text{rad/s}}$$

Q10.08.1 MCQ 2008 2M Ans: A

The given compensators are

$$C_1 = 10\left(\frac{s+1}{s+10}\right), \qquad C_2 = \frac{1}{10}\left(\frac{s+10}{s+1}\right)$$

For C_1, $\omega_z = 1$ rad/s and $\omega_p = 10$ rad/s. Since $\omega_z < \omega_p$, C_1 is a phase lead compensator.
For C_2, $\omega_z = 10$ rad/s and $\omega_p = 1$ rad/s. Since $\omega_z > \omega_p$, C_2 is a phase lag compensator.

$$\boxed{(A)\ \ C_1 \text{ is a lead compensator and } C_2 \text{ is a lag compensator}}$$

Q10.07.1 MCQ | 2007 | 2M | Ans: D | ● ● ●

The uncompensated system is

$$G(s) = \frac{900}{s(s+1)(s+9)} = \frac{900}{(j\omega)(1+j\omega)(9+j\omega)}$$

Phase crossover frequency (ω_{pc}):

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{9}\right)$$

At $\omega = \omega_{pc}$,

$$-90^\circ - \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{9}\right) = -180^\circ$$

$$\tan^{-1}(\omega) + \tan^{-1}\left(\frac{\omega}{9}\right) = 90^\circ$$

Using $\tan(A+B) = \infty \quad \Rightarrow \quad 1 - \tan A \tan B = 0$

$$1 - \omega\left(\frac{\omega}{9}\right) = 0$$

$$\frac{\omega^2}{9} = 1$$

$$\omega_{pc} = 3 \text{ rad/s}$$

Given that the gain crossover frequency of the compensated system is equal to the phase crossover frequency of the uncompensated system, $(\omega_{gc})_{\text{comp}} = 3$ rad/s
The required phase margin is $(\text{PM})_{\text{comp}} = 45^\circ$

$$\text{PM} = 180^\circ + \phi$$

$$45^\circ = 180^\circ + \angle G(j\omega_{gc})$$

$$\angle G(j\omega_{gc}) = -135^\circ$$

But the phase of the uncompensated system at 3 rad/s is, $\angle G(j3) = -180^\circ$. Therefore, the compensator should provide a phase lead of

$$(-135^\circ) - (-180^\circ) = 45^\circ$$

At the gain crossover frequency,

$$|G(j\omega)|_{\text{comp}} = 0\,\text{dB}$$

$$|G(j\omega)|_{\text{comp}} = X + |G(j\omega)|_{\text{uncomp}}$$

$$|G(j\omega)|_{\text{uncomp}} = 20\log_{10}900 - 20\log_{10}\omega - 20\log_{10}\sqrt{1+\omega^2} - 20\log_{10}\sqrt{1+\left(\frac{\omega}{9}\right)^2}$$

At $\omega = 3$ rad/s,

$$|G(j3)|_{\text{uncomp}} = 20\log_{10}900 - 20\log_{10}3 - 20\log_{10}\sqrt{10} - 20\log_{10}\sqrt{\frac{10}{9}} = 59.08 - 9.54 - 10 - 19.54 = 20\,\text{dB}$$

$$0 = X + 20$$

$$X = -20\,\text{dB}$$

Hence, the compensator should provide an attenuation of 20 dB.

$$\boxed{\begin{array}{c}(D)\ \text{a lag-lead compensator that provides an attenuation of 20 dB}\\ \text{and a phase lead of } 45^\circ \text{ at the frequency of 3 rad/s}\end{array}}$$

Q10.05.1 MCQ 2005 2M Ans: C ● ● ○

The open-loop transfer function is

$$G(s) = \frac{K + 0.366s}{s(s+1)}$$

The Phase Margin is 60° for the gain crossover frequency $\omega_{gc} = 1$ rad/s

$$\text{PM} = 180^\circ + \angle G(j\omega_{gc})$$

$$60^\circ = 180^\circ + \angle G(j1)$$

$$\angle G(j1) = -120^\circ$$

At $\omega = 1$ rad/s,

$$\angle G(j1) = \tan^{-1}\left(\frac{0.366}{K}\right) - 90^\circ - \tan^{-1}(1)$$

$$\tan^{-1}\left(\frac{0.366}{K}\right) - 90^\circ - 45^\circ = -120^\circ$$

$$\tan^{-1}\left(\frac{0.366}{K}\right) = 15^\circ$$

$$\frac{0.366}{K} = \tan 15^\circ = 0.2679$$

$$K = \frac{0.366}{0.2679} = 1.366$$

$$\boxed{(C)\ 1.366}$$

Q10.03.1 MCQ 2003 1M Ans: A ● ○ ○

The given lead compensator is

$$G_c(s) = K\left(\frac{1 + \frac{s}{a}}{1 + \frac{s}{b}}\right)$$

Comparing with the standard form of a phase lead compensator,

$$G_c(s) = K\left(\frac{1 + \frac{s}{\omega_z}}{1 + \frac{s}{\omega_p}}\right), \qquad \omega_z < \omega_p$$

we obtain $\omega_z = a$, and $\omega_p = b$
For a phase lead compensator,

$$\omega_z < \omega_p$$

$$a < b$$

$$\boxed{(A)\ \ a < b}$$

Q10.00.1 MCQ 2000 2M Ans: D ● ● ○

The given compensator is

$$D(s) = \frac{0.5s + 1}{0.05s + 1} = \frac{1 + \frac{s}{2}}{1 + \frac{s}{20}}$$

Therefore, $\omega_z = 2$ rad/s, and $\omega_p = 20$ rad/s
Since $\omega_z < \omega_p$, it is a phase lead compensator.
The maximum phase lead occurs at, $\omega_m = \sqrt{\omega_z \omega_p} = \sqrt{2 \times 20} = \sqrt{40} = 6.32$ rad/s
The maximum phase lead is, $\phi_m = \sin^{-1}\left(\frac{\omega_p - \omega_z}{\omega_p + \omega_z}\right) = \sin^{-1}\left(\frac{20-2}{20+2}\right) \approx 55°$

$$\boxed{(D)\ \text{None of the above}}$$

Q10.98.1 MCQ 1998 1M Ans: B ● ○ ○

Phase lead compensation increases the bandwidth and improves the damping ratio of the system.
An increase in bandwidth makes the system respond faster, thereby decreasing the rise time.
An increase in damping ratio reduces the peak overshoot.
Therefore, phase lead compensation decreases both the rise time and the overshoot.

$$\boxed{(B)\ \text{Decrease both rise time and overshoot}}$$

Q10.97.1 MCQ 1997 1M Ans: B ● ○ ○

Introducing an integral action in the forward path adds a pole at the origin. Consequently, the system type increases by one. An increase in system type improves the steady-state accuracy of the system. In particular, the steady-state error for a step input becomes zero.

$$\boxed{(B)\ \text{system with no steady-state error}}$$

Q10.94.1 MCQ 1994 1M Ans: B ● ○ ○

A lead compensator is also called a phase-lead compensator.

Its standard form is $G_c(s) = \frac{s+z}{s+p}, \quad z < p$

This means the zero lies closer to the origin, the pole lies farther from the origin.
Now checking the options:

Option A:
Pole is closer to the origin than the Zero. This corresponds to a lag compensator.

Option B:
Zero is closer to the origin than the Zero. Hence, this represents a lead compensator.

Option C:
Two Zeros are present between two Poles. The pole-zero pair nearer to origin acts as pole dominant, hence lag compensator. The pole-zero pair farther to the origin acts as zero dominant, hence lead compensator. Therefore, this entire arrangement is called as lag-lead compensator.

Option D:
Two Poles are present between two Zeros. The pole-zero pair nearer to origin acts as zero dominant, hence lead compensator. The pole-zero pair farther to the origin acts as pole dominant, hence lag compensator. Therefore, this entire arrangement is called as lead-lag compensator.

Therefore, the correct option is,

$$\boxed{\text{(B)}}$$

DEBUG INFO

Chapter X

Controllers and Compensators

Total Questions : 18

MCQ : 15

MSQ : 0

NAT : 3

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